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Leonard et al.

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(54) **MODULAR EXTRACELLULAR SENSOR
ARCHITECTURE FOR REGULATING
GENES**

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C12N 9/22 (2006.01)

C12N 15/11 (2006.01)

(52) **U.S. Cl.**
CPC **C12N 9/22** (2013.01); **C12N 15/11**
(2013.01); **C12N 2310/20** (2017.05)

(58) **Field of Classification Search**
None
See application file for complete search history.

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Assistant Examiner — Catherine Konopka

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(57) **ABSTRACT**

Disclosed are systems, components of systems, and methods
for sensing extracellular ligands and/or modulating expres-
sion of an endogenous or exogenous gene in a cell. In some
embodiments, the disclosed systems and methods comprise
or utilize first and second exogenous extracellular sensors
and/or nucleic acid sequences encoding the first and second
exogenous extracellular sensors, wherein the first exogenous
extracellular sensor comprises: a) a ligand binding domain;
b) a transmembrane domain; c) a protease cleavage site; and
d) a functional domain comprising an RNA-binding subdo-
main fused to a transcription regulatory subdomain; and the
second exogenous extracellular sensor comprises: e) a
ligand binding domain, f) a transmembrane domain, and g)
a protease domain that cleaves the protease cleavage site of
the first exogenous extracellular receptor.

7 Claims, 51 Drawing Sheets

Specification includes a Sequence Listing.

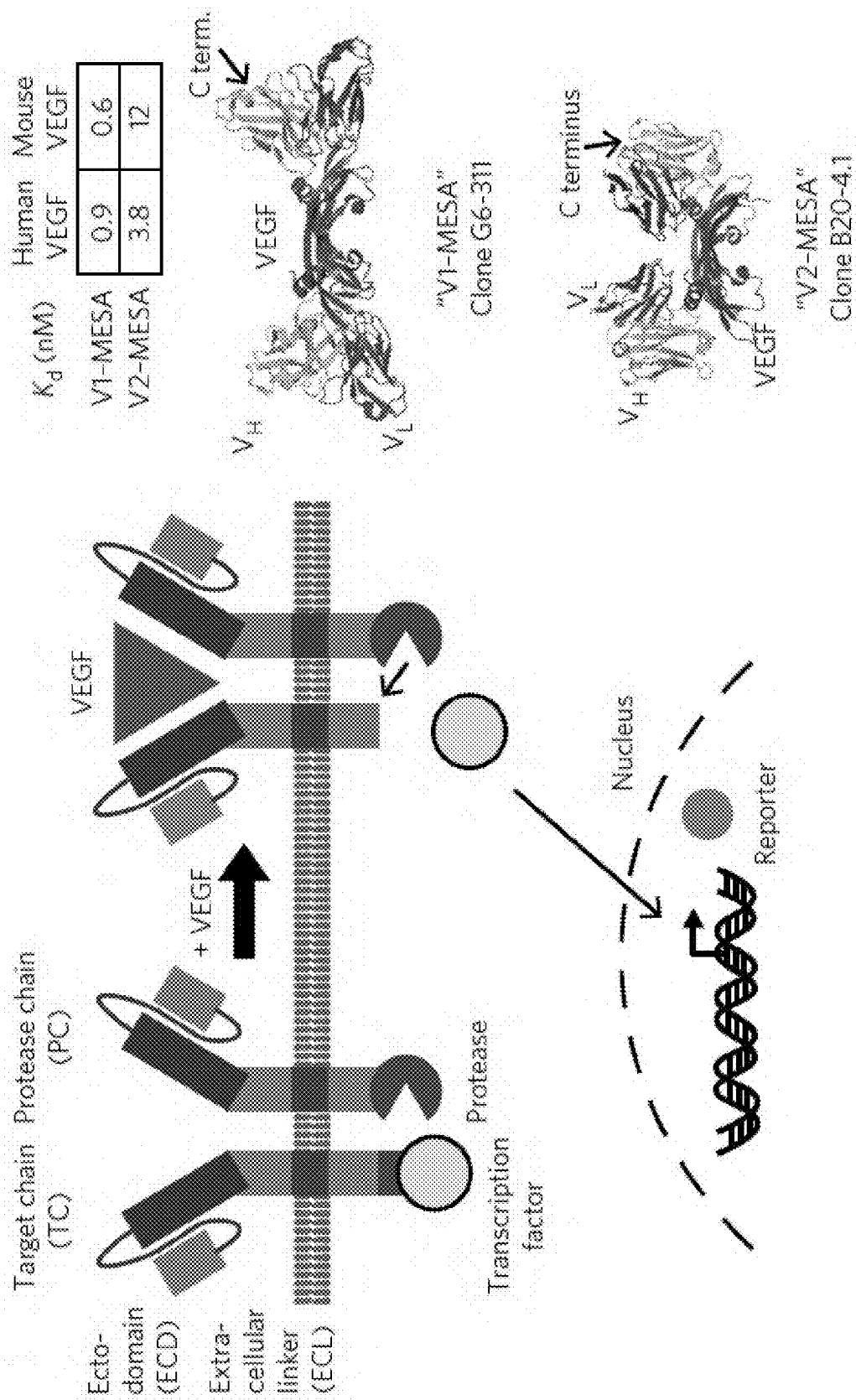
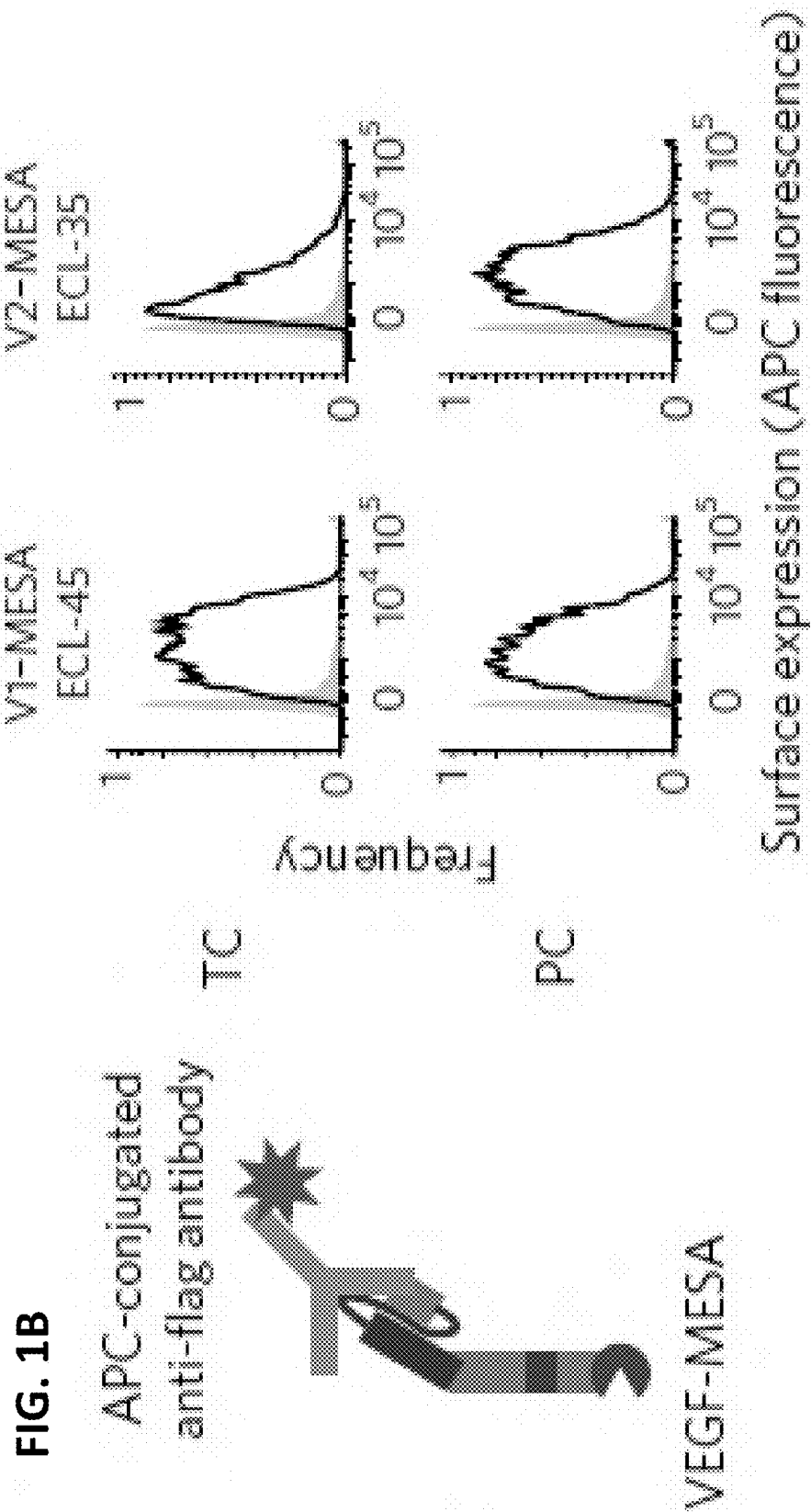


FIG. 1A



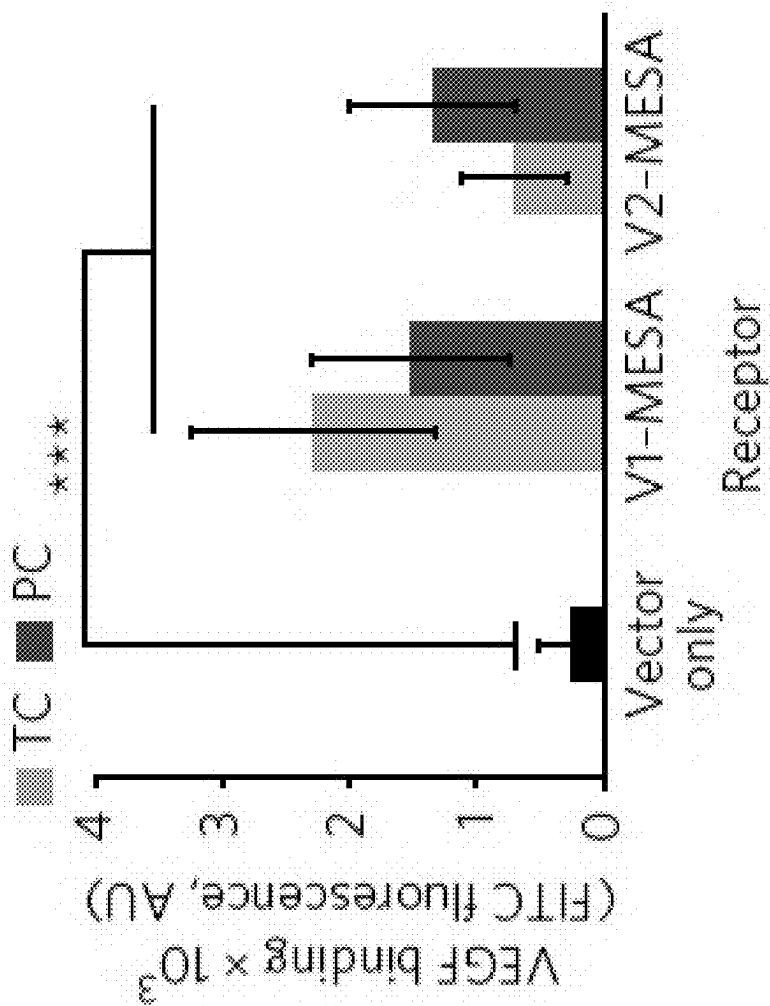
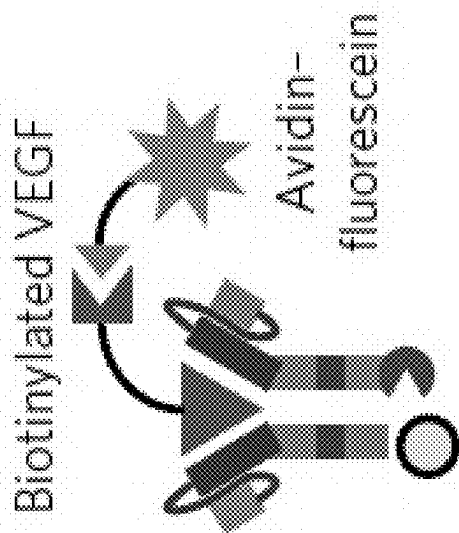
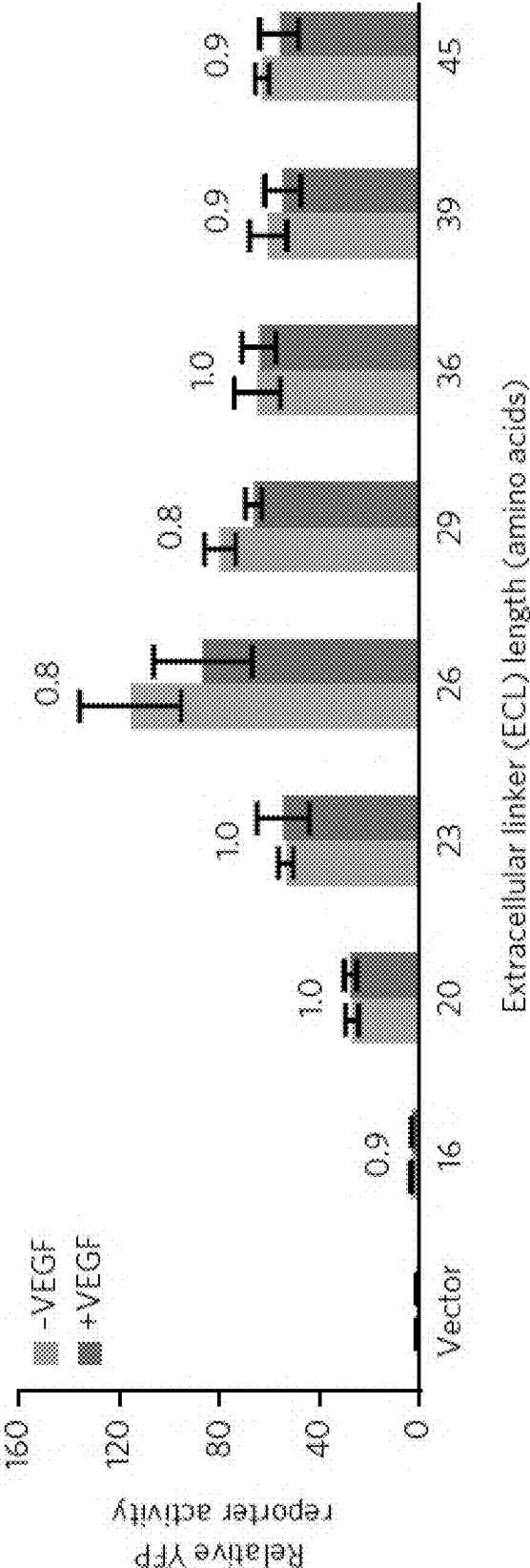


FIG. 1C



V1-MESA

FIG. 1D



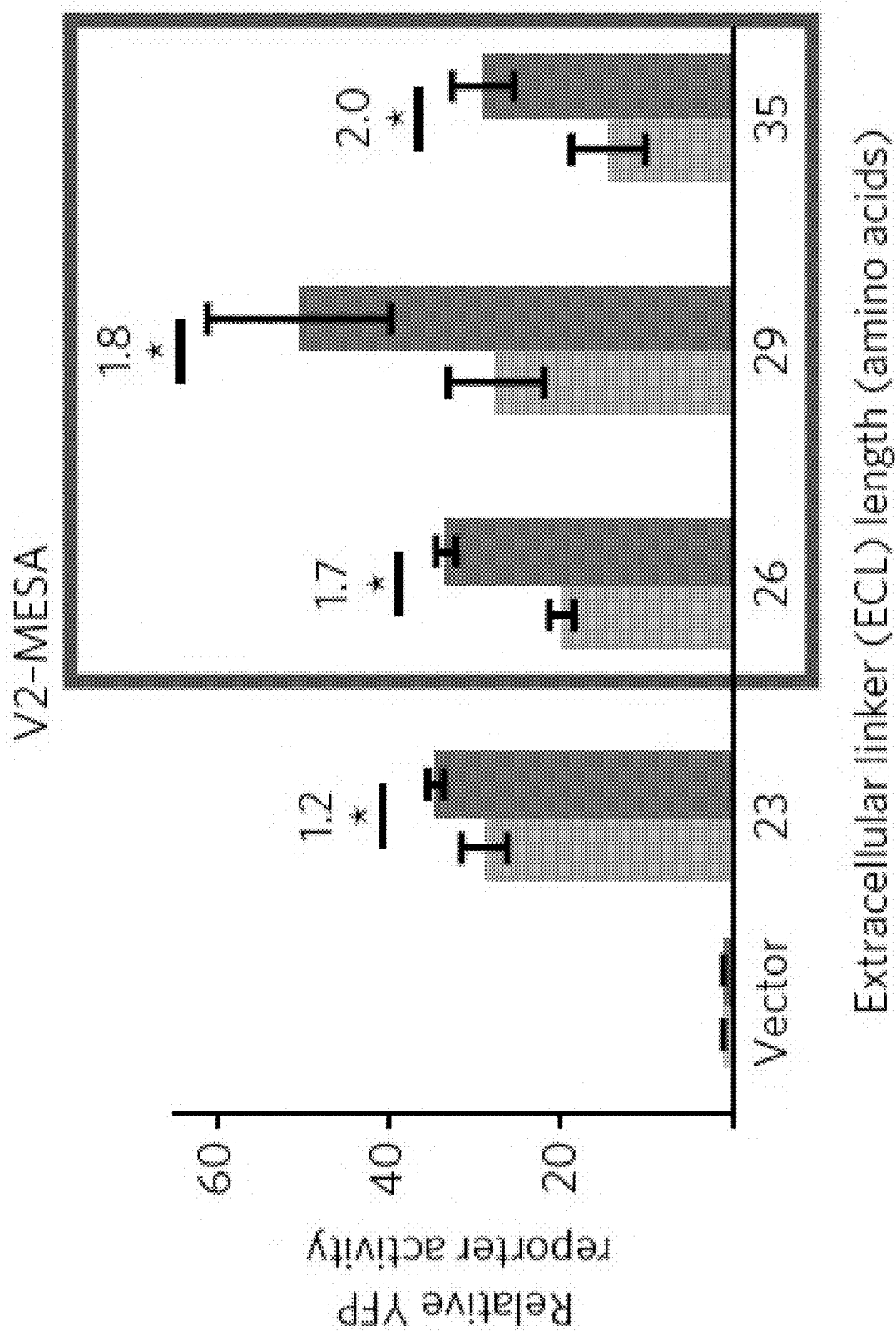


FIG. 1E

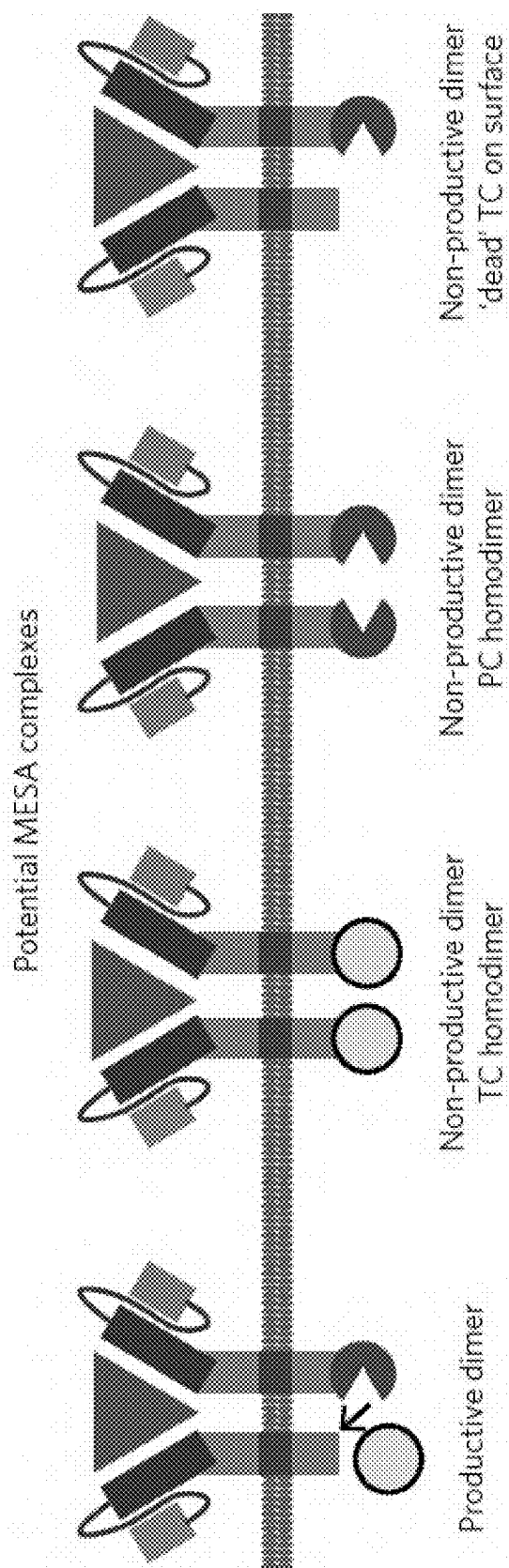


FIG. 2A

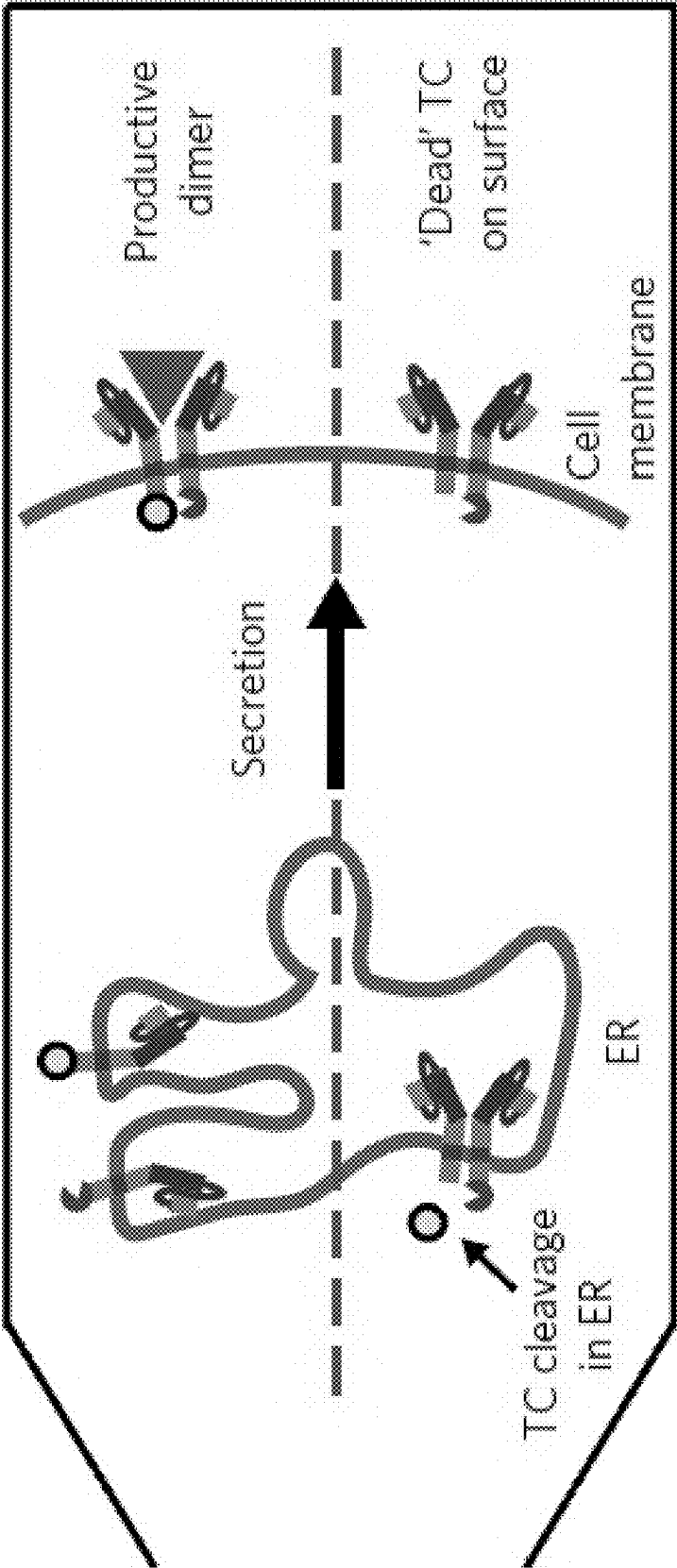


FIG. 2B

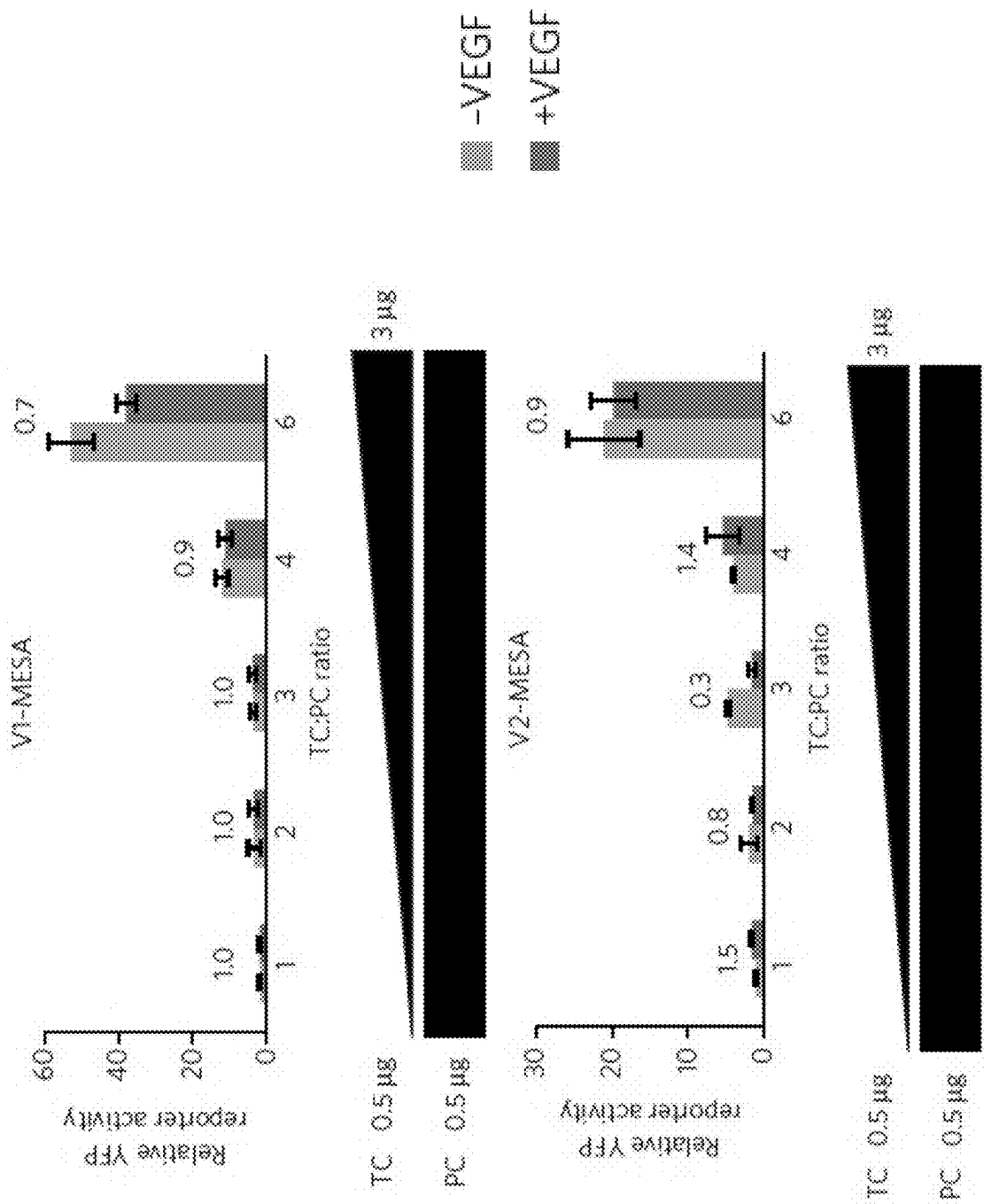


FIG. 2C

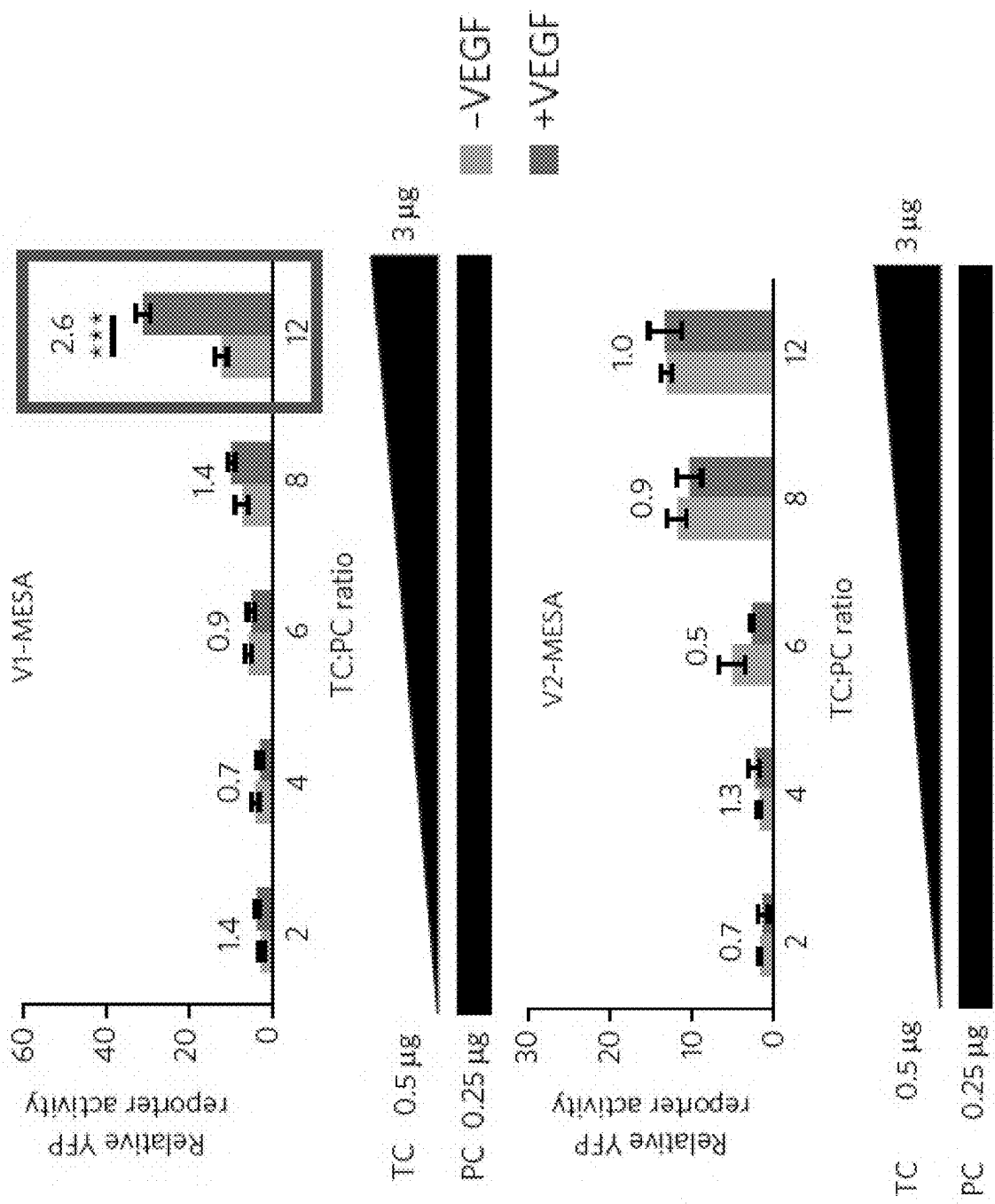


FIG. 2D

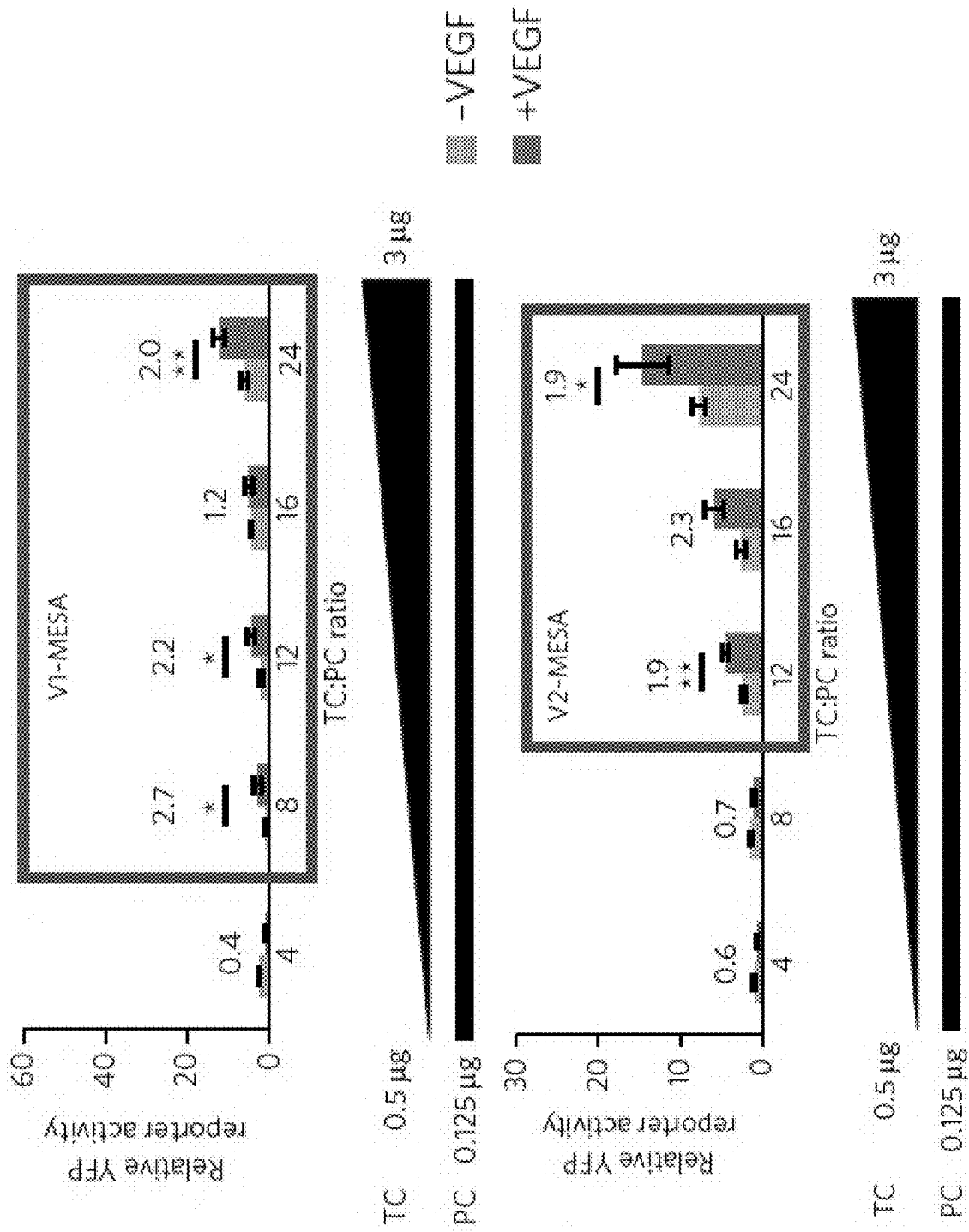
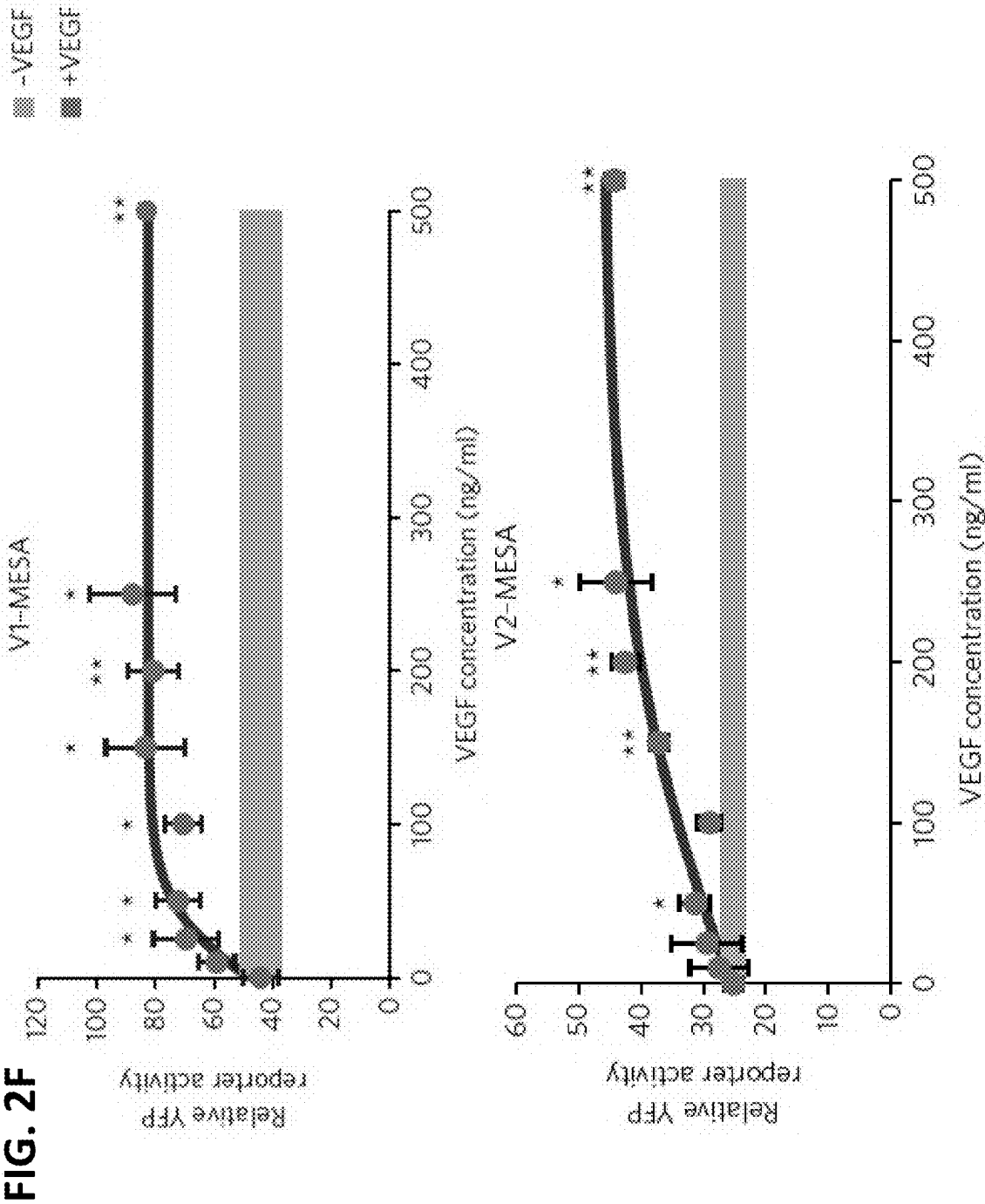


FIG. 2E



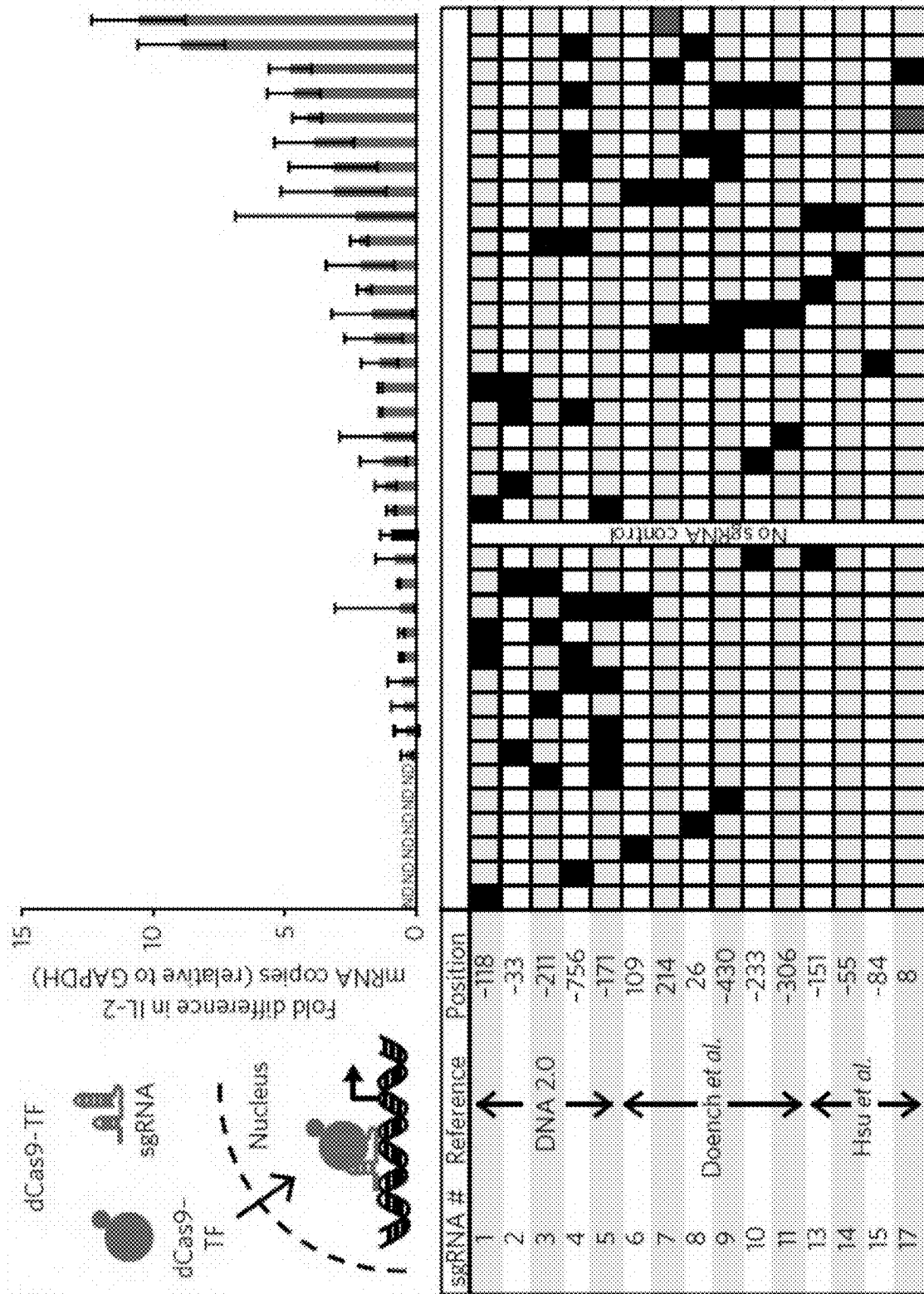
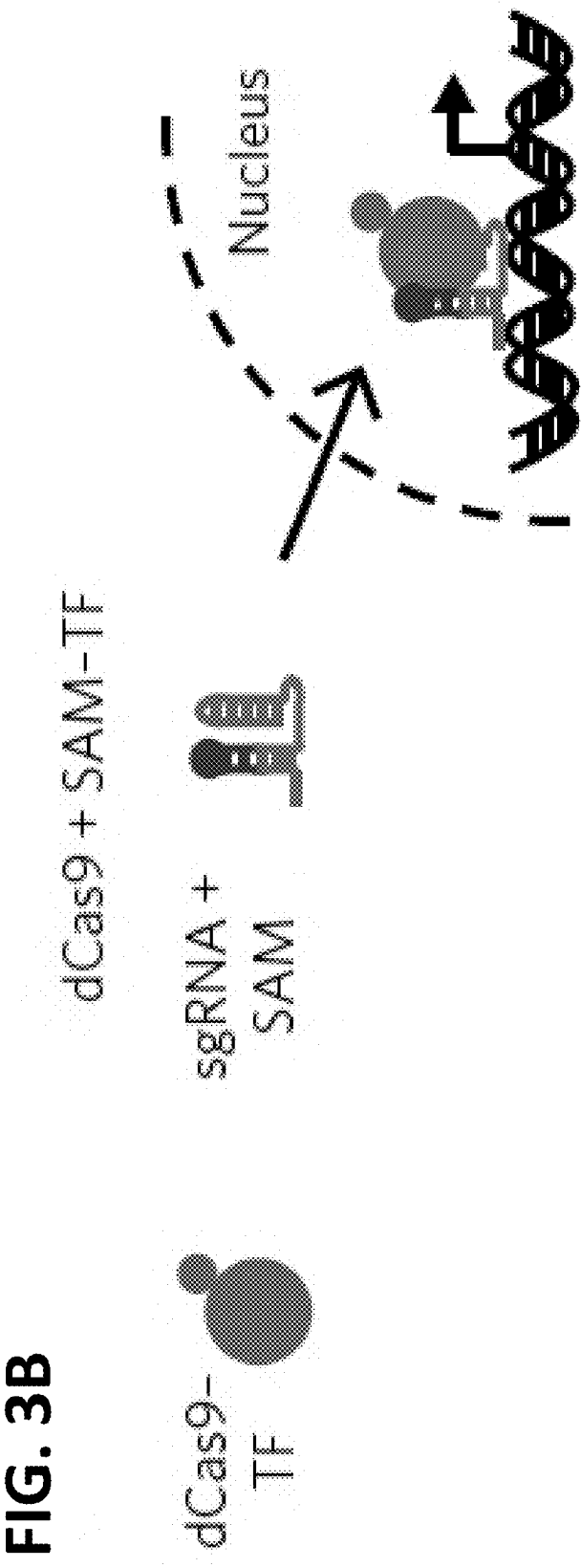


FIG. 3B



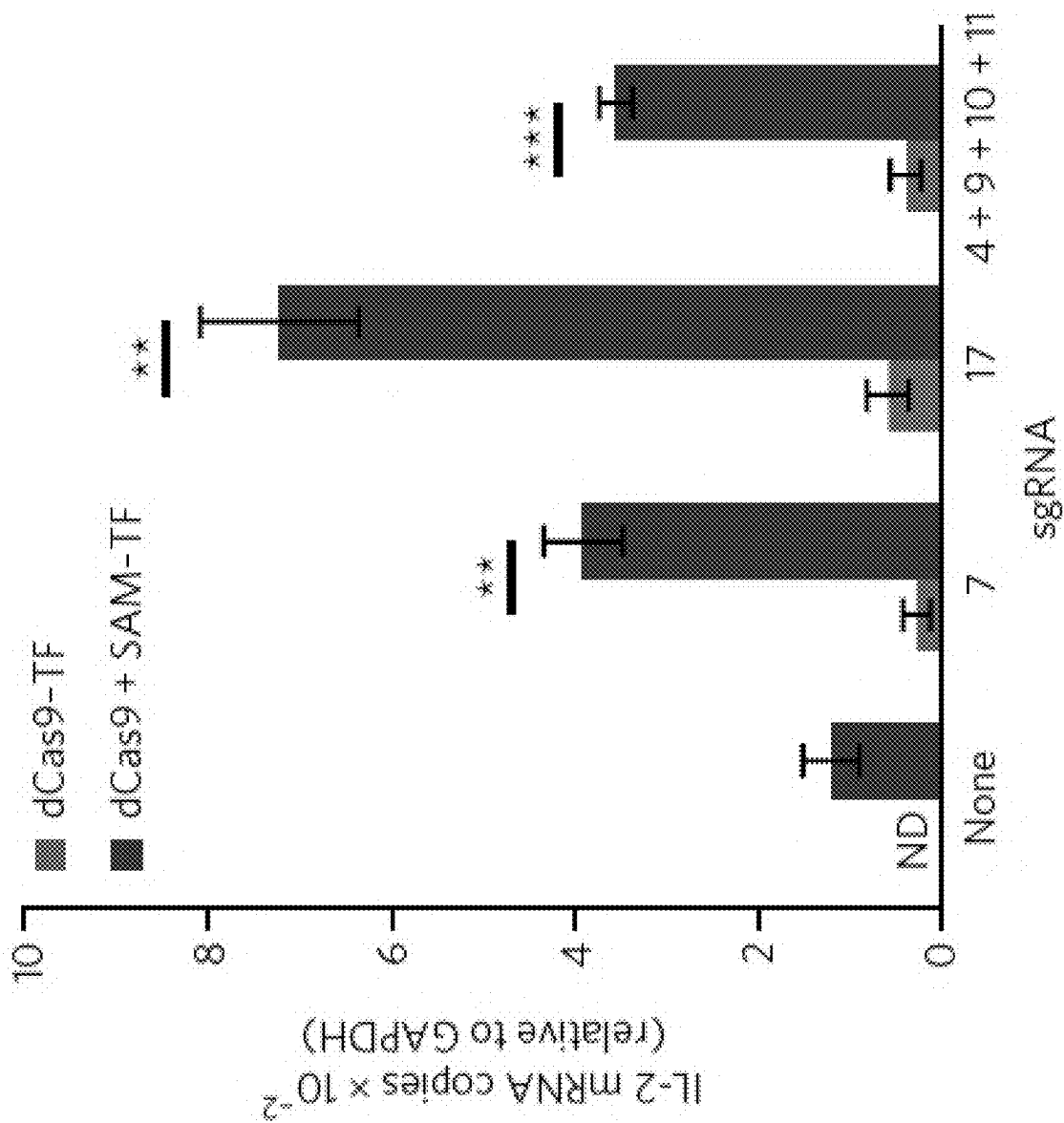


FIG. 3C

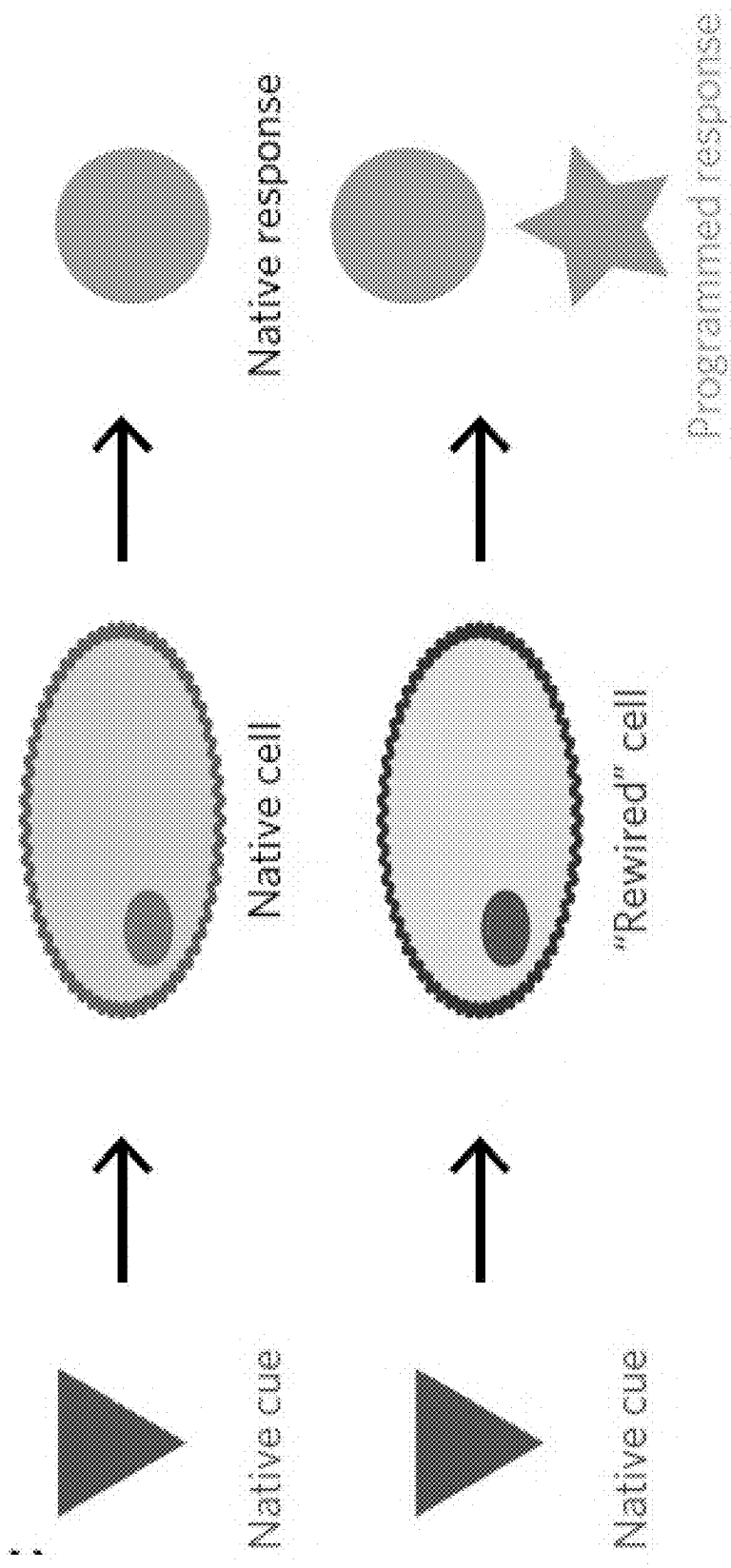


FIG. 3D

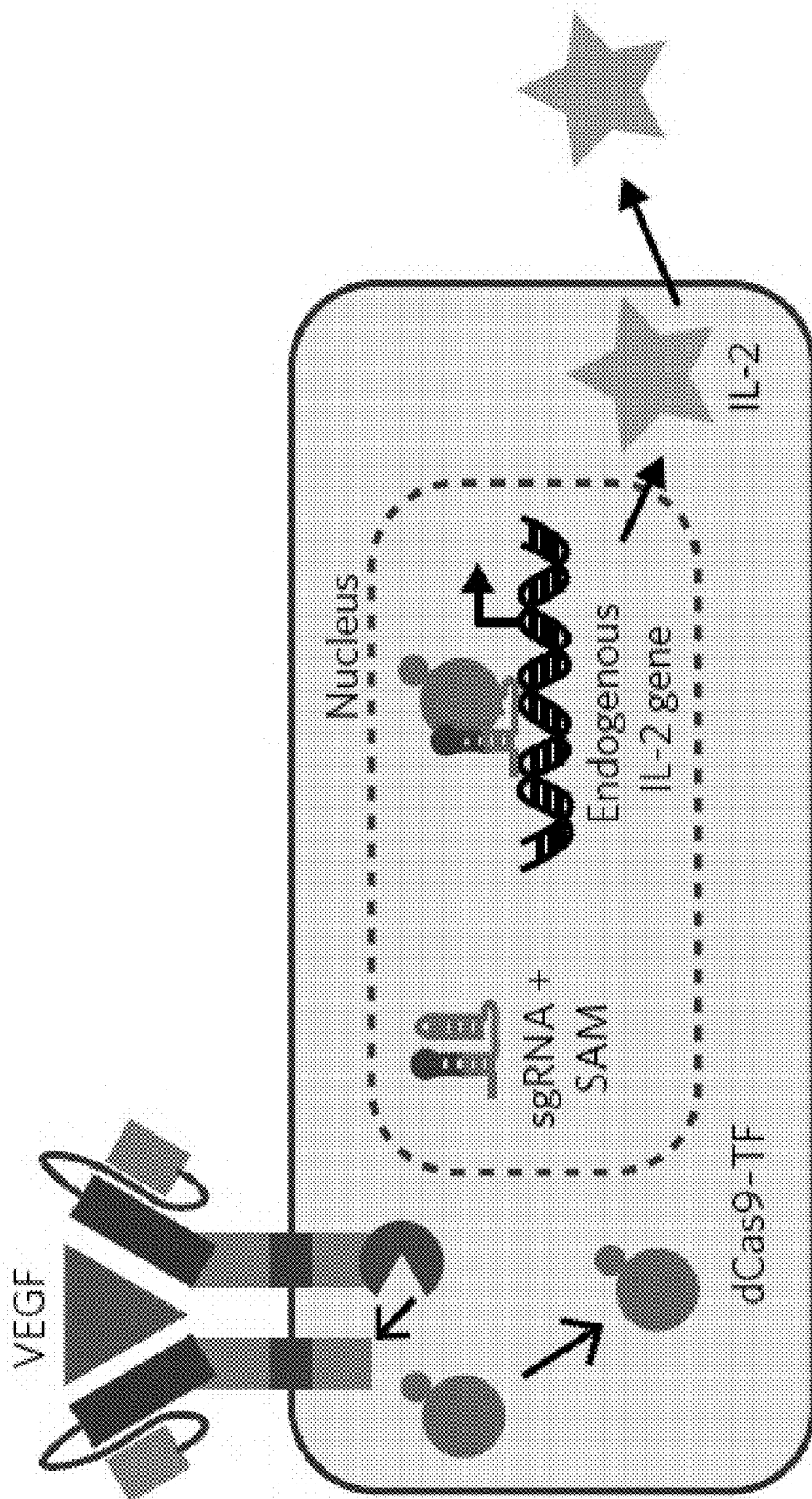


FIG. 3E

FIG. 3F

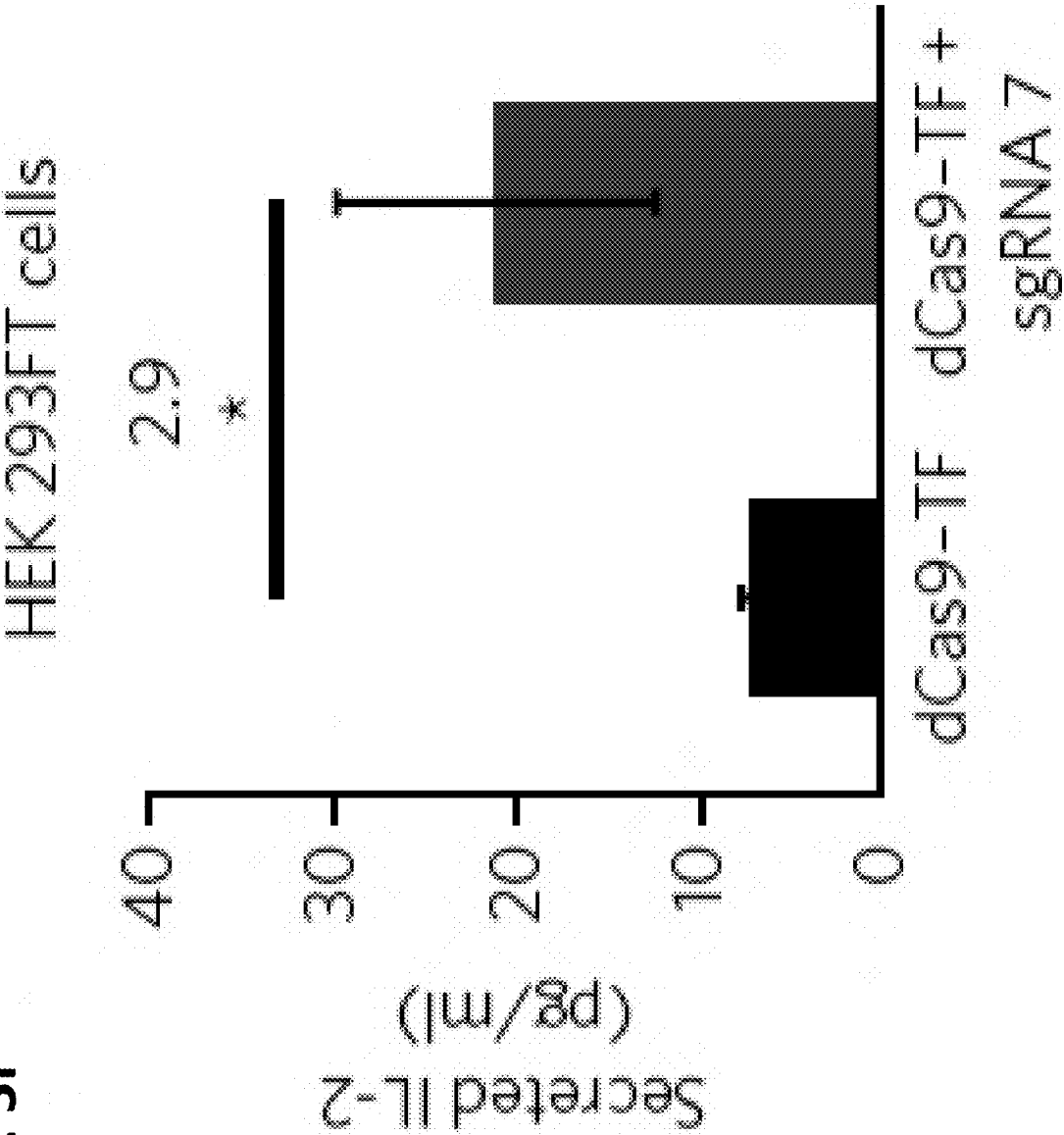


FIG. 3G

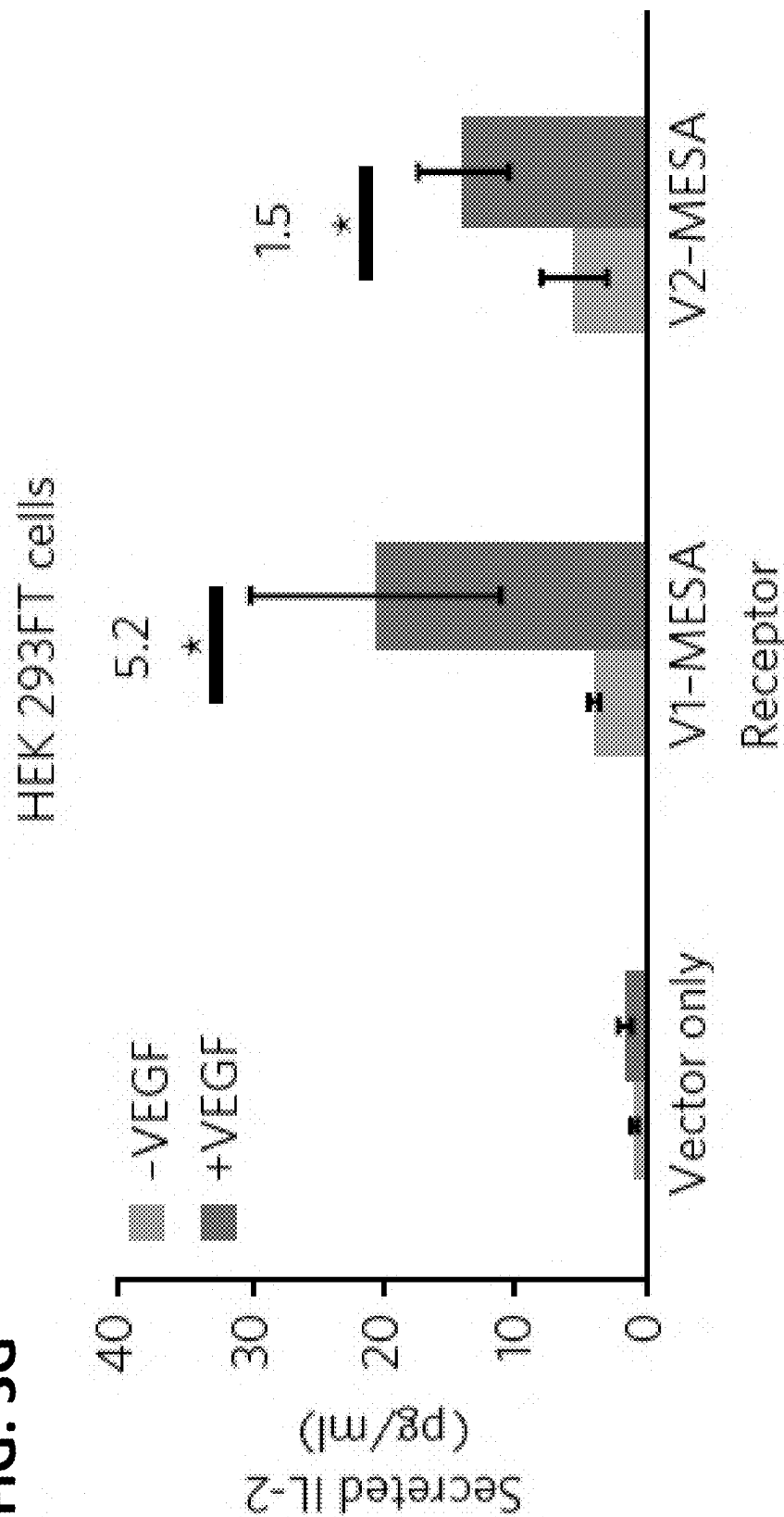


FIG. 3H

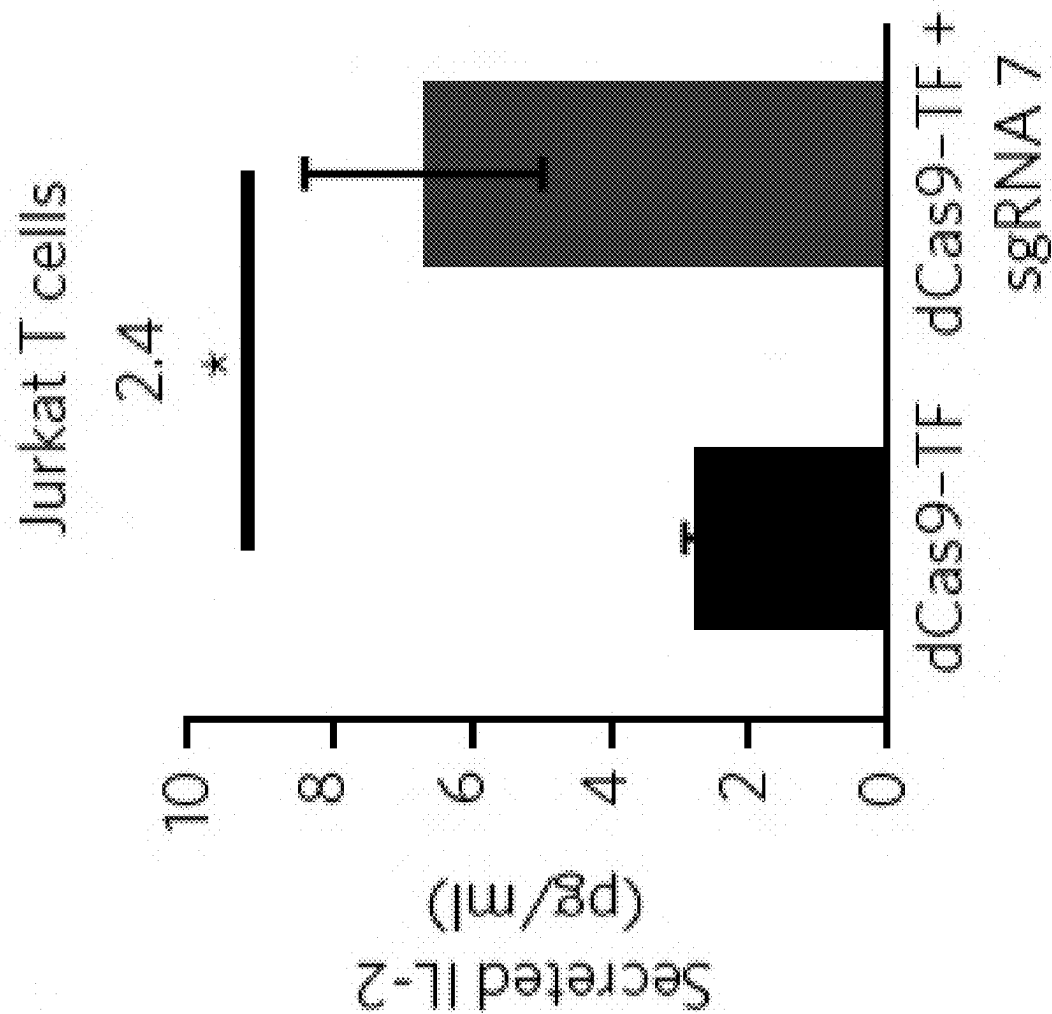
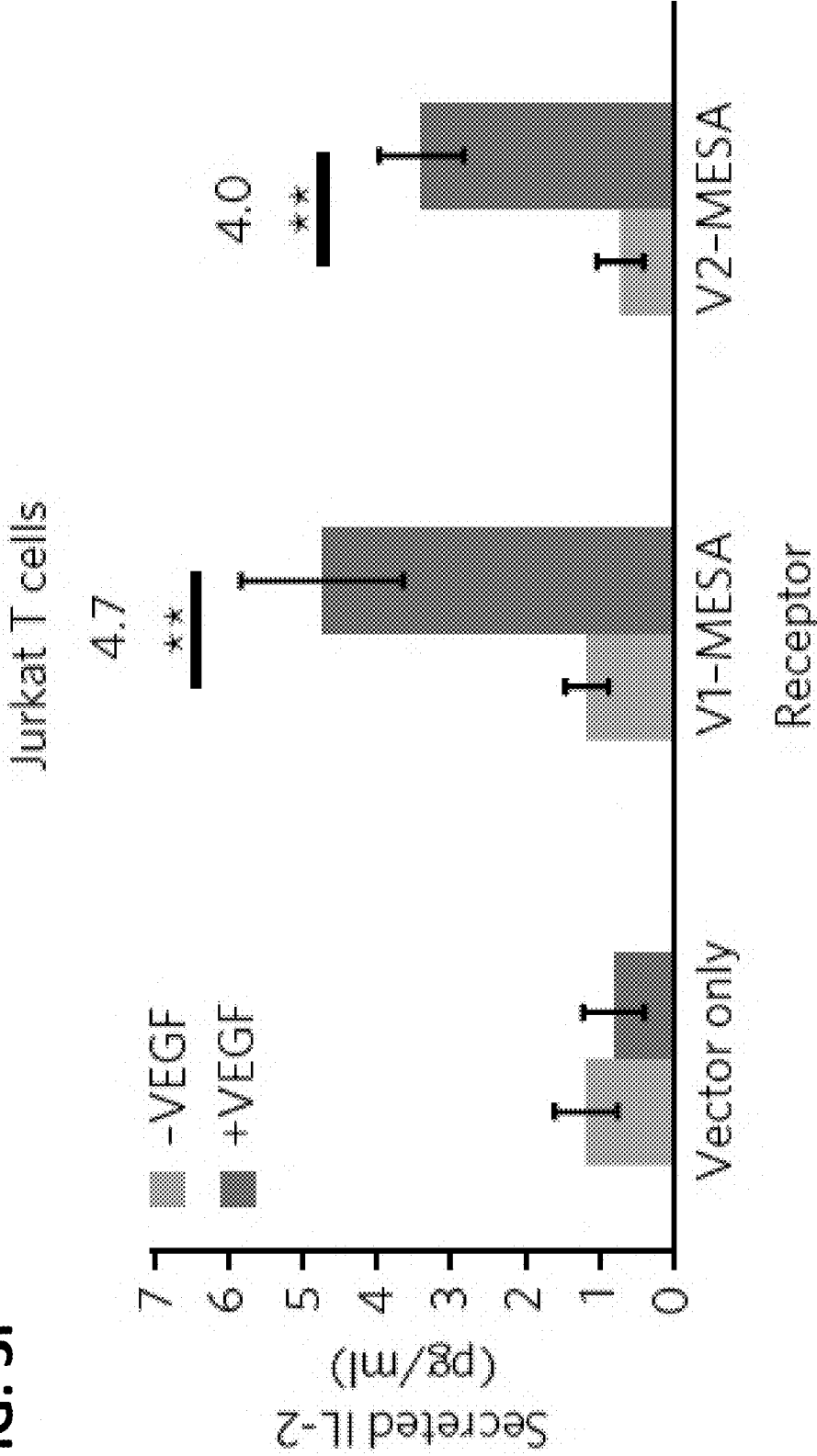


FIG. 3I



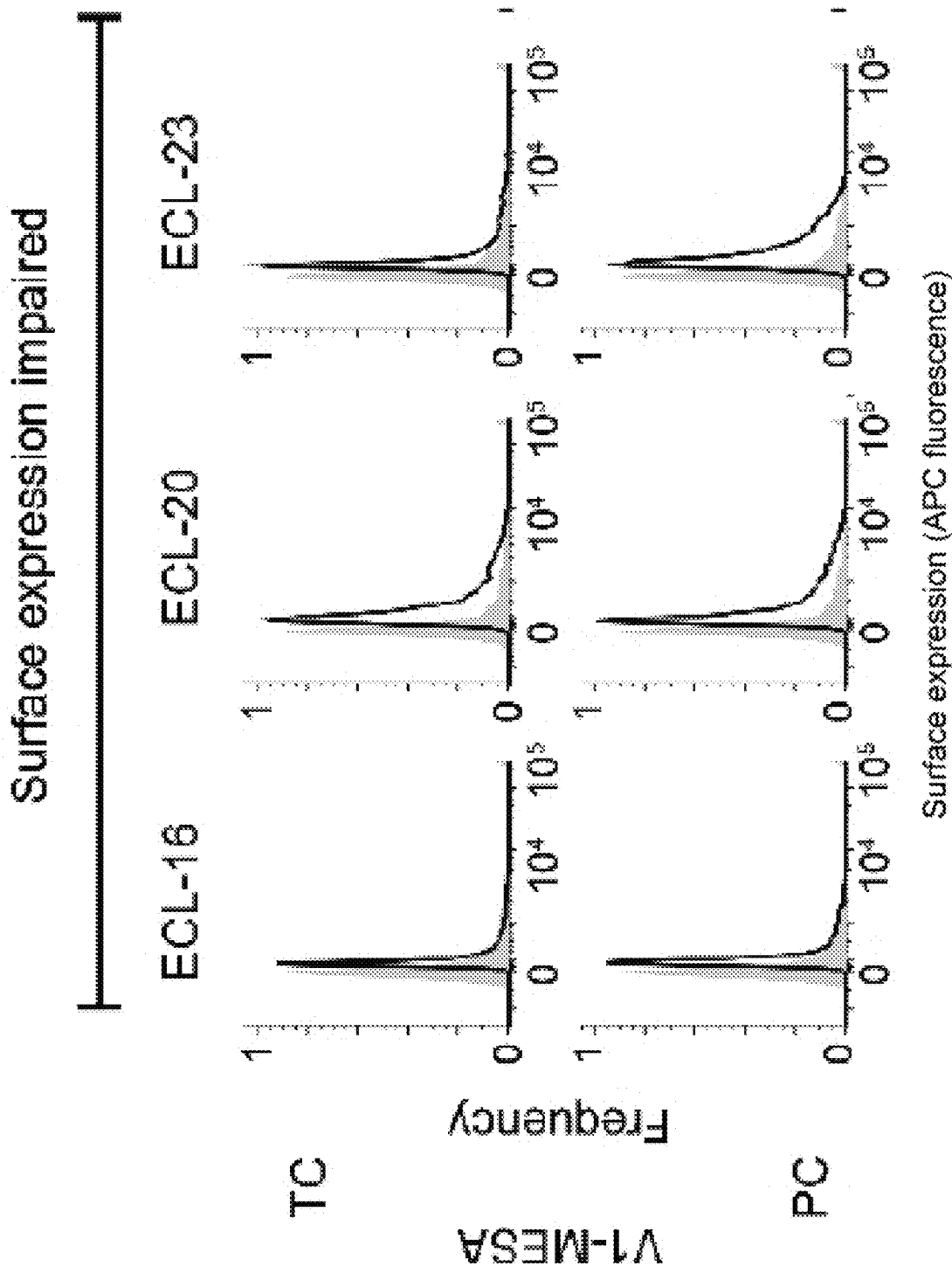


FIG. 4A

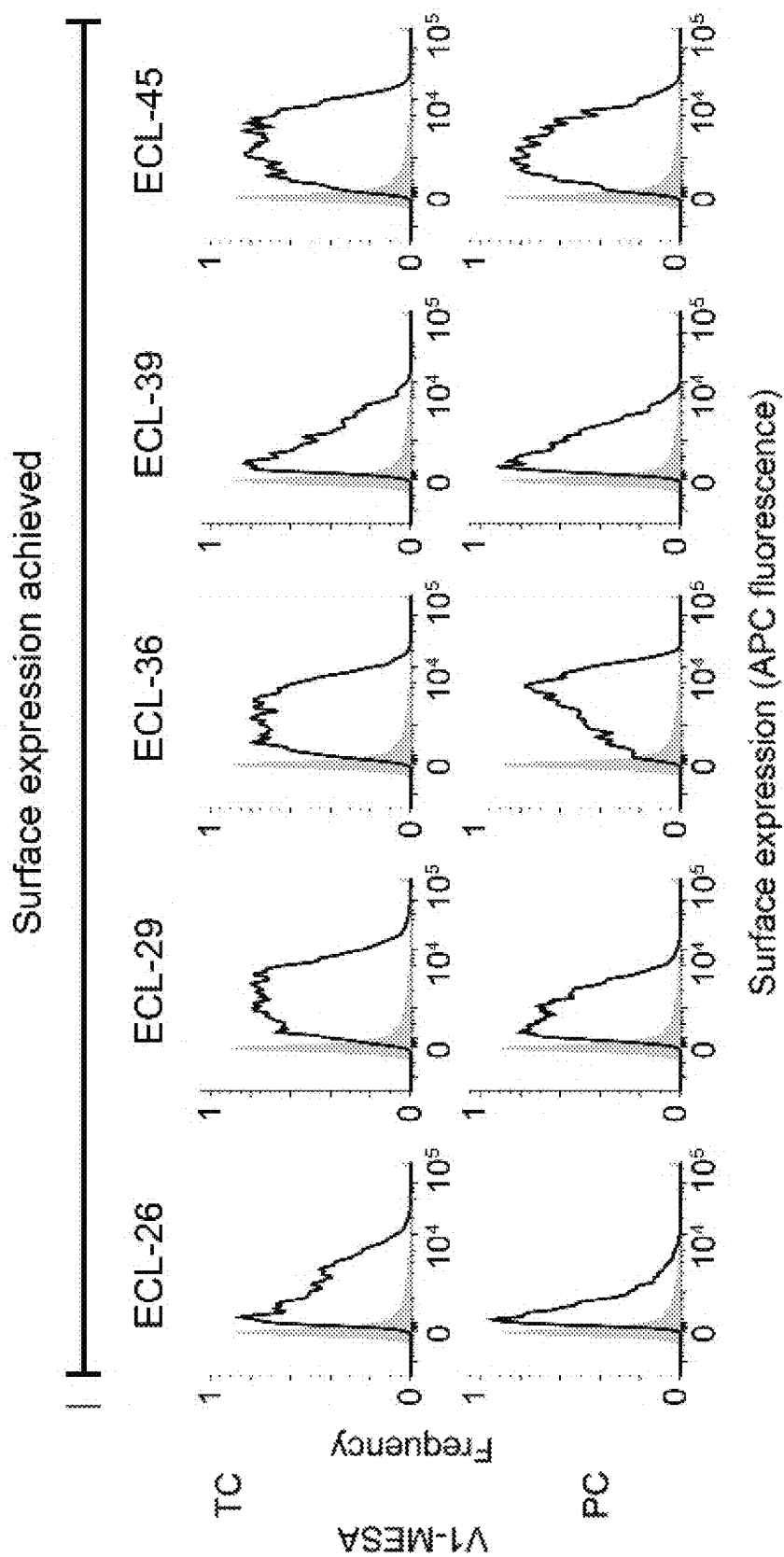


FIG. 4B

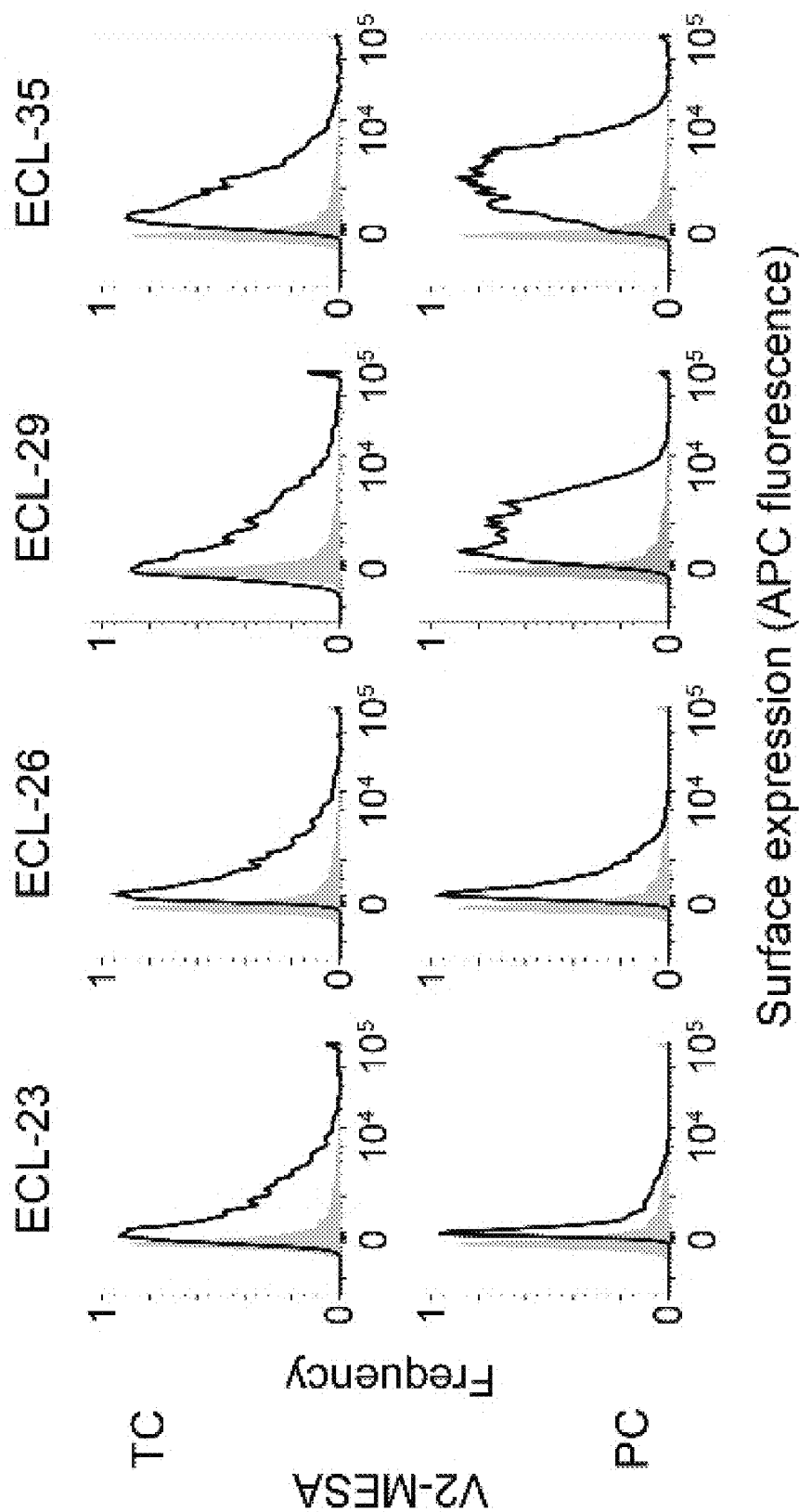


FIG. 4C

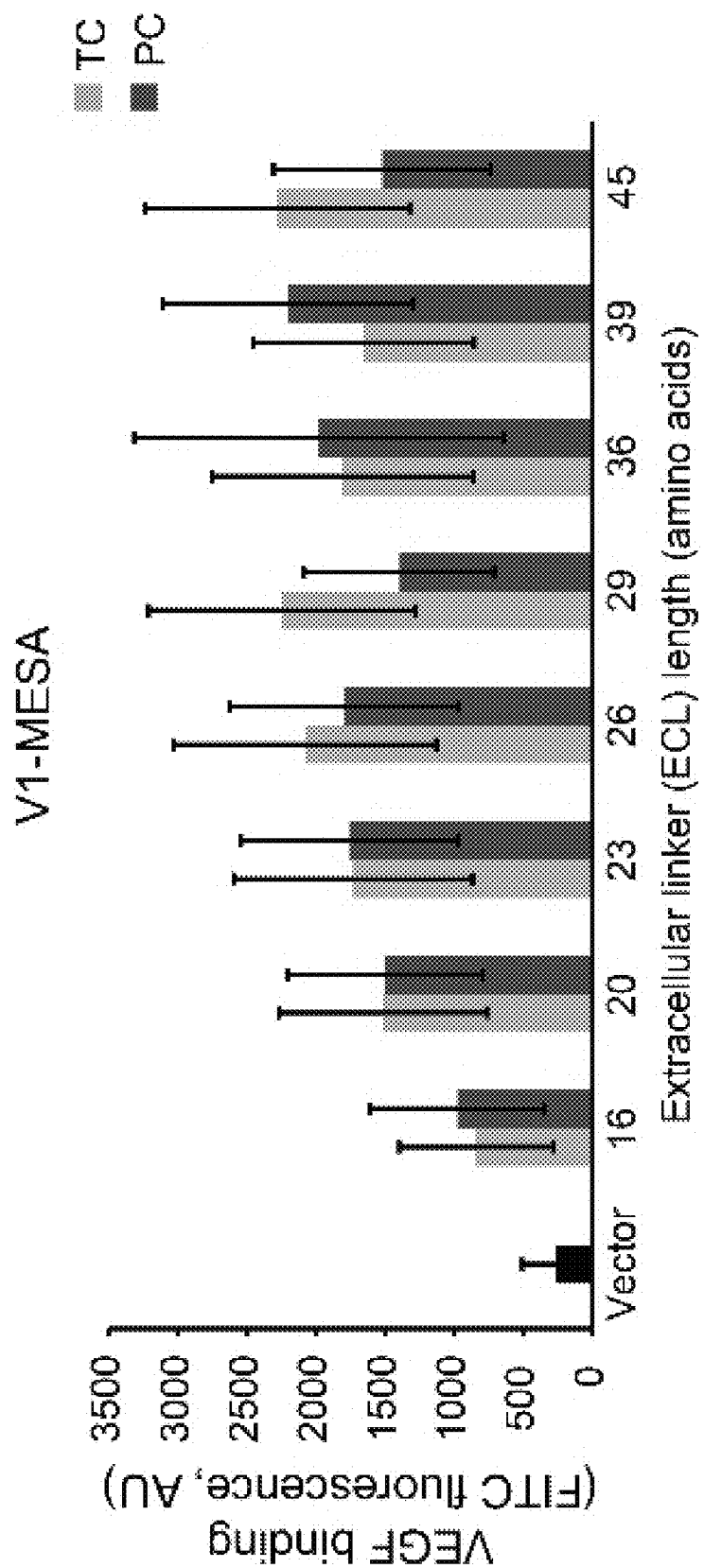


FIG. 4D

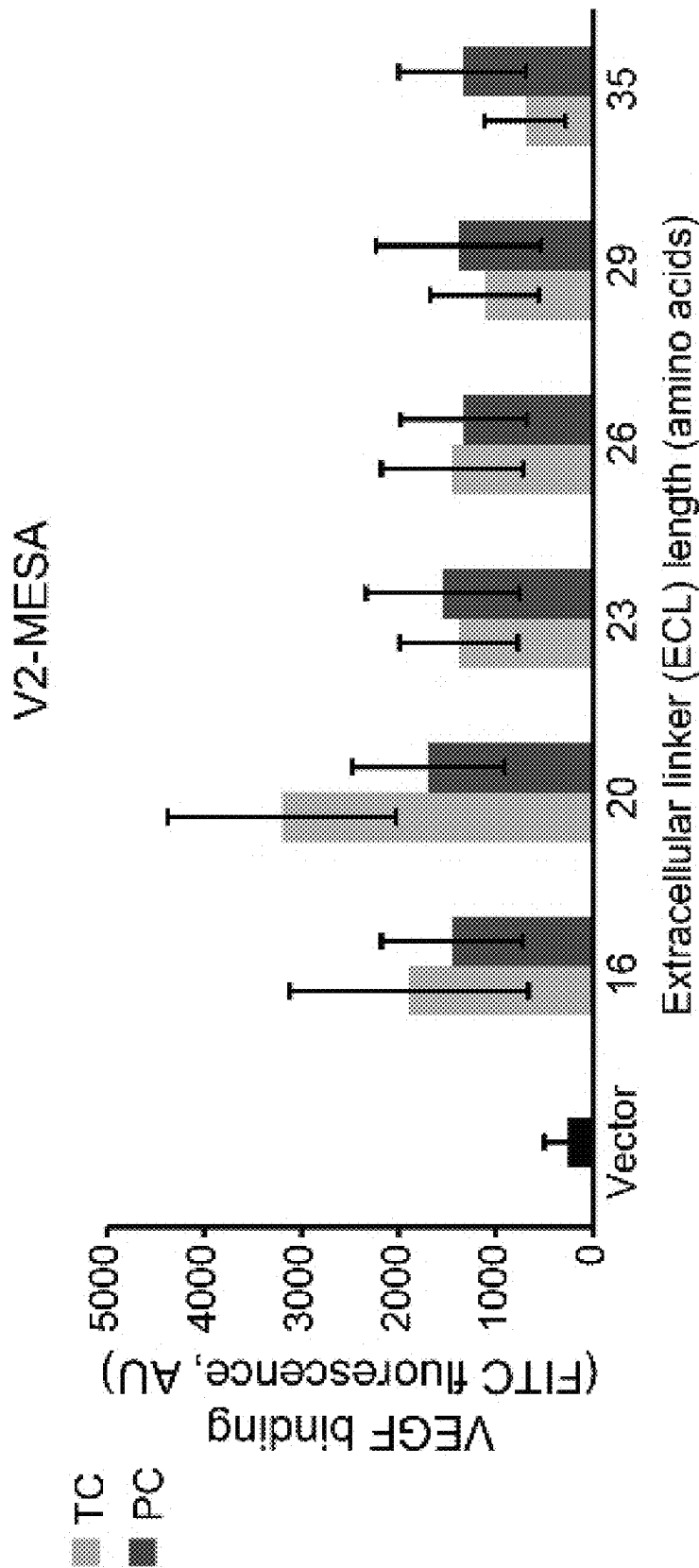


FIG. 4E

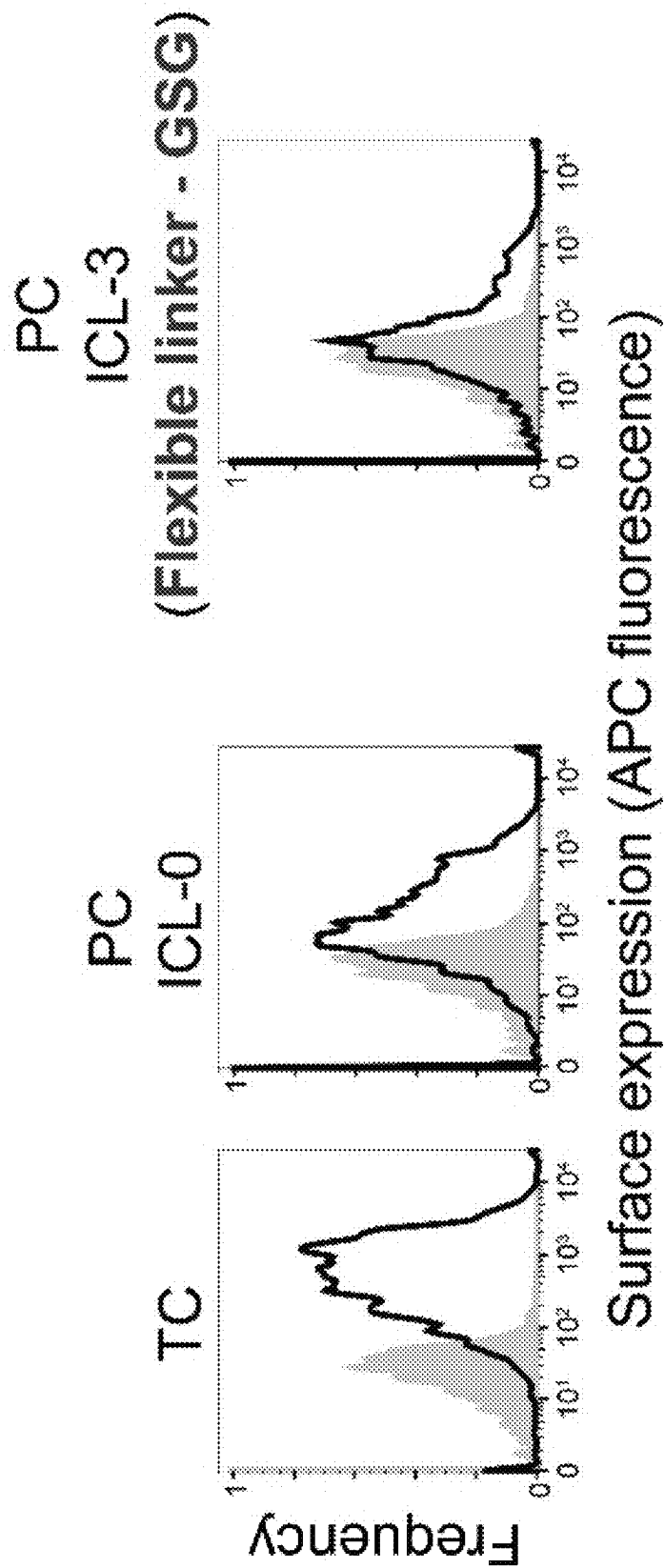
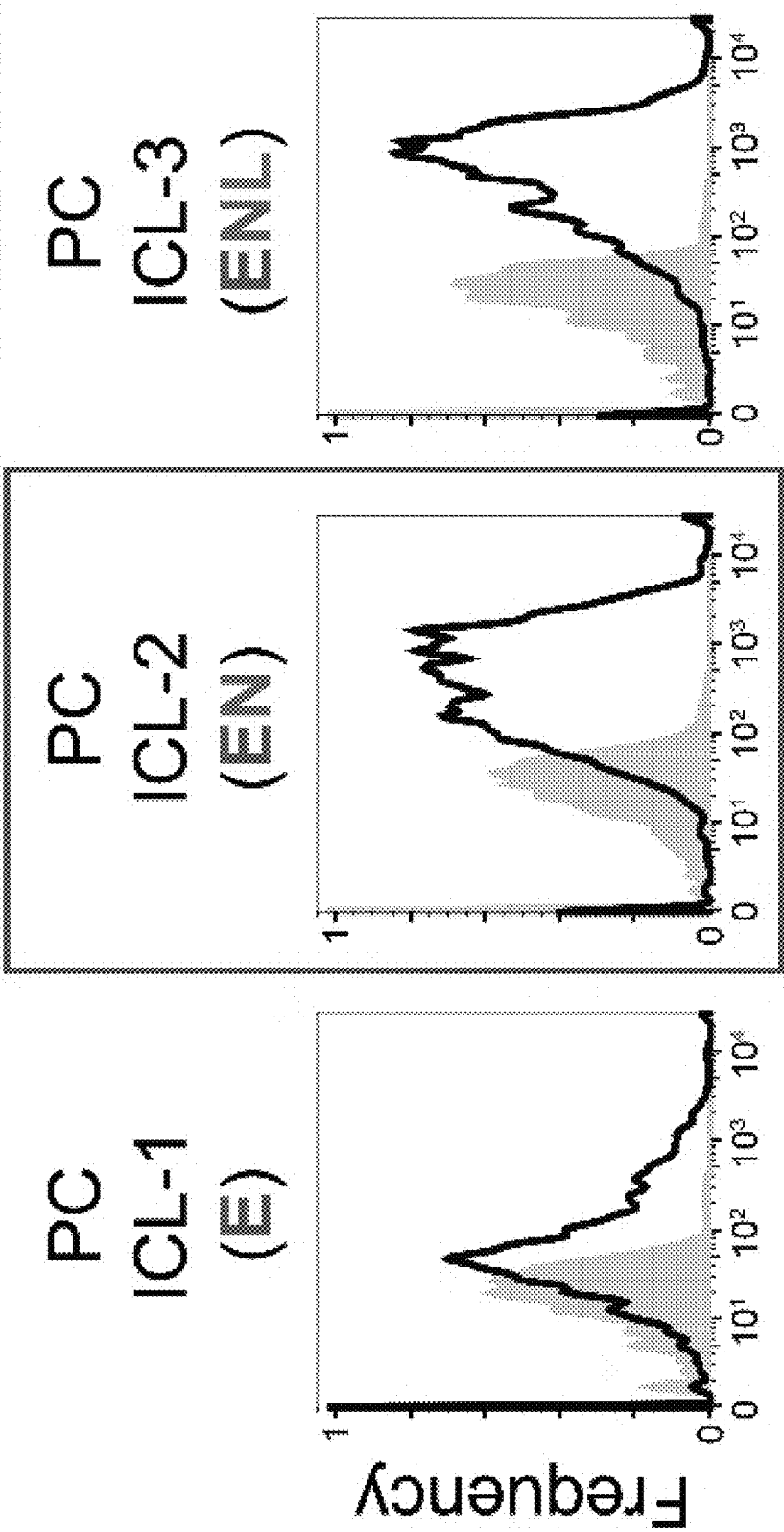


FIG. 5A

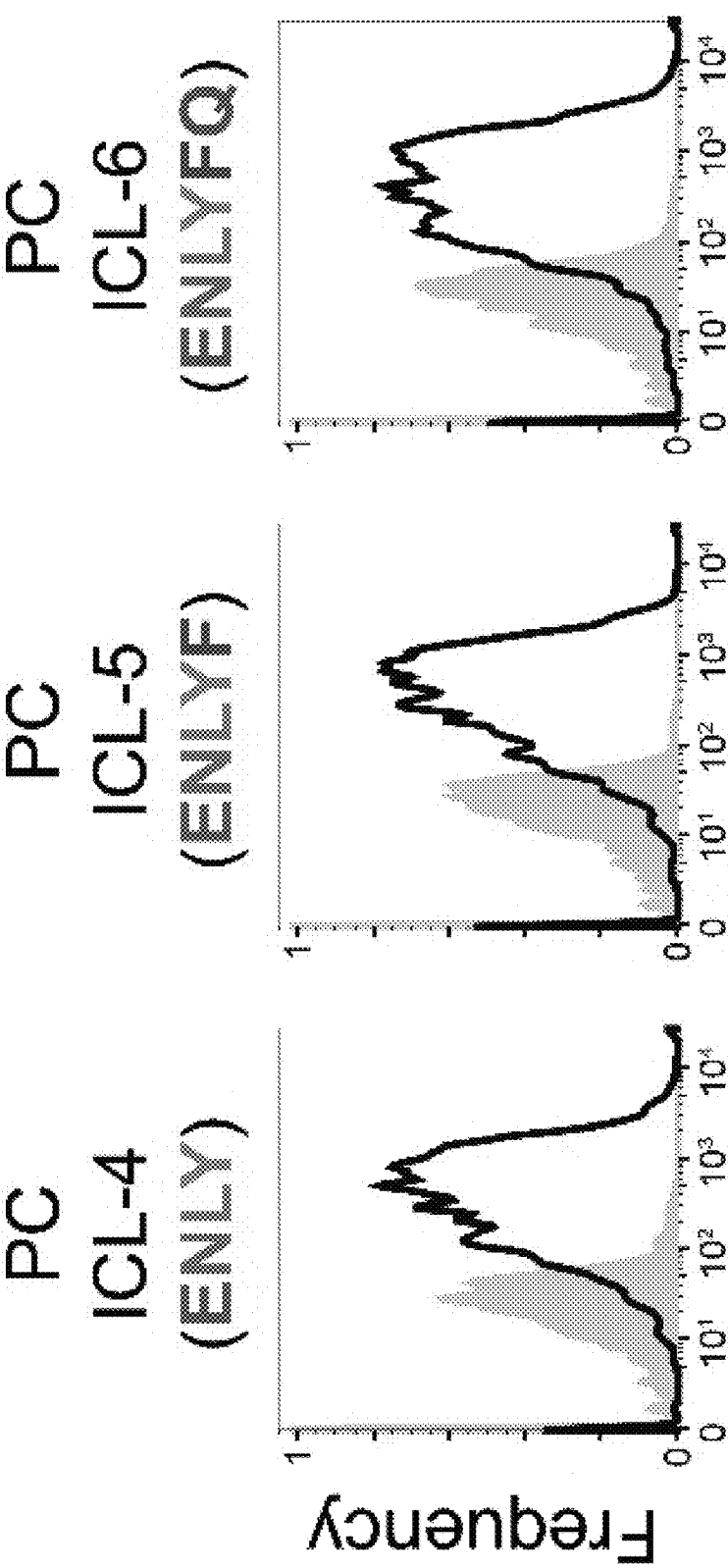
TEV cleavage
sequence:
ENLYFQG



Surface expression (APC fluorescence)

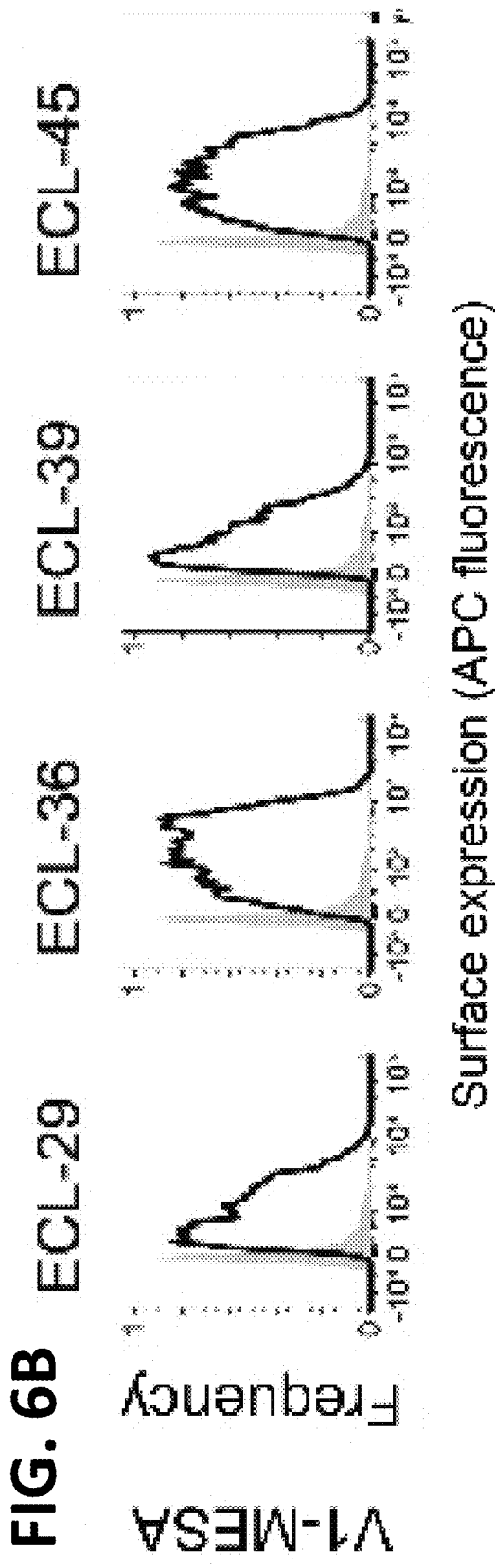
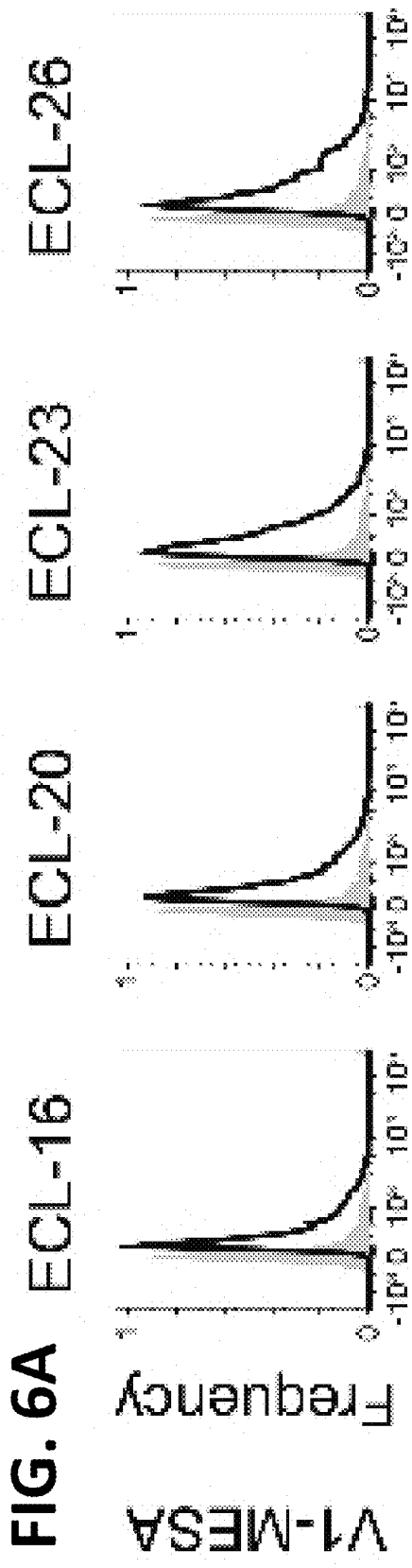
FIG. 5B

TEV cleavage
sequence:
ENLYFQG



Surface expression (APC fluorescence)

FIG. 5C



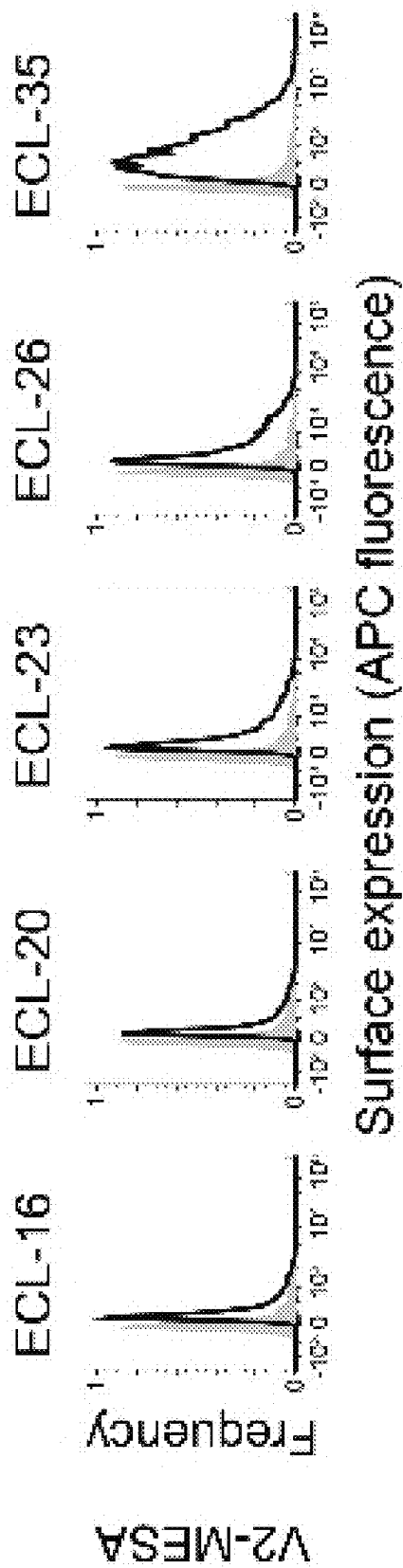


FIG. 6C

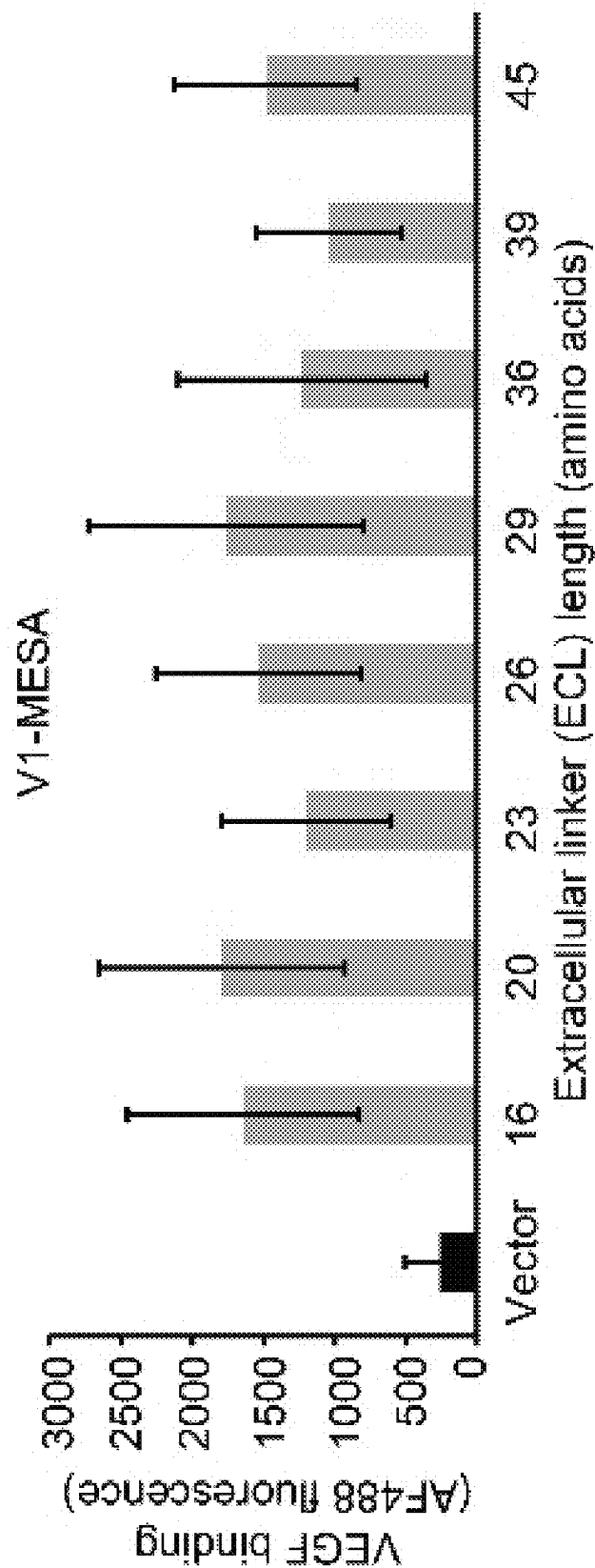


FIG. 6D

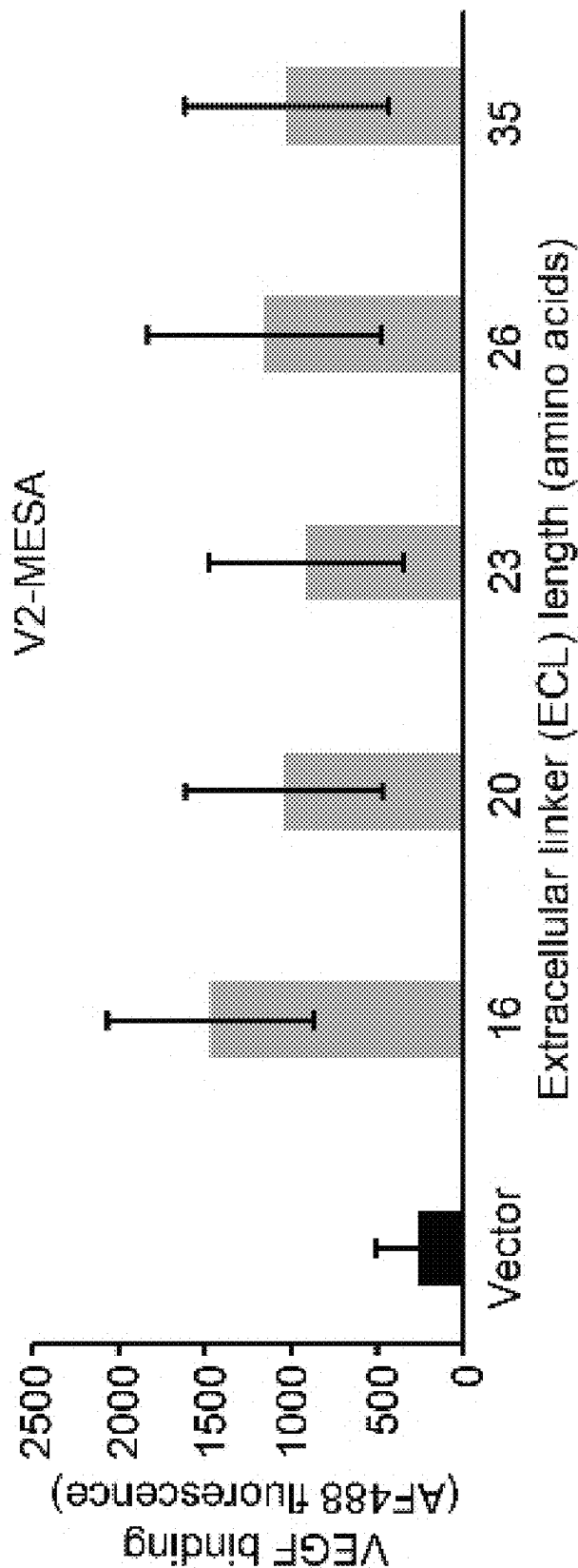


FIG. 6E

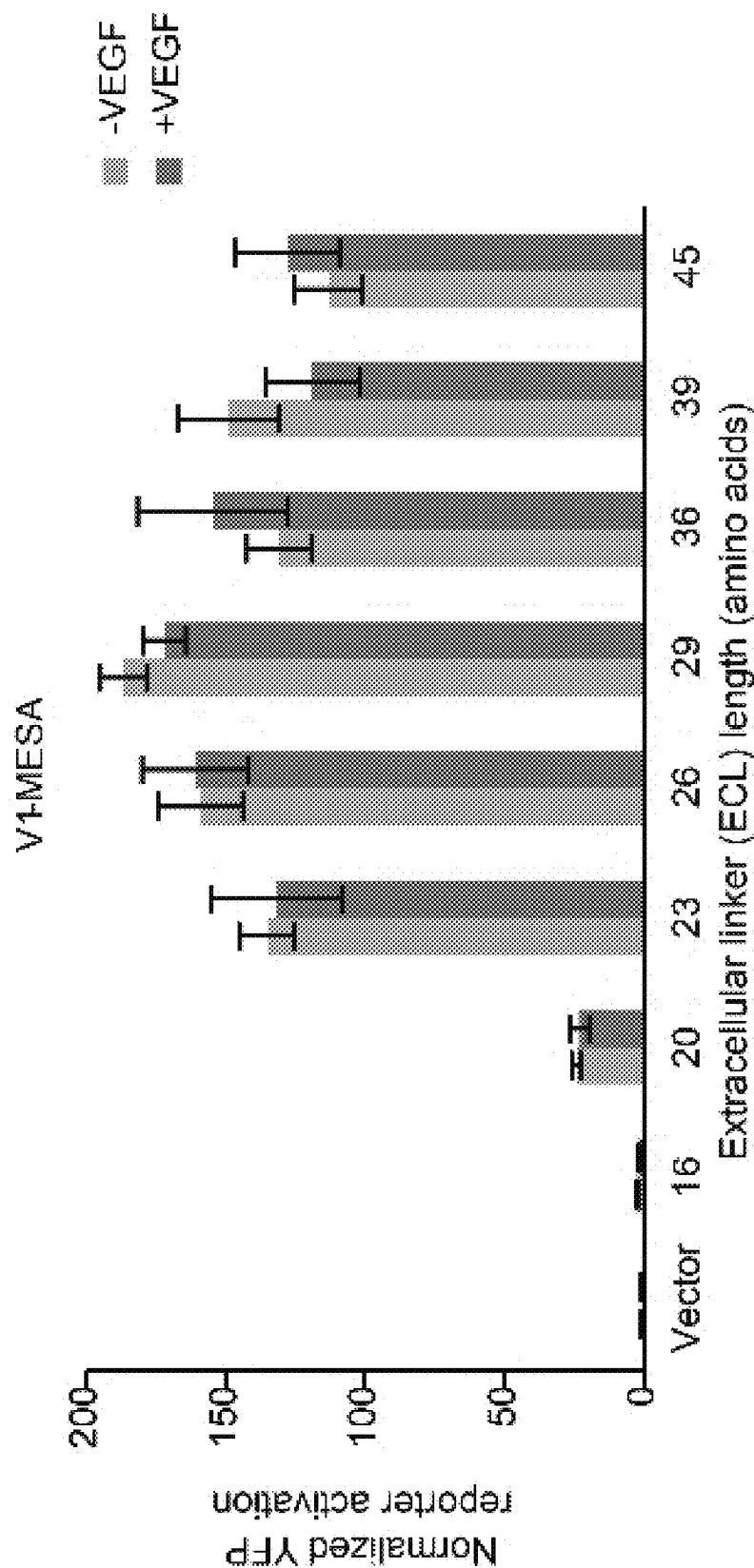


FIG. 6F

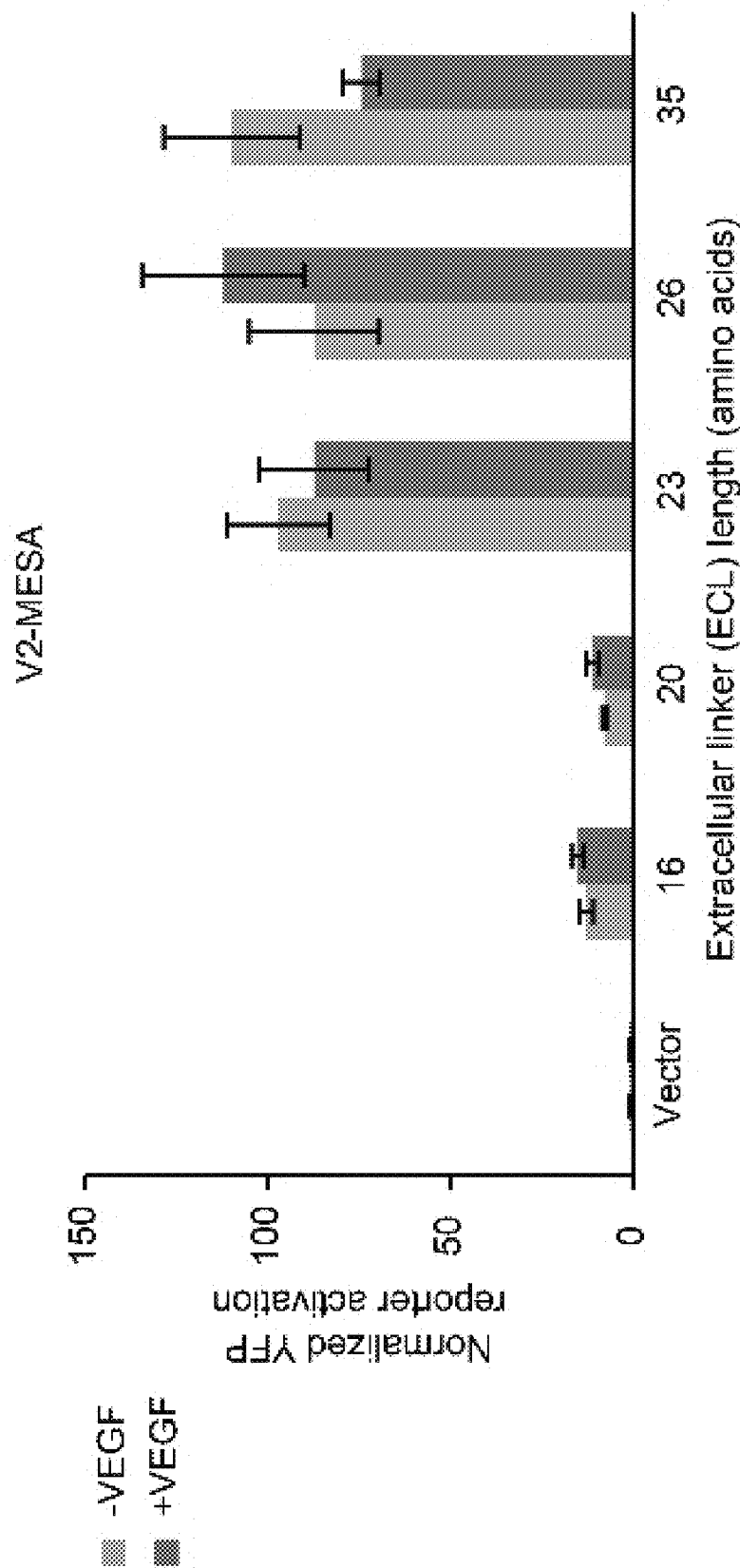
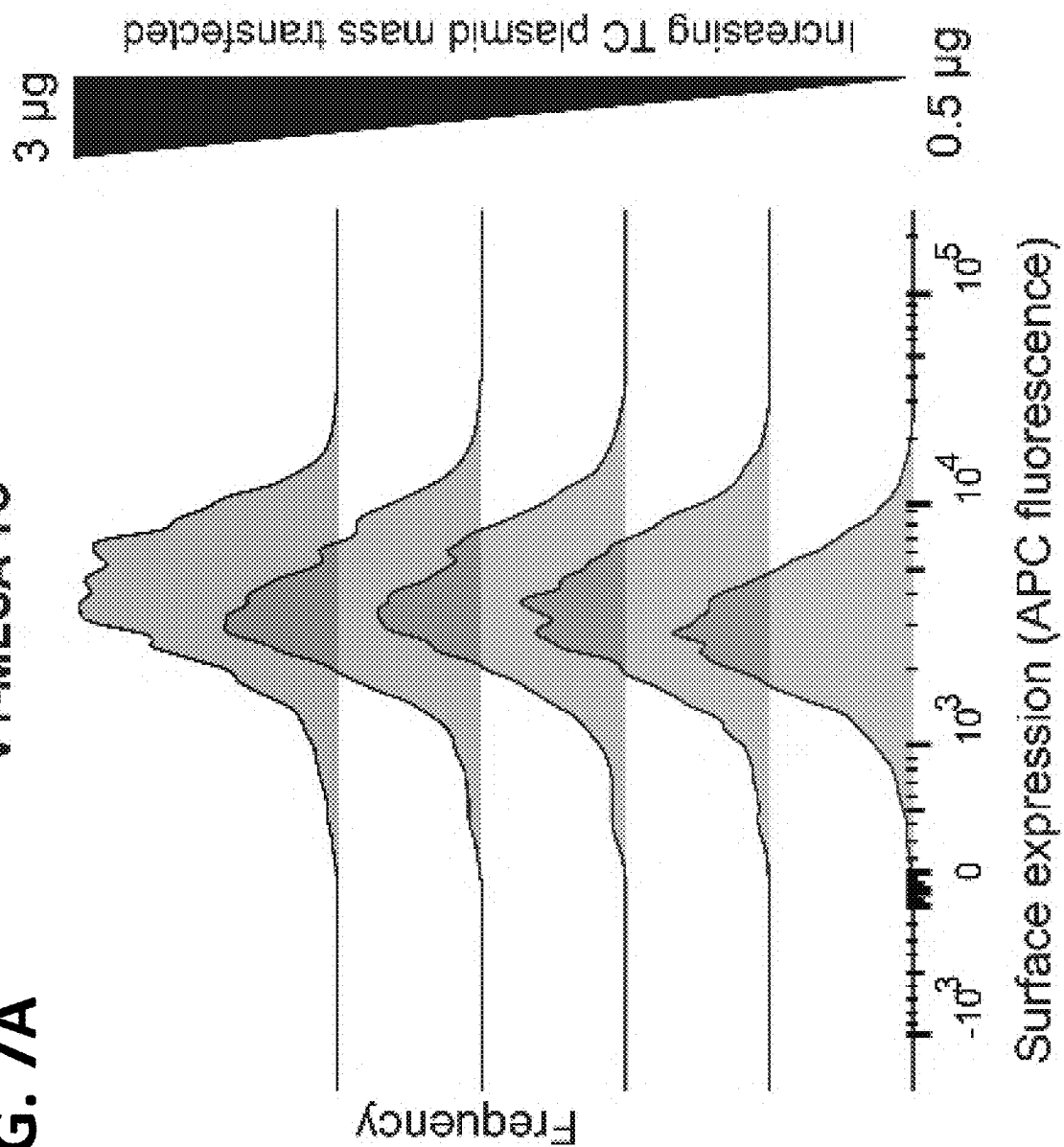


FIG. 6G

V1-MESA TC

FIG. 7A



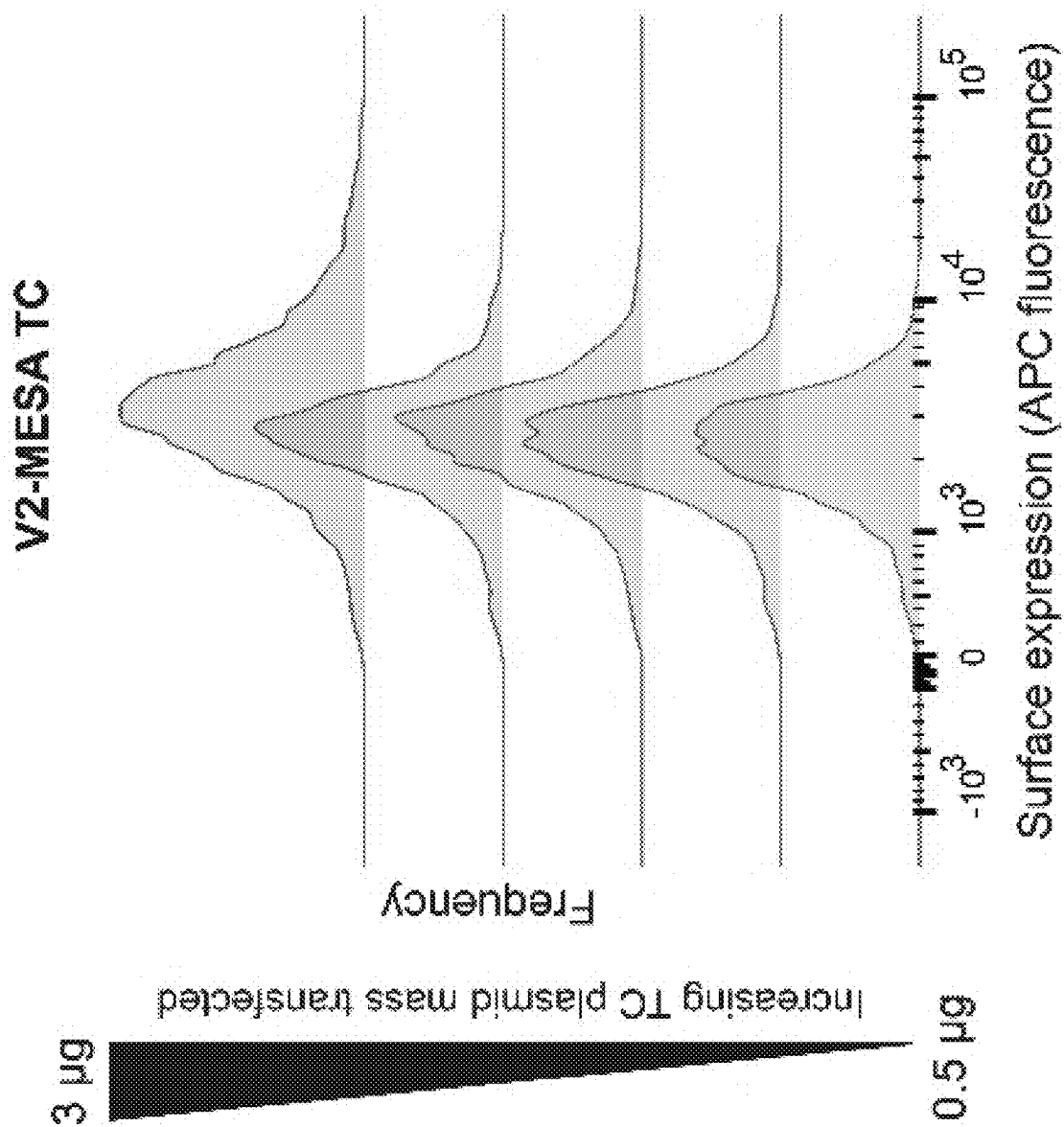


FIG. 7B

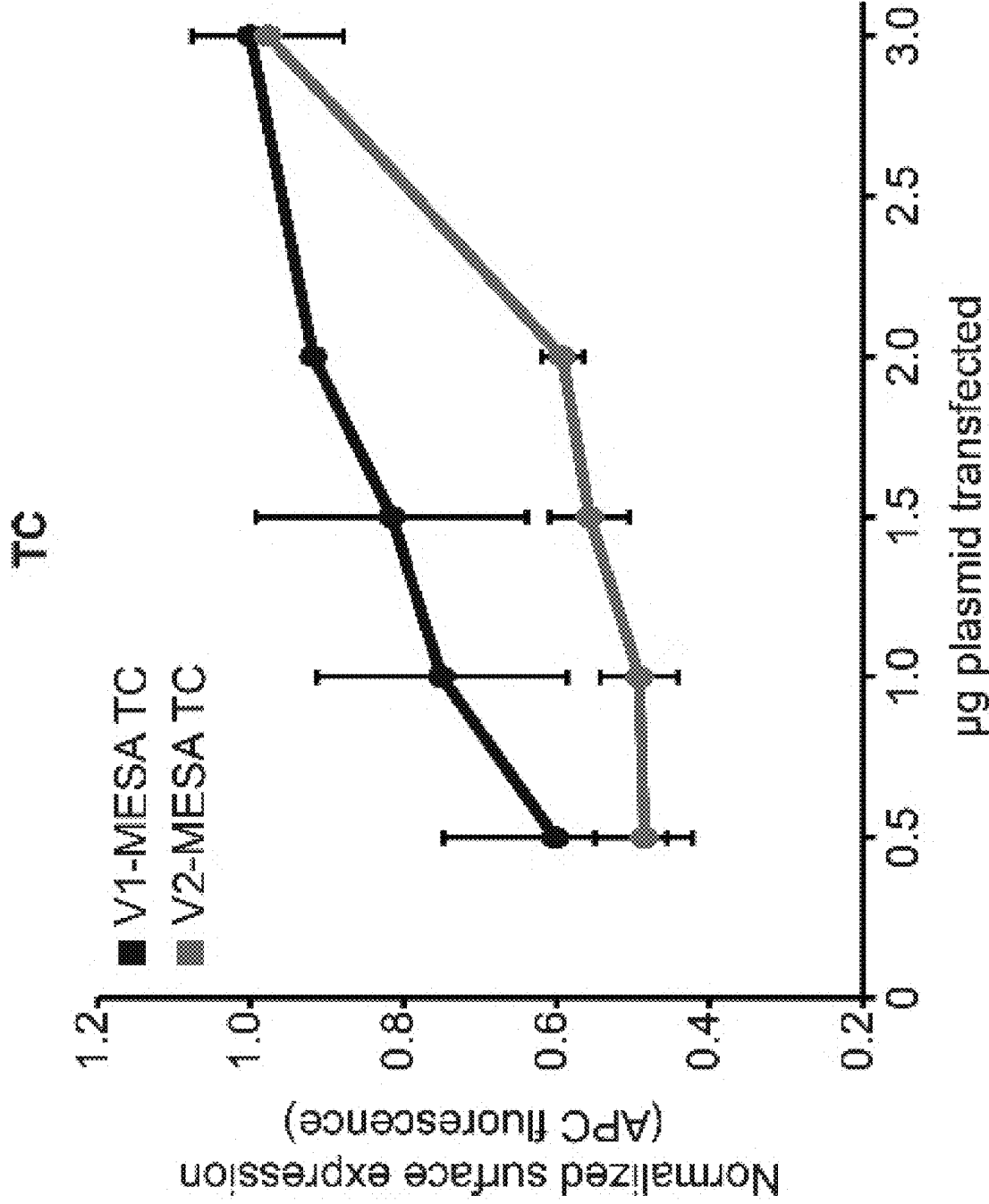
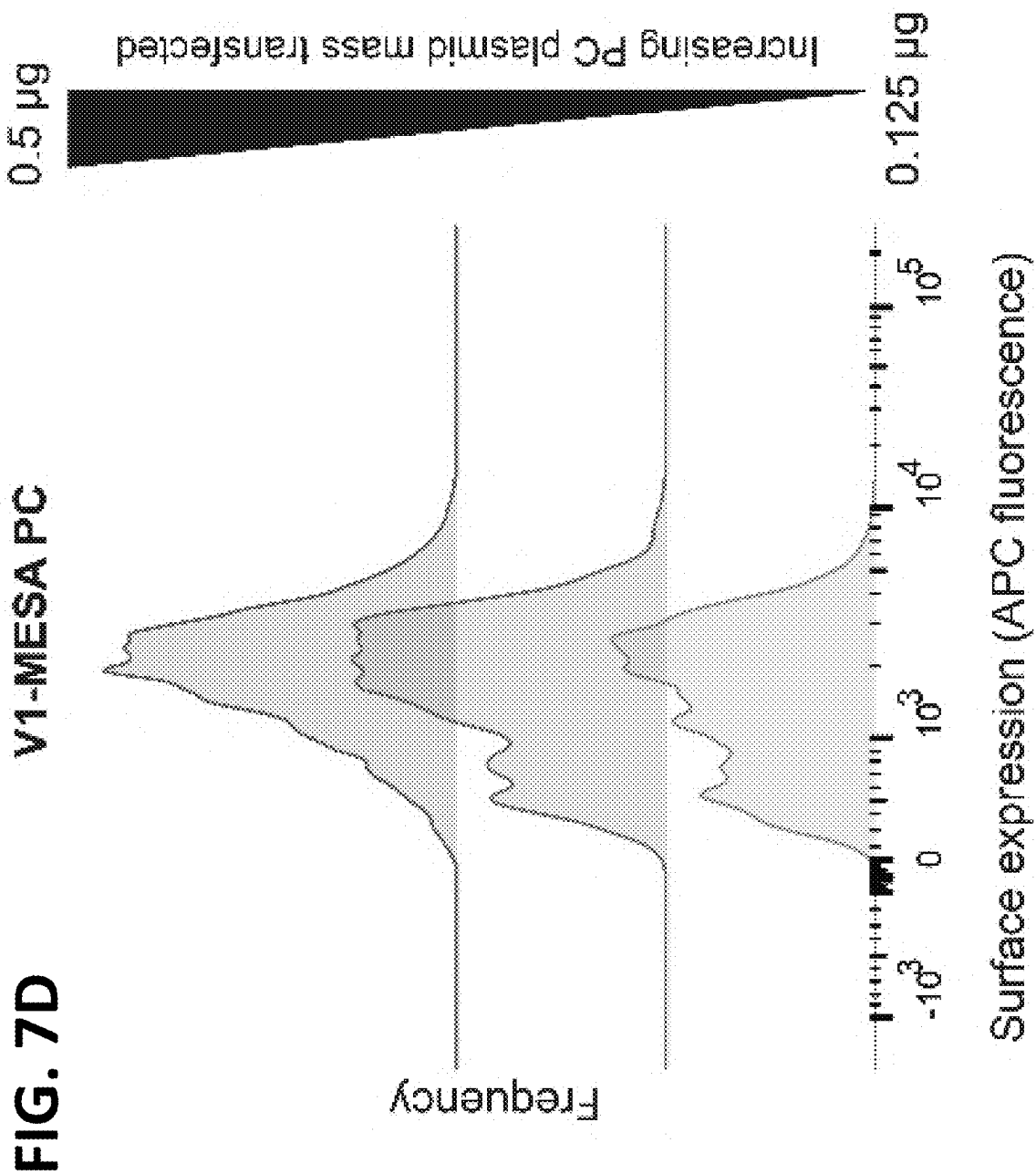


FIG. 7C



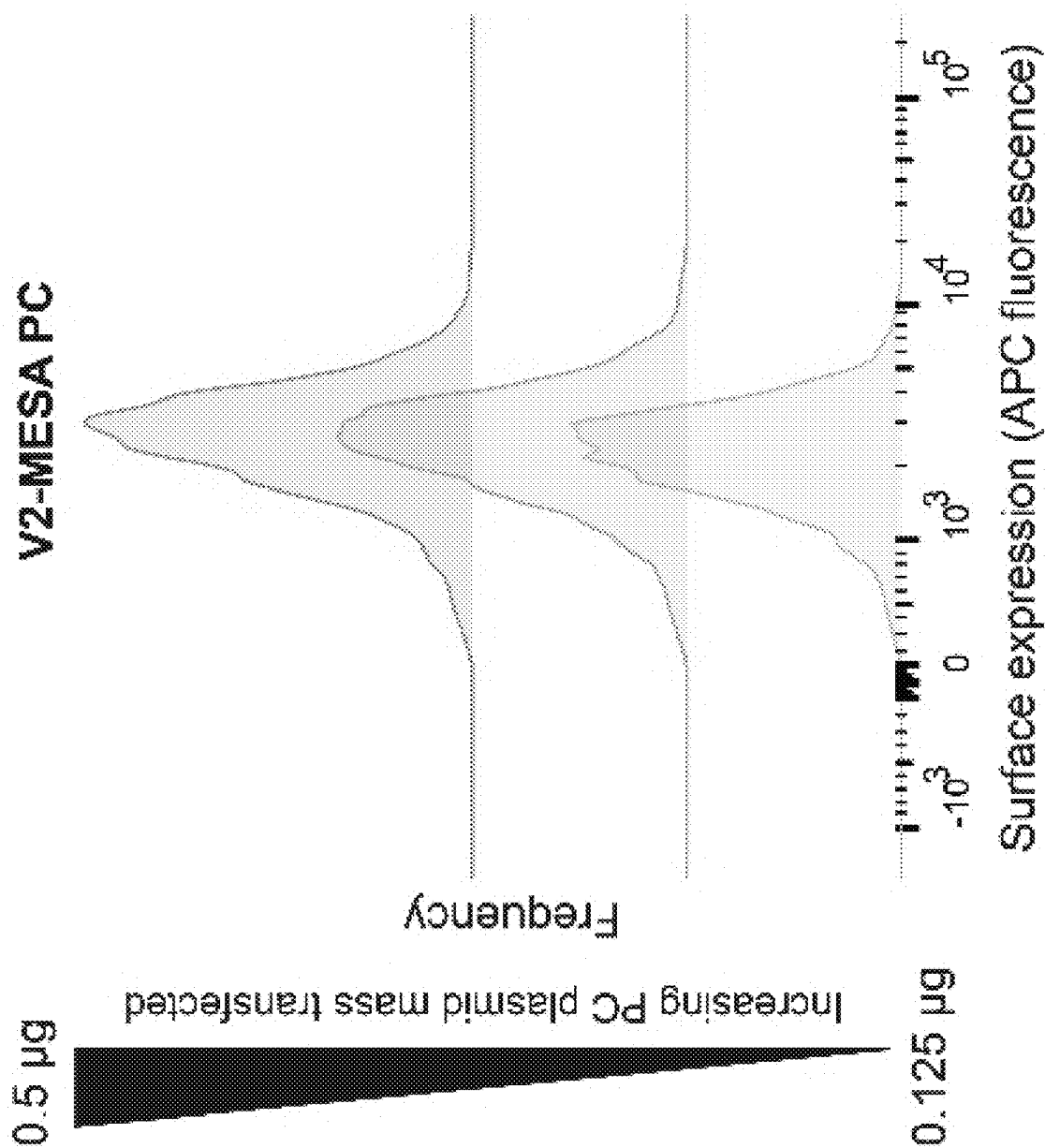


FIG. 7E

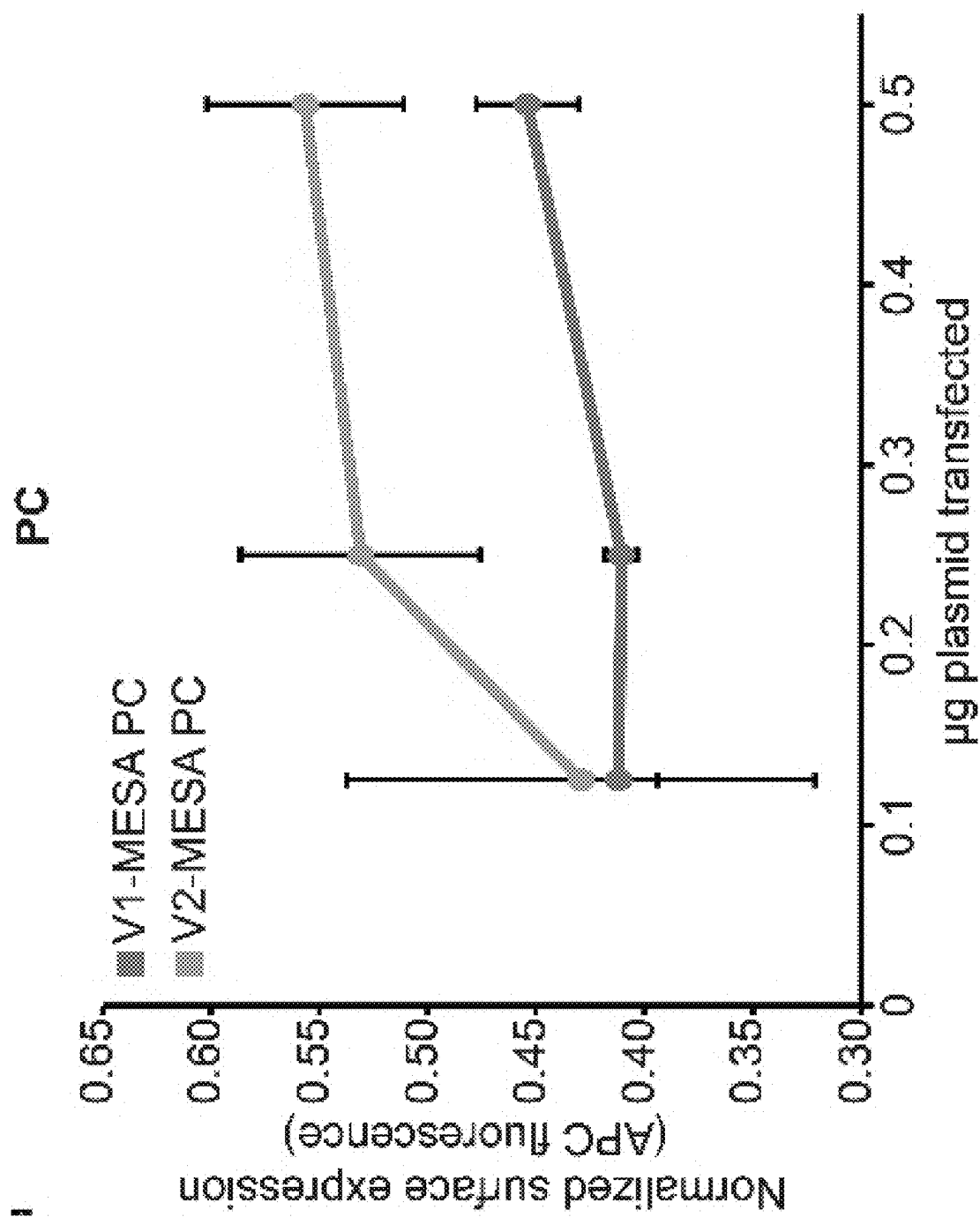
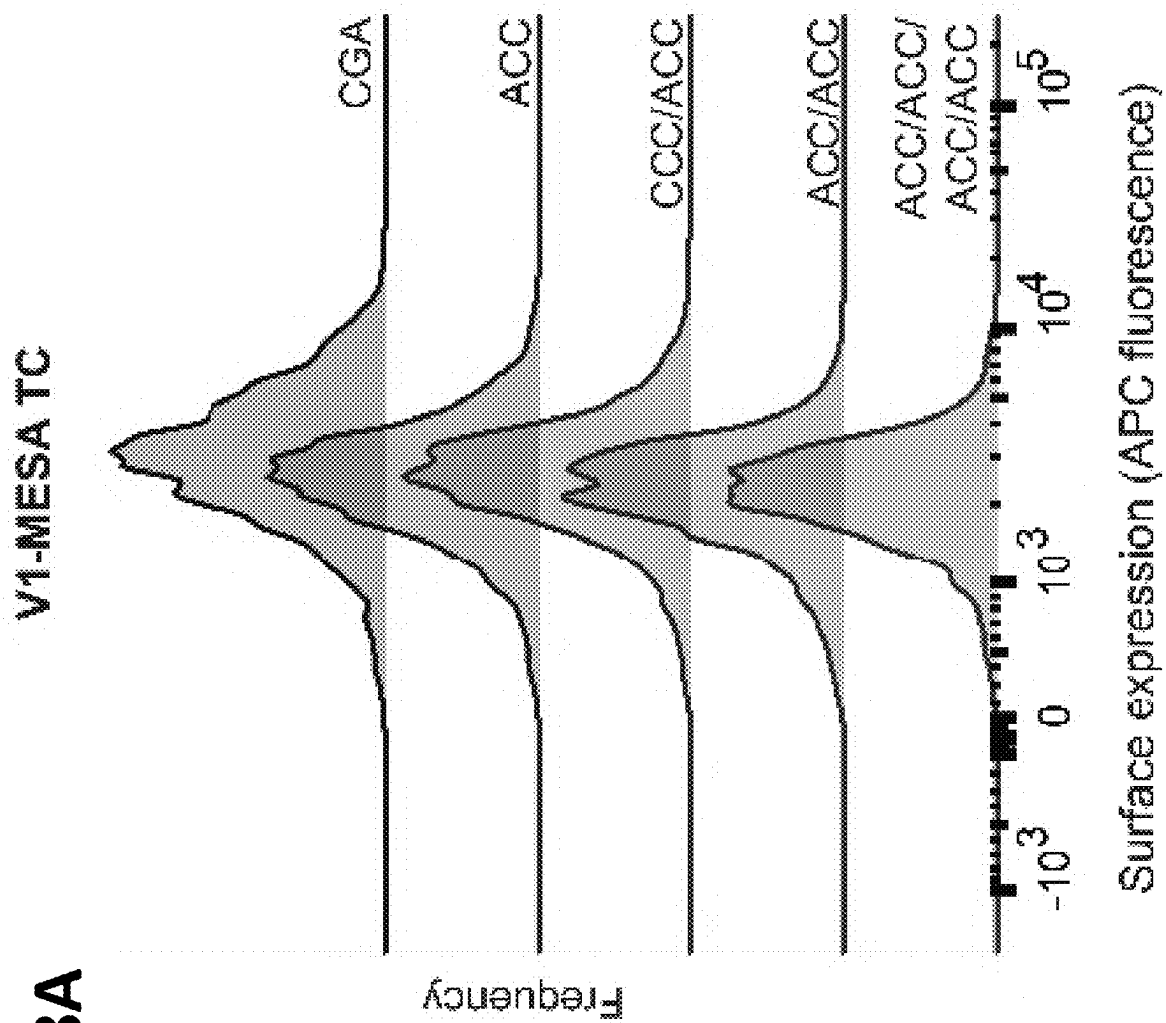


FIG. 7F

FIG. 8A



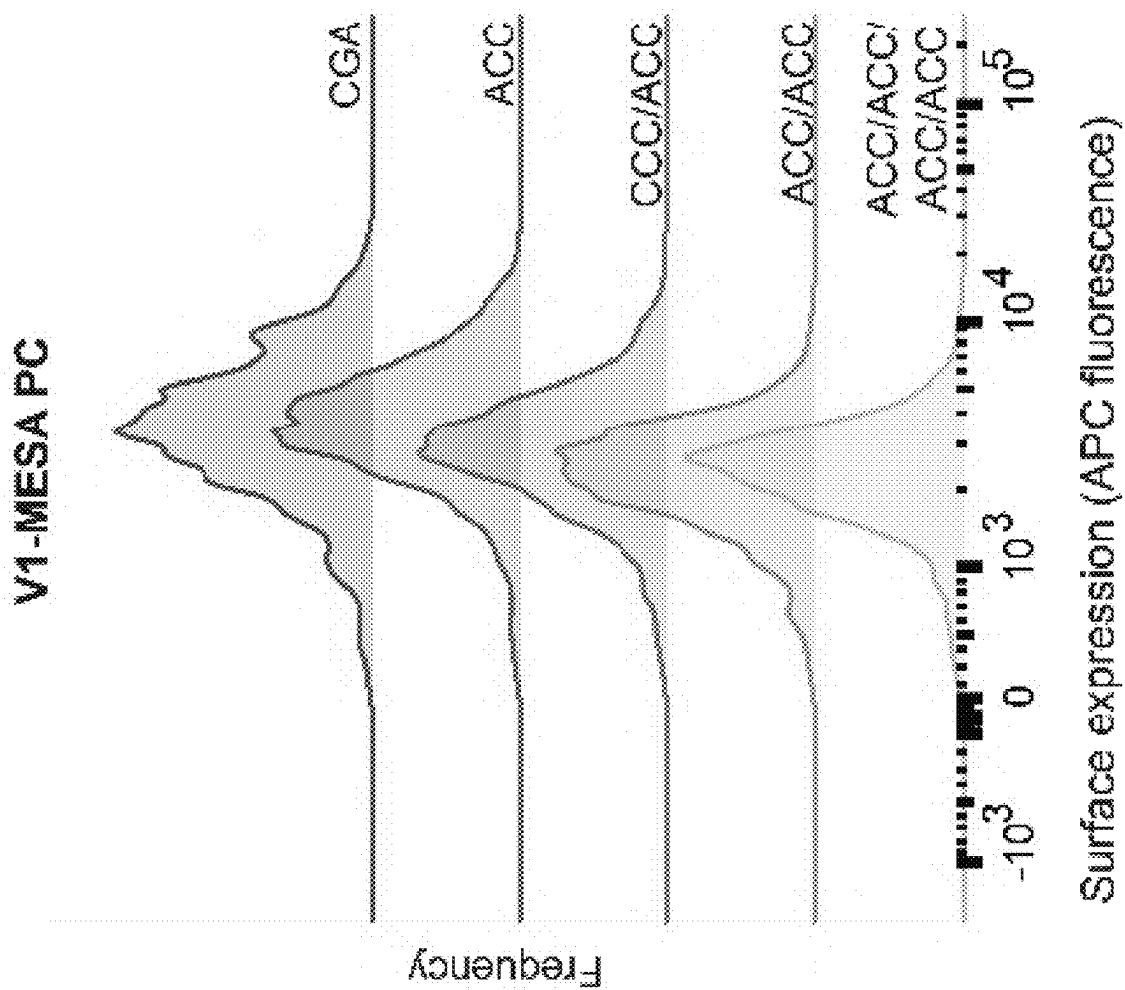


FIG. 8B

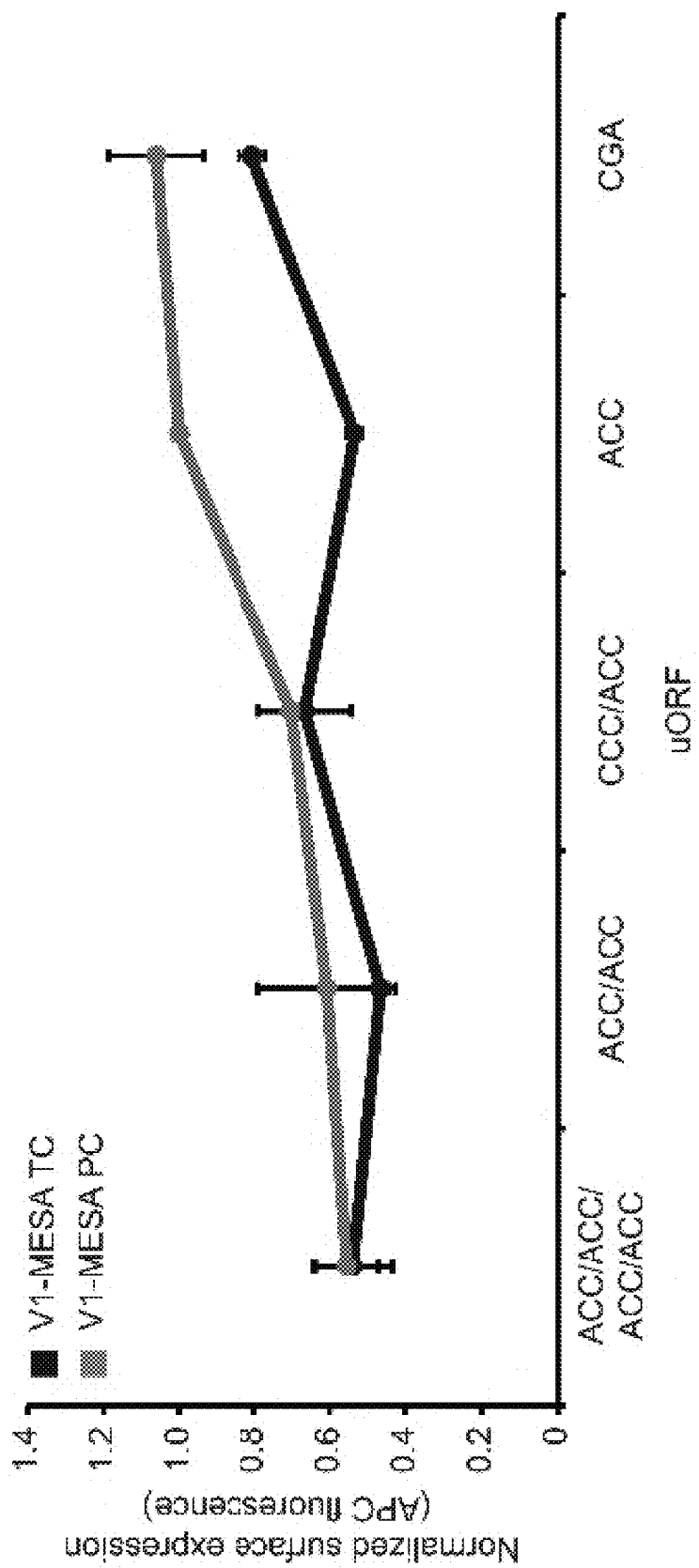


FIG. 8C

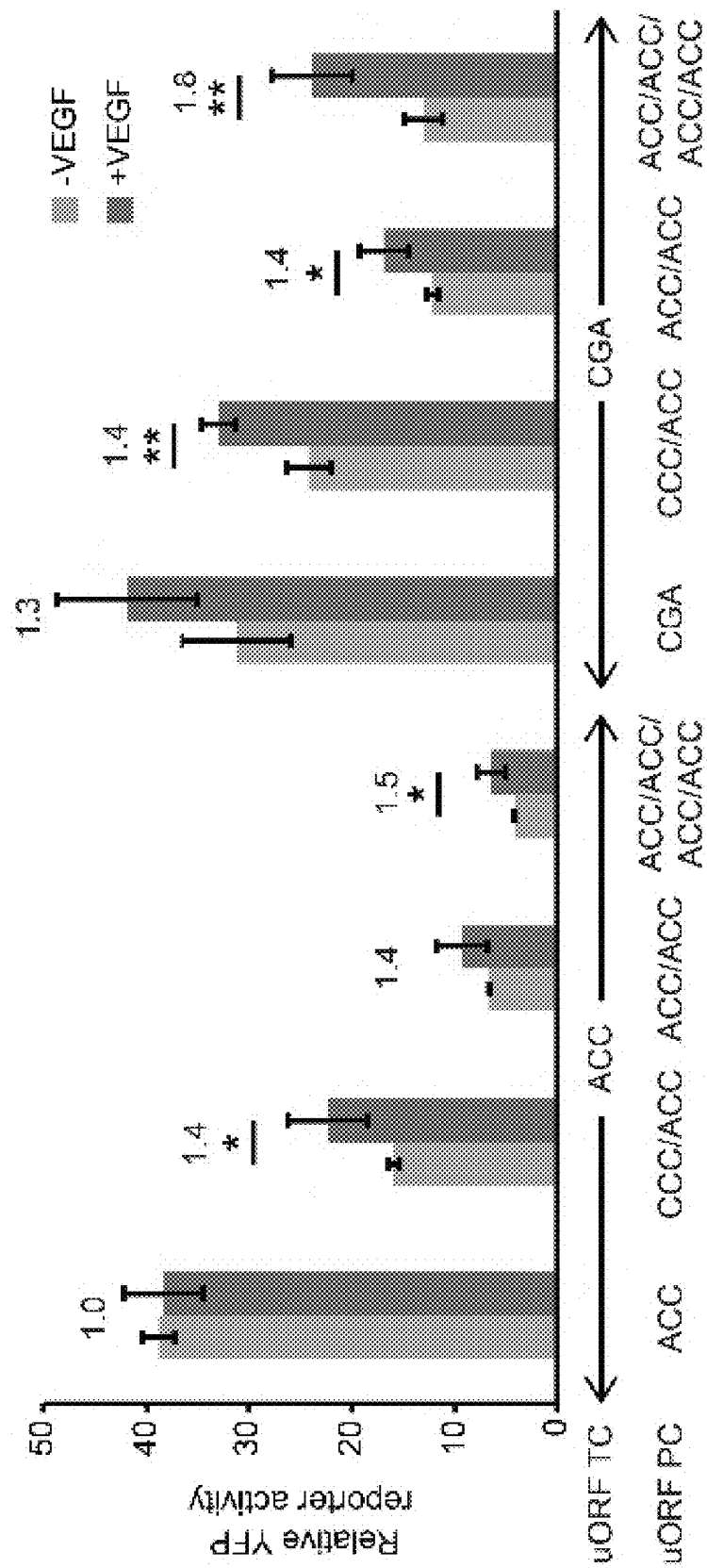


FIG. 8D

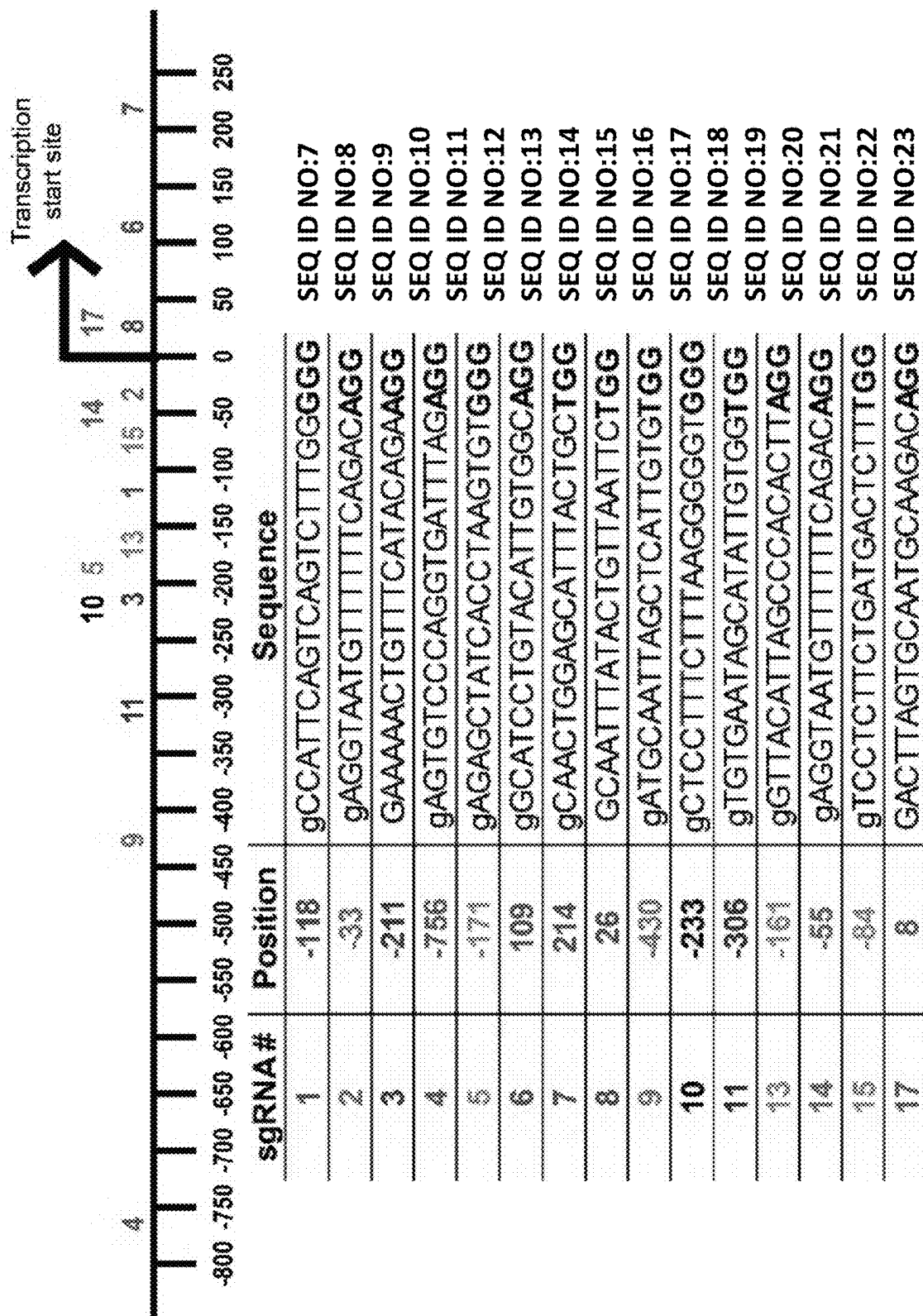


Figure 9

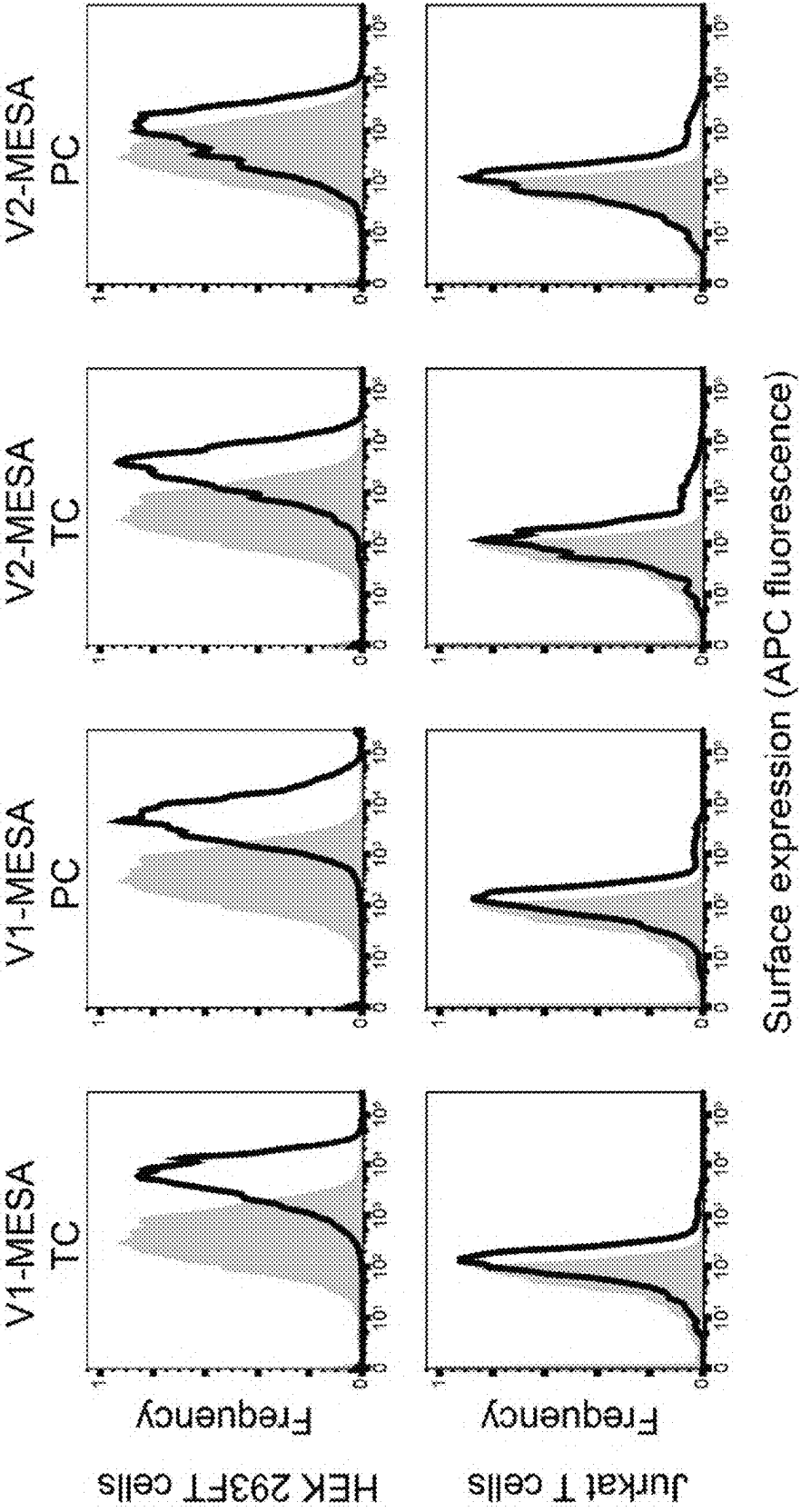


Figure 10

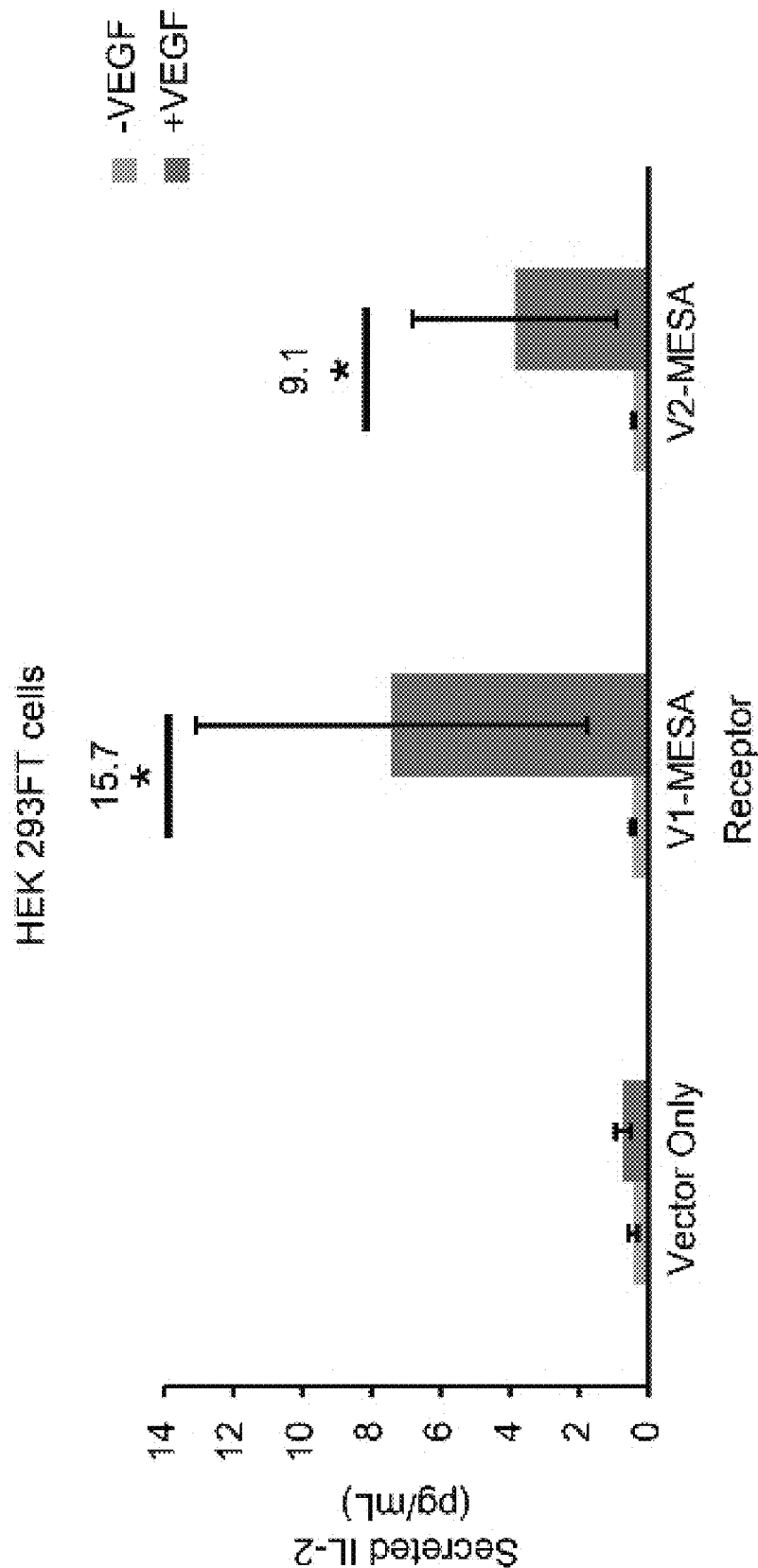


FIG. 11A

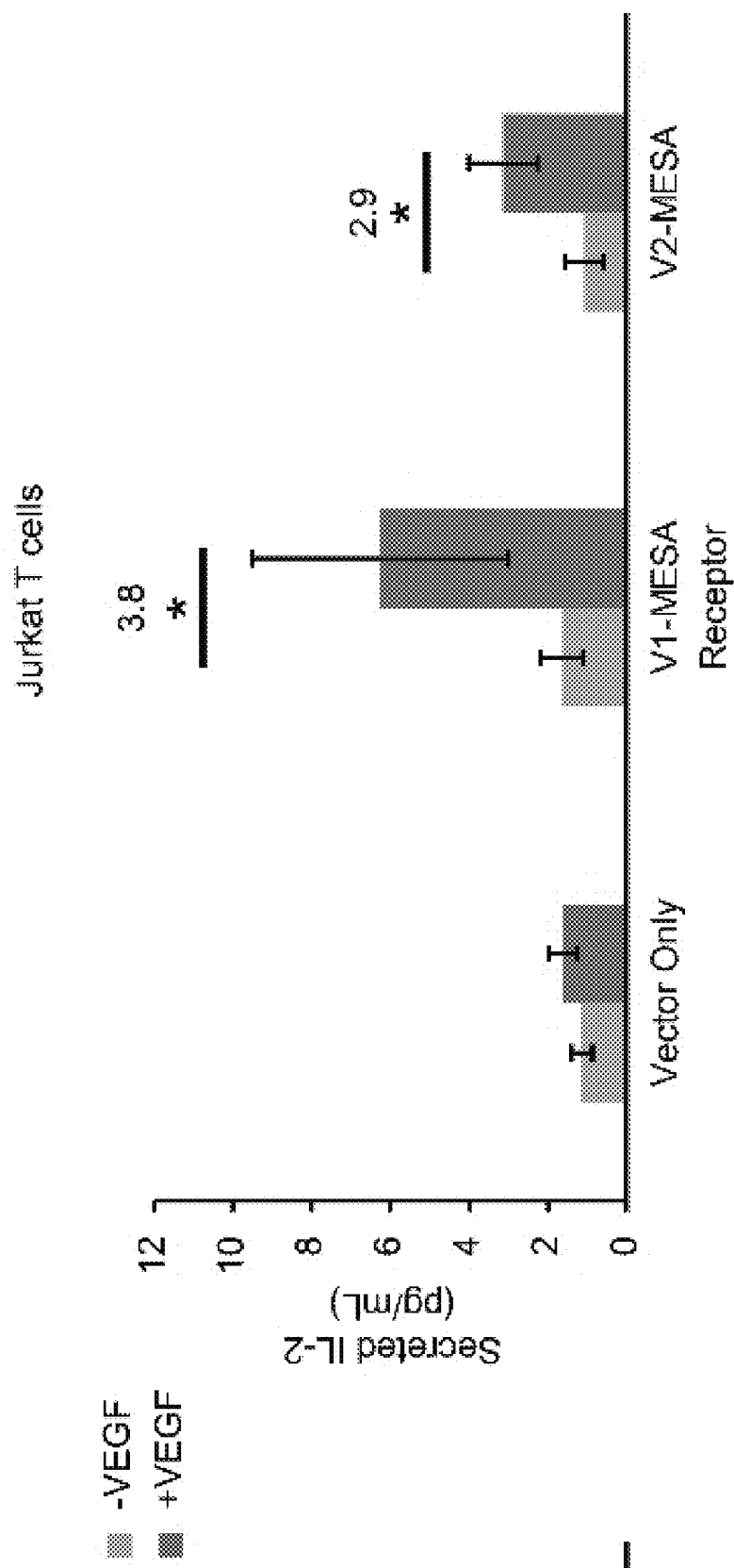


FIG. 11B

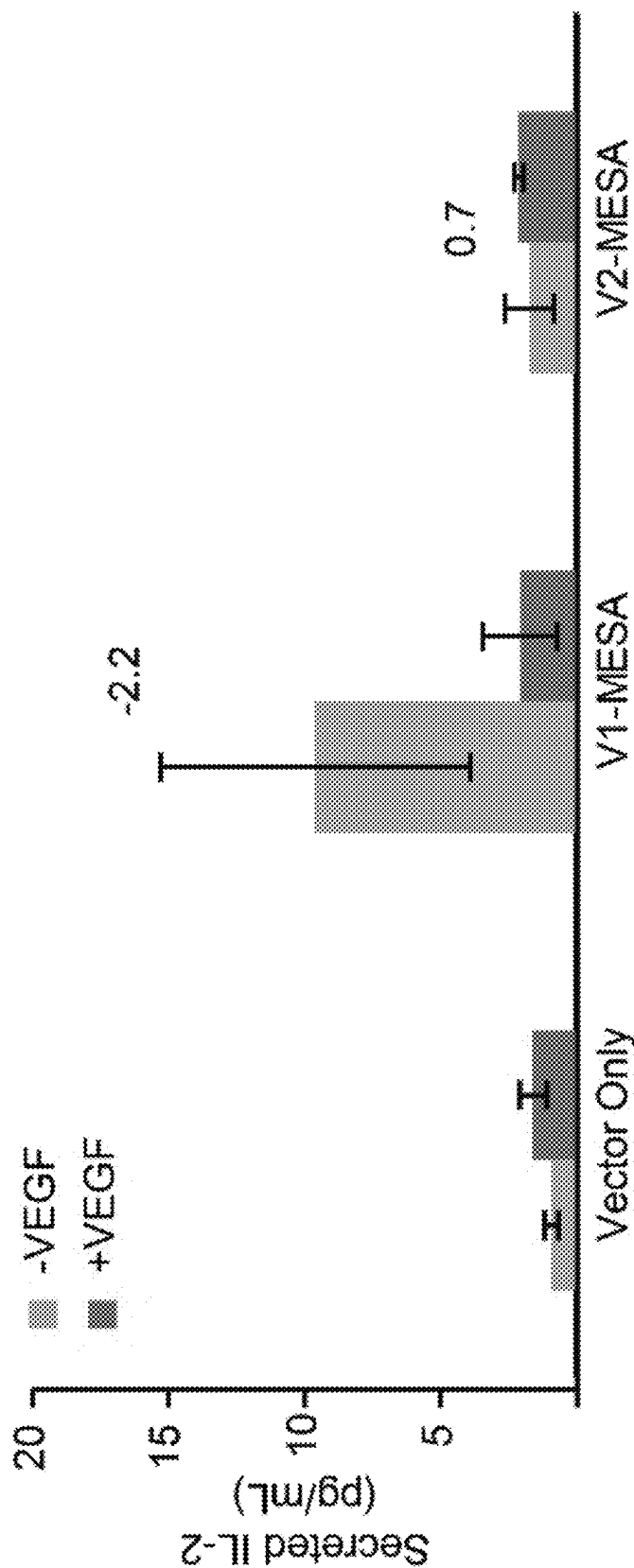


FIG. 12A

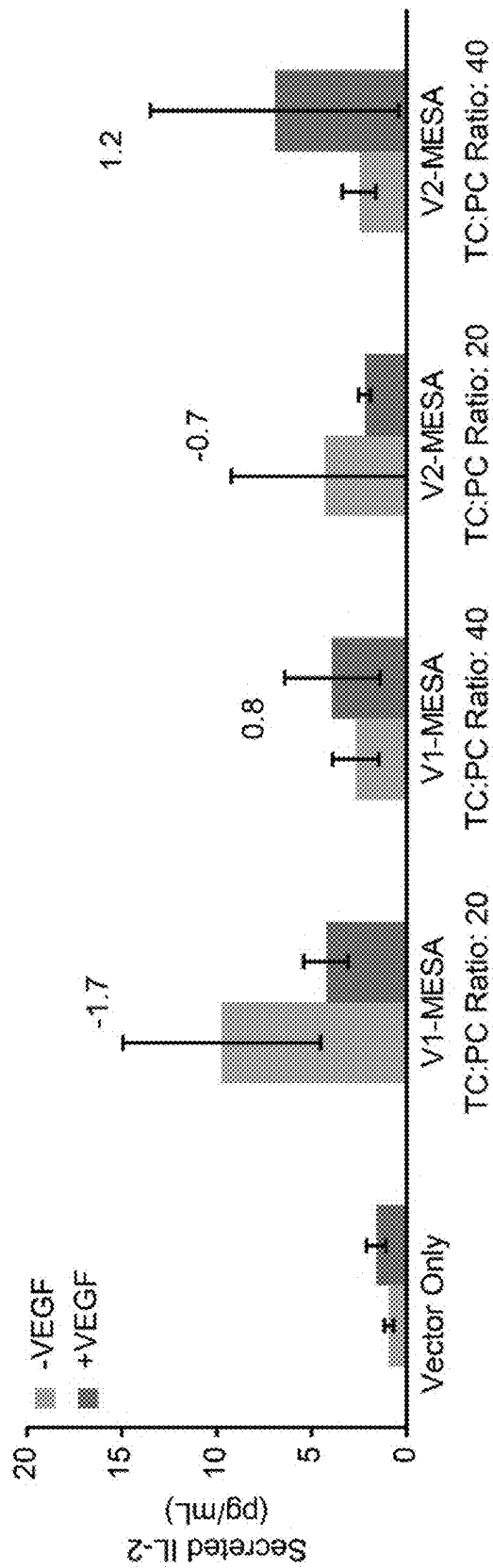


FIG. 12B

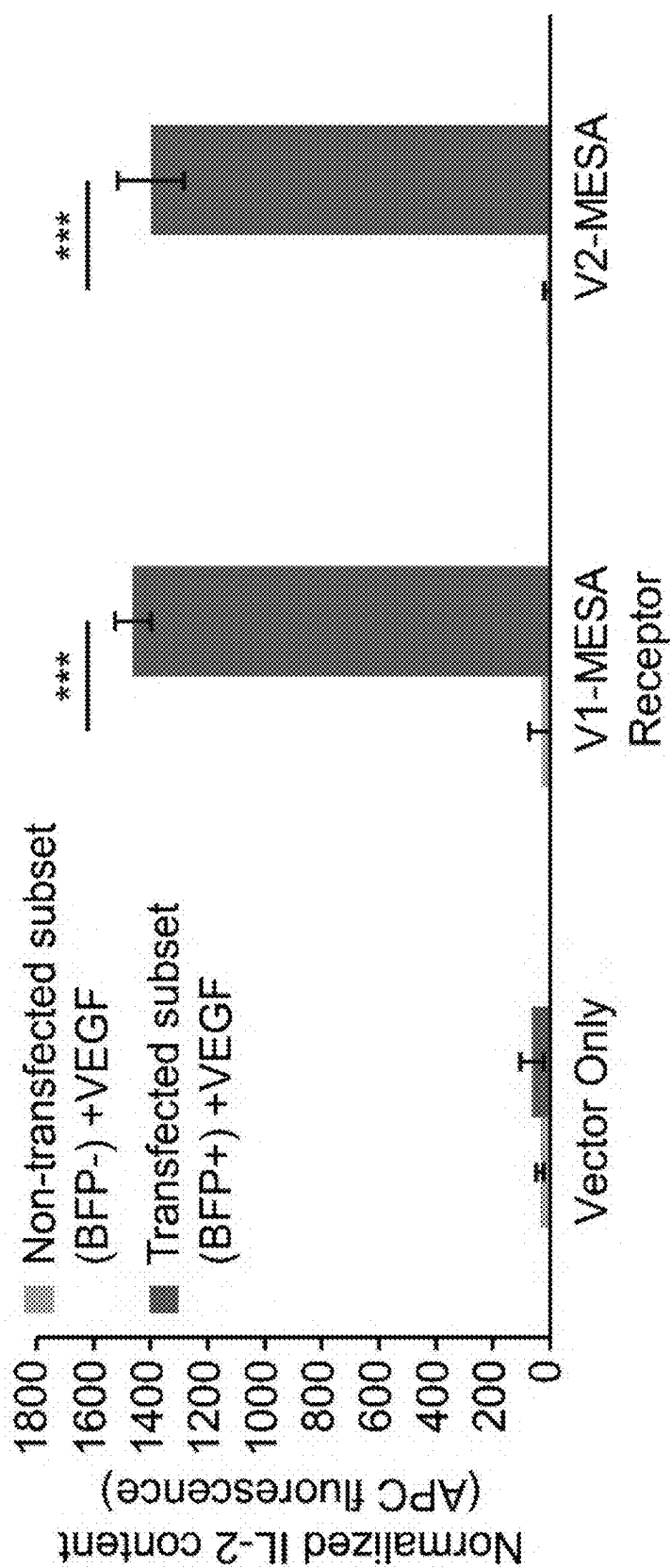


FIG. 13

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MODULAR EXTRACELLULAR SENSOR ARCHITECTURE FOR REGULATING GENES

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS

This application claims the benefit of priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 62/411,895, filed on Oct. 24, 2016, the content of which is incorporated herein by reference in its entirety.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

This invention was made with government support under W911NF-11-2-0066 awarded by the Army Research Office (DARPA). The government has certain rights in the invention.

BACKGROUND

The present invention provides modular extracellular sensors, nucleic acids encoding such sensors, cells expressing such sensors, and methods of employing such sensors and cells for sensing extracellular ligands and/or modulating expression of a gene in a cell when the ligand for the sensors is present.

SUMMARY

Disclosed are systems, compounds, compositions, and methods for sensing extracellular ligands and/or modulating expression of genes in a cell including endogenous genes and exogenous genes. In some embodiments, the disclosed systems and methods comprise or utilize: i) an exogenous extracellular sensor, and/or ii) a nucleic acid sequence encoding the exogenous extracellular sensor, wherein the exogenous extracellular sensor comprises: a) a ligand binding domain, b) a transmembrane domain, c) a protease cleavage site, and d) a functional domain comprising an RNA-binding subdomain fused to a transcription regulatory subdomain. The exogenous extracellular sensor may be referred to as a recombinant fusion protein comprising a) the ligand binding domain, b) the transmembrane domain, c) the protease cleavage site, and d) the functional domain comprising an RNA-binding subdomain fused to a transcription regulatory subdomain, wherein one or more of the domains of the fusion protein may be heterologous to one or more other domains of the fusion protein.

Suitable RNA-binding subdomains of the functional domain of the exogenous extracellular sensor may include RNA-binding domains found in eukaryotic proteins and/or prokaryotic proteins. In some embodiments, the RNA-binding subdomain comprises a RNA-binding domain of a Cas9 protein. The RNA-binding subdomain of the functional domain may bind an RNA that confers DNA recognition to the functional domain and/or that targets the functional domain to a DNA target such as a transcription promoter. Suitable RNAs bound by the RNA-binding subdomain of the functional domain may include small guide RNAs (sgRNAs) that may be utilized to target the functional domain to a DNA target.

In some embodiments, the disclosed systems and methods comprise or utilize: i) first and second exogenous extracellular sensors, and/or ii) one or more nucleic acid sequences encoding the first and second exogenous extracellular sen-

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sors, wherein the first exogenous extracellular sensor comprises: a) a ligand binding domain, b) a transmembrane domain, c) a protease cleavage site, and d) a functional domain comprising an RNA-binding subdomain fused to a transcription regulatory subdomain; and wherein the second exogenous extracellular sensor comprises: e) a ligand binding domain, f) a transmembrane domain, and g) a protease domain. In systems where the first and second exogenous extracellular sensors are present in a cell as membrane proteins, the protease domain of the second exogenous extracellular sensor may recognize and cleave the protease cleavage site of the first exogenous extracellular sensor, for example, when the first exogenous extracellular sensor and the second exogenous extracellular sensor bind to an extracellular ligand via the ligand binding domain of the first exogenous extracellular sensor and the ligand binding domain of the second exogenous extracellular sensor, respectively. In some embodiments, the first exogenous extracellular sensor and the second exogenous extracellular sensor bind to the same ligand, and when the first exogenous extracellular sensor and the second exogenous extracellular sensor bind to the same ligand, the protease domain of the second exogenous extracellular sensor is positioned near the protease cleavage site of the first exogenous extracellular sensor to facilitate cleavage of the protease cleavage site of the first exogenous extracellular sensor by the protease domain of the second exogenous extracellular sensor. Through this cleavage, the functional domain of the first exogenous extracellular sensor comprising an RNA-binding subdomain fused to a transcription regulatory subdomain then may be released from the first exogenous extracellular sensor and may migrate to a targeted gene, for example, via a sgRNA bound by the RNA-binding subdomain of the functional domain. Thus released and targeted, the functional domain may function to modulate expression of the targeted gene.

The disclosed systems and methods comprising and/or utilizing the disclosed extracellular sensors may be utilized for sensing extracellular ligands and/or modulating expression of an endogenous or exogenous gene. In some embodiments, the disclosed systems and methods further may comprise or utilize an RNA or a vector that expresses an RNA that targets the functional domain of an extracellular sensor to a target gene. In the disclosed systems and methods, optionally the RNA-binding subdomain is a Cas9 RNA-binding domain and the RNA is a small guide RNA (sgRNA) that is bound by the RNA-binding subdomain of the functional domain and that targets the functional domain to a target gene whose transcription is modulated by the transcription regulatory subdomain of the functional domain.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1A-1E. VEGF-MESA receptor development. FIG. 1A Upper: schematic of the VEGF-MESA mechanism, by which ligand binding-induced dimerization results in trans-cleavage and release of a previously sequestered transcription factor. Lower: crystal structures of two Fabs indicating domains within each Fab binding to each of two epitopes on VEGF³⁷. Table inset: reported equilibrium dissociation constants for Fabs binding to human and mouse VEGF²². FIG. 1B Cell-surface expression of representative VEGF-MESA constructs was evaluated by flow cytometry in cells transfected with vectors expressing VEGF-MESA versus vector-only transfected controls. Variants shown here differ in the lengths (in amino acids) of the non-structured extracellular

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linkers (ECL). Gray area of graphs represents vector-only transfected controls. This experiment is representative of two independent experiments. FIG. 1C VEGF-binding to VEGF-MESA was quantified using flow cytometry. Bar heights correspond to mean fluorescence intensities of samples comprising at least 5000 transfected cells, and each error bar represents one standard deviation. This experiment is representative of two independent experiments. FIG. 1D VEGF-inducible signaling by V1-MESA was quantified by flow cytometry. "Reporter activity" was calculated by quantifying mean fluorescence intensity (MFI) of YFP in transfected cells, then normalizing this value to that calculated for the internal control (cells transfected with reporter-only, the value of which was set to one). Numbers above the bars represents fold induction, calculated by dividing the reporter signaling in the presence of VEGF by the reporter signaling in the absence of VEGF. Experiments were conducted in biological triplicate, each experiment is representative of at least three independent biological experiments, and error bars represent one standard deviation. (* $p \leq 0.05$, ** $p \leq 0.001$). FIG. 1E VEGF-inducible signaling by V2-MESA was quantified by flow cytometry. "Reporter activity" was calculated by quantifying mean fluorescence intensity (MFI) of YFP in transfected cells, then normalizing this value to that calculated for the internal control (cells transfected with reporter-only, the value of which was set to one). The boxed area highlights constructs exhibiting significant VEGF-inducible signaling. Numbers above the bars represents fold induction, calculated by dividing the reporter signaling in the presence of VEGF by the reporter signaling in the absence of VEGF. Experiments were conducted in biological triplicate, each experiment is representative of at least three independent biological experiments, and error bars represent one standard deviation. (* $p \leq 0.05$, ** $p \leq 0.001$)

FIG. 2A, FIG. 2B FIG. 2C, FIG. 2D, FIG. 2E, and FIG. 2F. VEGF-MESA receptor implementation. FIG. 2A This cartoon illustrates potential MESA complexes expected to form upon VEGF binding-induced dimerization. FIG. 2B This cartoon illustrates the proposed mechanism by which 'dead' TC complexes may form due to transient receptor contact during trafficking or on the cell surface. FIG. 2C, FIG. 2D, and FIG. 2E Both overall expression levels and ratios of the two VEGF-MESA chains was varied by dosing in different amounts of plasmid encoding each of the two MESA chains (TC and PC), as indicated in both the table (plasmid ratio) and ramps (plasmid masses) placed under each panel. Boxes highlight the regimes in which VEGF-inducible receptor signaling was observed. Numbers above the bars represents fold induction and data were analyzed as described in FIG. 1. FIG. 2F Dose response of VEGF-MESA. Based upon results shown in FIG. 2C, FIG. 2D, and FIG. 2E, VEGF-MESA chains were expressed at a TC:PC ratio of 24 (3 μ g TC plasmid and 0.125 μ g PC plasmid per well). Horizontal bars highlight the 0 ng/mL VEGF sample, to which each test sample was compared, with the thickness indicating one standard deviation. Sigmoidal lines were added to guide the eye. Data were analyzed as described in FIG. 1. (* $p \leq 0.05$, ** $p \leq 0.01$).

FIG. 3A, FIG. 3B, FIG. 3C, FIG. 3D, FIG. 3E, FIG. 3F, FIG. 3G, FIG. 3H, and FIG. 3I. Rewiring of cellular input-output using MESA. FIG. 3A Identification of sgRNAs that confer dCas9-TF-mediated induction of IL-2 expression. sgRNAs were screened individually and in combination and IL-2 mRNA expression was measured by qPCR. IL-2 mRNA copy number was normalized to GAPDH mRNA copy number, and this ratio was normalized to that calcu-

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lated for cells transfected with only dCas9-TF (black bar). Labeled squares in the table indicate sgRNA that, individually, confer substantial IL-2 expression and were selected for further analysis. mRNA concentrations were evaluated in technical triplicate, and results are representative of three independent biological experiments. FIG. 3B and FIG. 3C Evaluation of SAM system³⁴ for enhancing dCas9-TF-induced IL-2 gene expression. Data were analyzed as in FIG. 3A. FIG. 3D and FIG. 3E These cartoons summarize the proposed scheme for rewiring a cell to generate a non-natural function, whereby exposure to a canonically immunosuppressive cue (VEGF) results in the production of a canonically immunostimulatory factor in response (IL-2). FIG. 3F-I IL-2 secretion driven by soluble dCas9-TF was evaluated in HEK 293FT cells (FIG. 3F) and Jurkat T cells (FIG. 3G) and quantified by ELISA. VEGF-inducible secretion of IL-2 by cells expressing VEGF-MESA-dCas9-TF at a TC:PC ratio of 10 and sgRNA 7 was evaluated in HEK 293FT cells (FIG. 3H) and Jurkat T cells (FIG. 3I). Each sample was quantified in technical triplicate and data shown are representative of three independent biological experiments (see FIG. 11). Numbers above the bars represents fold induction, as described in FIG. 1. Error bars represent one standard deviation. (* $p \leq 0.05$, ** $p \leq 0.01$).

FIG. 4A, FIG. 4B, FIG. 4C, FIG. 4D, and FIG. 4E. Surface expression and VEGF binding of VEGF-MESAs. FIG. 4A, FIG. 4B, and FIG. 4C Cell-surface expression of VEGF-MESA was evaluated by flow cytometry (See Methods and FIG. 1 for measurement details). Variants shown here differ in the lengths (in amino acids) of the non-structured extracellular linkers (ECL). Gray areas of graphs represent vector-only transfected controls. This experiment is representative of two independent experiments. FIG. 4D and FIG. 4E VEGF-binding to all reported VEGF-MESA clones was quantified using flow cytometry (see Methods and FIG. 1 for measurement details). Bar heights corresponds to sample mean fluorescence intensities and each error bar represents one standard deviation.

FIG. 5A-FIG. 5C. Impact of intracellular linker on cell surface expression. Cellsurface staining of various TC and PC constructs was quantified using immunohistochemistry and flow cytometry (see Methods and FIG. 1 for measurement details). Gray areas of graphs represent vector only transfected cell controls. FIG. 5A The two left panels indicate the original VEGF-MESA constructs, where cell surface expression for the TC was much greater than for PC, when the PC lacked an intracellular linker (ICL) between transmembrane and protease domains. PC constructs were evaluated with a 3 amino acid flexible linker between the transmembrane domain and protease (right). FIG. 5B and FIG. 5C Sequential addition of amino acids derived from the cleavage sequence. The box in FIG. 5B indicates that no further improvements in surface expression were observed for linkers beyond this length; this PC design was used for all subsequent experiments in this study.

FIG. 6A, FIG. 6B, FIG. 6C, FIG. 6D, FIG. 6E, FIG. 6F, and FIG. 6G. Evaluation of VEGF-MESA with G cleavage sequence on the TC. FIG. 6A, FIG. 6B, and FIG. 6C Cell-surface staining of VEGF-MESA was quantified using immunohistochemistry and flow cytometry (see Methods and FIG. 1 for measurement details). Gray areas of graphs represent vector only transfected cell controls. FIG. 6D and FIG. 6E VEGF-binding to all reported VEGF-MESA clones was quantified using flow cytometry (see Methods). Bar heights corresponds to sample mean fluorescence intensities and each error bar represents one standard deviation. FIG. 6F and FIG. 6G VEGF-induced reporter activation for

VEGF-MESA with 'G' cleavage sequence on the TC was quantified using flow cytometry. See FIG. 1 for measurement details.

FIG. 7A, FIG. 7B, FIG. 7C, FIG. 7D, FIG. 7E, and FIG. 7F. VEGF-MESA surface expression levels as a function of plasmid dose. Cell surface expression of VEGF-MESA was evaluated for the range of plasmid doses evaluated in FIG. 2 for the TC (FIG. 7A, FIG. 7B, FIG. 7C) and PC (FIG. 7D, FIG. 7E, FIG. 7F) using flow cytometry (See Methods and FIG. 1 for measurement details). This experiment was conducted in biological duplicate with the histograms (FIG. 7A, FIG. 7B, FIG. 7D) showing a representative example and the line graphs FIG. 7C and FIG. 7F quantifying the normalized average expression level observed across experiments. All samples were normalized to a vector only control, and to compare between the biological replicates, the highest value for each experiment (3 g plasmid for V1-MESA TC) was set to unity. Error bars represent one standard deviation.

FIG. 8A, FIG. 8B, FIG. 8C, and FIG. 8D. Tuning MESA expression levels by varying protein translation initiation rate. Protein translation initiation rate was varied by utilizing a range of synthetic upstream open reading frames (uORF). The three letter codes above denote the last three nucleotides of the Kozak sequence utilized for that particular sample. Samples with a '/' have an additional uORF added in the 5'UTR—i.e., CCC/ACC has an ACC Kozak sequence directly before the start codon for VEGF-MESA as well as one uORF in the 5' UTR, in which the last three nucleotides of the Kozak sequence are CCC. The ACC Kozak sequence is the consensus sequence that was used for the constructs in FIGS. 1, 2, and 3. FIG. 8A, FIG. 8B, FIG. 8C Cell-surface expression of uORF-containing V1-MESA constructs was evaluated utilizing flow cytometry. This experiment was conducted in biological duplicate with the histograms FIG. 8A and FIG. 8B showing a representative example and the line graph FIG. 8C quantifying the normalized average expression level observed across experiments. All samples were normalized to a vector only control, and to compare between the biological replicates, the highest value for each experiment (CGA for V1-MESA PC) was set to unity. FIG. 8D VEGF-inducible signaling was quantified by flow cytometry for uORF-containing V1-MESA constructs. The two uORFs that conferred the highest levels of surface expression, CGA and ACC, were utilized for the TC, and these plasmids were transfected at a 1:1 ratio with one of various plasmids encoding the PC. Numbers above the bar represent fold induction, and data were analyzed as described in FIG. 1. Error bars represent one standard deviation. (* $p \leq 0.05$, ** $p \leq 0.01$).

FIG. 9. Location and sequence of IL-2 sgRNAs. Schematic of locations for all IL-2 sgRNAs evaluated. Locations represent distance upstream or downstream from the transcription start site. Full sgRNA targeting sequence are listed in the table. Uppercase letters represent direct base pairing interactions with the target DNA, lowercase letters represent deviations from the target sequence. Sequences that start with a lowercase 'g' were designed in this manner to enable transcription from a U6 promoter.

FIG. 10. Surface staining of VEGF-MESA with dCas9-TF. Cell-surface staining of VEGF-MESA expressed in either HEK 293FT cells or Jurkat T cells was quantified using immunohistochemistry and flow cytometry (see Methods and FIG. 1 for measurement details). Gray areas of graphs represent vector only transfected cell controls.

FIG. 11A and FIG. 11B. Biological repeats for VEGF-MESA with dCas9-TF. FIG. 11A and FIG. 11B HEK 293FT

cells or Jurkat T cells were transfected to transiently express VEGF-MESA with dCas9-TF and sgRNA 7, and IL-2 protein production was quantified by ELISA. TC:PC ratio was set to 10 and refers to relative amounts of TC and PC plasmids included in the transfection. Numbers above the bars represent fold induction (See FIG. 1). See FIG. 3 for measurement details. (* $p \leq 0.05$).

FIG. 12A and FIG. 12B. Investigating alternative TC:PC ratios and choice of sgRNA for VEGF-MESA with dCas9-TF. FIG. 12A HEK 293FT cells were transfected to transiently express VEGFMESA with dCas9-TF and sgRNA 17, and IL-2 protein production was quantified by ELISA. TC:PC ratio was set to 10 and refers to relative amounts of TC and PC plasmids included in the transfection. Numbers above the bars represent fold induction (See FIG. 1). See FIG. 3 for measurement details. FIG. 12B HEK 293FT cells were transfected to transiently express VEGF-MESA with dCas9-TF and sgRNA 7, and IL-2 protein production was quantified by ELISA. TC:PC ratio refers to relative amounts of TC and PC plasmids included in the transfection. Numbers above the bars represent fold induction (See FIG. 1). See FIG. 3 for measurement details.

FIG. 13. Investigating the IL-2 producing subset of Jurkat T cells. Jurkat T cells were transfected with VEGF-MESA at a TC:PC ratio of 10 and sgRNA 7 and treated with VEGF and Brefeldin A (the latter blocks protein secretion), for 24 h. IL-2 production was evaluated by labeling intracellular IL-2 in fixed cells utilizing an anti-IL-2 antibody and then co-evaluating both IL-2 labeling and expression of a transfection control (BFP) by flow cytometry. Each sample was analyzed by dividing cells into two subsets based upon the expression (or non-expression) of the transfection control (BFP), and the mean level of anti-IL-2 labeling (APC) was quantified. Each sample was conducted in biological triplicate, and error bars represent one standard deviation. (* $p \leq 0.001$).

DETAILED DESCRIPTION

The present invention is described herein using several definitions, as set forth below and throughout the application.

Unless otherwise specified or indicated by context, the terms "a", "an", and "the" mean "one or more." For example, "a receptor," "ligand," and "complex" should be interpreted to mean "one or more receptors," "one or more ligands," and "one or more complexes," respectively.

As used herein, "about," "approximately," "substantially," and "significantly" will be understood by persons of ordinary skill in the art and will vary to some extent on the context in which they are used. If there are uses of these terms which are not clear to persons of ordinary skill in the art given the context in which they are used, "about" and "approximately" will mean plus or minus $\leq 10\%$ of the particular term and "substantially" and "significantly" will mean plus or minus $> 10\%$ of the particular term.

As used herein, the terms "include" and "including" have the same meaning as the terms "comprise" and "comprising" in that these latter terms are "open" transitional terms that do not limit claims only to the recited elements succeeding these transitional terms. The term "consisting of," while encompassed by the term "comprising," should be interpreted as a "closed" transitional term that limits claims only to the recited elements succeeding this transitional term. The term "consisting essentially of," while encompassed by the term "comprising," should be interpreted as a "partially closed" transitional term which permits additional elements

succeeding this transitional term, but only if those additional elements do not materially affect the basic and novel characteristics of the claim.

The disclosed technology relates to “extracellular sensors.” As disclosed herein, an “extracellular sensor” is a molecule or a system of molecules that can be used to bind to a ligand and provide a detectable response based on binding the ligand. In some cases, “extracellular sensor” may be referred to as “molecular switches.” Extracellular sensors are disclosed in the art. (See, e.g., Daringer et al., “Modular Extracellular Sensor Architecture for Engineering Mammalian Cell-based Devices,” Nichole M. Daringer, Rachel M. Dudek, Kelly A. Schwarz, and Josh N. Leonard, ACS Synth. Biol. 2014, 3, 892-902, published Feb. 25, 2014; WO 2013/022739, published on Feb. 14, 2013; and U.S. Publication No. 2014-0234851; the contents of which are incorporated herein by reference in their entireties).

As contemplated herein, a “ligand-binding protein” is a macromolecule, typically a protein, which binds to a ligand. For example, a ligand-binding protein may include a receptor for a ligand or a portion of a receptor for a ligand, for example, where the receptor is a membrane protein and the ligand-binding protein comprises the extracellular portion of the receptor that binds an extracellular ligand.

The disclosed extracellular sensors may be utilized for sensing an extracellular ligand and providing a molecular signal to a cell when the ligand is sensed. The molecular signal may result in modulating expression from an endogenous promoter in the cell or an exogenous promoter in the cell. As used herein, “modulating expression” may include “activating expression” or “de-repressing expression,” and/or “modulating expression” may include “repressing expression” and/or “inhibiting expression.”

Reference is made herein to nucleic acid and nucleic acid sequences. The terms “nucleic acid” and “nucleic acid sequence” refer to a nucleotide, oligonucleotide, polynucleotide (which terms may be used interchangeably), or any fragment thereof. These phrases also refer to DNA or RNA of genomic or synthetic origin (which may be single-stranded or double-stranded and may represent the sense or the antisense strand).

Reference also is made herein to peptides, polypeptides, proteins and compositions comprising peptides, polypeptides, and proteins. As used herein, a polypeptide and/or protein is defined as a polymer of amino acids, typically of length ≥ 100 amino acids (Garrett & Grisham, Biochemistry, 2nd edition, 1999, Brooks/Cole, 110). A peptide is defined as a short polymer of amino acids, of a length typically of 20 or less amino acids, and more typically of a length of 12 or less amino acids (Garrett & Grisham, Biochemistry, 2nd edition, 1999, Brooks/Cole, 110).

As disclosed herein, exemplary peptides, polypeptides, proteins may comprise, consist essentially of, or consist of any reference amino acid sequence disclosed herein, or variants of the peptides, polypeptides, and proteins may comprise, consist essentially of, or consist of an amino acid sequence having at least about 80%, 90%, 95%, 96%, 97%, 98%, or 99% sequence identity to any amino acid sequence disclosed herein, for example SEQ ID NOs:1-17. Variant peptides, polypeptides, and proteins may include peptides, polypeptides, and proteins having one or more amino acid substitutions, deletions, additions and/or amino acid insertions relative to a reference peptide, polypeptide, or protein. Also disclosed are nucleic acid molecules that encode the disclosed peptides, polypeptides, and proteins (e.g., polynucleotides that encode any of the peptides, polypeptides, and proteins disclosed herein and variants thereof).

The term “amino acid,” includes but is not limited to amino acids contained in the group consisting of alanine (Ala or A), cysteine (Cys or C), aspartic acid (Asp or D), glutamic acid (Glu or E), phenylalanine (Phe or F), glycine (Gly or G), histidine (His or H), isoleucine (Ile or I), lysine (Lys or K), leucine (Leu or L), methionine (Met or M), asparagine (Asn or N), proline (Pro or P), glutamine (Gln or Q), arginine (Arg or R), serine (Ser or S), threonine (Thr or T), valine (Val or V), tryptophan (Trp or W), and tyrosine (Tyr or Y) residues. The term “amino acid residue” also may include amino acid residues contained in the group consisting of homocysteine, 2-Aminoadipic acid, N-Ethylasparagine, 3-Aminoadipic acid, Hydroxylysine, β -alanine, β -Amino-propionic acid, allo-Hydroxylysine acid, 2-Aminobutyric acid, 3-Hydroxyproline, 4-Aminobutyric acid, 4-Hydroxyproline, piperidinic acid, 6-Aminocaproic acid, Isodesmosine, 2-Aminoheptanoic acid, allo-Isoleucine, 2-Aminoisobutyric acid, N-Methylglycine, sarcosine, 3-Aminoisobutyric acid, N-Methylisoleucine, 2-Aminopimelic acid, 6-N-Methyllysine, 2,4-Diaminobutyric acid, N-Methylvaline, Desmosine, Norvaline, 2,2'-Diaminopimelic acid, Norleucine, 2,3-Diaminopropionic acid, Ornithine, and N-Ethylglycine. Typically, the amide linkages of the peptides are formed from an amino group of the backbone of one amino acid and a carboxyl group of the backbone of another amino acid.

The amino acid sequences contemplated herein may include conservative amino acid substitutions relative to a reference amino acid sequence. For example, a variant peptides, polypeptides, and proteins as contemplated herein may include conservative amino acid substitutions relative to an amino acid sequence of a reference peptide, polypeptide, or protein. “Conservative amino acid substitutions” are those substitutions that are predicted to interfere least with the properties of the reference peptide, polypeptide, or protein. In other words, conservative amino acid substitutions substantially conserve the structure and the function of the reference peptide, polypeptide, or protein. The following table provides a list of exemplary conservative amino acid substitutions.

Table of Conservative Amino Acid Substitutions

Original Residue	Conservative Substitution
Ala	Gly, Ser
Arg	His, Lys
Asn	Asp, Gln, His
Asp	Asn, Glu
Cys	Ala, Ser
Gln	Asn, Glu, His
Glu	Asp, Gln, His
Gly	Ala
His	Asn, Arg, Gln, Glu
Ile	Leu, Val
Leu	Ile, Val
Lys	Arg, Gln, Glu
Met	Leu, Ile
Phe	His, Met, Leu, Trp, Tyr
Ser	Cys, Thr
Thr	Ser, Val
Trp	Phe, Tyr
Tyr	His, Phe, Trp
Val	Ile, Leu, Thr

“Non-conservative amino acid substitutions” are those substitutions that are predicted to interfere most with the properties of the reference peptide, polypeptide, or protein. For example, a non-conservative amino acid substitution

might replace a basic amino acid at physiological pH such as Arg, His, or Lys, with a non-basic or acidic amino acid at physiological pH such as Asp or Glu. A non-conservative amino acid substitution might replace a non-polar amino acid at physiological pH such as Ala, Gly, Ile, Leu, Phe, or Val, with a polar amino acid at physiological pH such as Arg, Asp, Glu, His, or Lys.

The peptides, polypeptides, and proteins disclosed herein may be modified to include non-amino acid moieties. Modifications may include but are not limited to carboxylation (e.g., N-terminal carboxylation via addition of a di-carboxylic acid having 4-7 straight-chain or branched carbon atoms, such as glutaric acid, succinic acid, adipic acid, and 4,4-dimethylglutaric acid), amidation (e.g., C-terminal amidation via addition of an amide or substituted amide such as alkylamide or dialkylamide), PEGylation (e.g., N-terminal or C-terminal PEGylation via additional of polyethylene glycol), acylation (e.g., O-acylation (esters), N-acylation (amides), S-acylation (thioesters)), acetylation (e.g., the addition of an acetyl group, either at the N-terminus of the protein or at lysine residues), formylation lipoylation (e.g., attachment of a lipate, a C8 functional group), myristoylation (e.g., attachment of myristate, a C14 saturated acid), palmitoylation (e.g., attachment of palmitate, a C16 saturated acid), alkylation (e.g., the addition of an alkyl group, such as an methyl at a lysine or arginine residue), isoprenylation or prenylation (e.g., the addition of an isoprenoid group such as farnesol or geranylgeraniol), amidation at C-terminus, glycosylation (e.g., the addition of a glycosyl group to either asparagine, hydroxylysine, serine, or threonine, resulting in a glycoprotein). Distinct from glycation, which is regarded as a nonenzymatic attachment of sugars, polysialylation (e.g., the addition of polysialic acid), glypiation (e.g., glycosylphosphatidylinositol (GPI) anchor formation, hydroxylation, iodination (e.g., of thyroid hormones), and phosphorylation (e.g., the addition of a phosphate group, usually to serine, tyrosine, threonine or histidine).

Variants comprising deletions relative to a reference amino acid sequence or nucleotide sequence are contemplated herein. A "deletion" refers to a change in the amino acid or nucleotide sequence that results in the absence of one or more amino acid residues or nucleotides relative to a reference sequence. A deletion removes at least 1, 2, 3, 4, 5, 10, 20, 50, 100, or 200 amino acids residues or nucleotides. A deletion may include an internal deletion or a terminal deletion (e.g., an N-terminal truncation or a C-terminal truncation or both of a reference polypeptide or a 5'-terminal or 3'-terminal truncation or both of a reference polynucleotide).

Variants comprising a fragment of a reference amino acid sequence or nucleotide sequence are contemplated herein, for example fragments of an of SEQ ID NOs:1-17. A "fragment" is a portion of an amino acid sequence or a nucleotide sequence which is identical in sequence to but shorter in length than the reference sequence. A fragment may comprise up to the entire length of the reference sequence, minus at least one nucleotide/amino acid residue. For example, a fragment may comprise from 5 to 1000 contiguous nucleotides or contiguous amino acid residues of a reference polynucleotide or reference polypeptide, respectively. In some embodiments, a fragment may comprise at least 5, 10, 15, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 40, 50, 60, 70, 80, 90, 100, 150, 250, or 500 contiguous nucleotides or contiguous amino acid residues of a reference polynucleotide or reference polypeptide, respectively. Fragments may be preferentially selected from certain regions of

a molecule, for example the N-terminal region and/or the C-terminal region of a polypeptide or the 5'-terminal region and/or the 3' terminal region of a polynucleotide. The term "at least a fragment" encompasses the full length polynucleotide or full length polypeptide.

Variants comprising insertions or additions relative to a reference sequence are contemplated herein. The words "insertion" and "addition" refer to changes in an amino acid or nucleotide sequence resulting in the addition of one or more amino acid residues or nucleotides. An insertion or addition may refer to 1, 2, 3, 4, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 150, or 200 amino acid residues or nucleotides.

Fusion proteins and fusion polynucleotides also are contemplated herein. The disclosed extracellular sensors may be defined as a fusion protein. A "fusion protein" refers to a protein formed by the fusion of at least one peptide, polypeptide, protein or variant thereof as disclosed herein to at least one molecule of a heterologous peptide, polypeptide, protein or variant thereof. The heterologous protein(s) may be fused at the N-terminus, the C-terminus, or both termini. A fusion protein comprises at least a fragment or variant of the heterologous protein(s) that are fused with one another, preferably by genetic fusion (i.e., the fusion protein is generated by translation of a nucleic acid in which a polynucleotide encoding all or a portion of a first heterologous protein is joined in-frame with a polynucleotide encoding all or a portion of a second heterologous protein). The heterologous protein(s), once part of the fusion protein, may each be referred to herein as a "portion", "region" or "moiety" of the fusion protein. For example, where the fusion protein comprises at least a portion of a ligand binding protein, at least a portion of a transmembrane protein, at least a portion of a protease, the portions of the fusion protein may be referred to as "a ligand binding portion," "a transmembrane portion," and "a protease portion," respectively.

A fusion polynucleotide refers to the fusion of the nucleotide sequence of a first polynucleotide to the nucleotide sequence of a second heterologous polynucleotide (e.g., the 3' end of a first polynucleotide to a 5' end of the second polynucleotide). Where the first and second polynucleotides encode proteins, the fusion may be such that the encoded proteins are in-frame and results in a fusion protein. The first and second polynucleotide may be fused such that the first and second polynucleotide are operably linked (e.g., as a promoter and a gene expressed by the promoter as discussed below).

"Homology" refers to sequence similarity or, interchangeably, sequence identity, between two or more polypeptide sequences or polynucleotide sequences. Homology, sequence similarity, and percentage sequence identity may be determined using methods in the art and described herein.

The phrases "percent identity" and "% identity," as applied to polypeptide sequences, refer to the percentage of residue matches between at least two polypeptide sequences aligned using a standardized algorithm. Methods of polypeptide sequence alignment are well-known. Some alignment methods take into account conservative amino acid substitutions. Such conservative substitutions, explained in more detail above, generally preserve the charge and hydrophobicity at the site of substitution, thus preserving the structure (and therefore function) of the polypeptide. Percent identity for amino acid sequences may be determined as understood in the art. (See, e.g., U.S. Pat. No. 7,396,664, which is incorporated herein by reference in its entirety). A suite of commonly used and freely available sequence comparison algorithms is provided by the National Center for Biotechnology Information (NCBI) Basic Local Align-

ment Search Tool (BLAST) (Altschul, S. F. et al. (1990) *J. Mol. Biol.* 215:403-410), which is available from several sources, including the NCBI, Bethesda, Md., at its website. The BLAST software suite includes various sequence analysis programs including “blastp,” that is used to align a known amino acid sequence with other amino acid sequences from a variety of databases.

Percent identity may be measured over the length of an entire defined polypeptide sequence or may be measured over a shorter length, for example, over the length of a fragment taken from a larger, defined polypeptide sequence, for instance, a fragment of at least 15, at least 20, at least 30, at least 40, at least 50, at least 70 or at least 150 contiguous residues. Such lengths are exemplary only, and it is understood that any fragment length may be used to describe a length over which percentage identity may be measured.

A “variant” of a particular polypeptide sequence may be defined as a polypeptide sequence having at least 50% sequence identity to the particular polypeptide sequence over a certain length of one of the polypeptide sequences using blastp with the “BLAST 2 Sequences” tool available at the National Center for Biotechnology Information’s website. (See Tatiana A. Tatusova, Thomas L. Madden (1999), “Blast 2 sequences—a new tool for comparing protein and nucleotide sequences”, *FEMS Microbiol. Lett.* 174:247-250). In some embodiments a variant polypeptide may show, for example, at least 60%, at least 70%, at least 80%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% or greater sequence identity over a certain defined length relative to a reference polypeptide such as a polypeptide comprising an amino acid sequence of any of SEQ ID NOs:1-17.

A variant polypeptide may have substantially the same functional activity as a reference polypeptide. For example, a variant polypeptide may exhibit or more biological activities associated with binding a ligand and/or binding DNA at a specific binding site.

The terms “percent identity” and “% identity,” as applied to polynucleotide sequences, refer to the percentage of residue matches between at least two polynucleotide sequences aligned using a standardized algorithm. Such an algorithm may insert, in a standardized and reproducible way, gaps in the sequences being compared in order to optimize alignment between two sequences, and therefore achieve a more meaningful comparison of the two sequences. Percent identity for a nucleic acid sequence may be determined as understood in the art. (See, e.g., U.S. Pat. No. 7,396,664, which is incorporated herein by reference in its entirety). A suite of commonly used and freely available sequence comparison algorithms is provided by the National Center for Biotechnology Information (NCBI) Basic Local Alignment Search Tool (BLAST) (Altschul, S. F. et al. (1990) *J. Mol. Biol.* 215:403-410), which is available from several sources, including the NCBI, Bethesda, Md., at its website. The BLAST software suite includes various sequence analysis programs including “blastn,” that is used to align a known polynucleotide sequence with other polynucleotide sequences from a variety of databases. Also available is a tool called “BLAST 2 Sequences” that is used for direct pairwise comparison of two nucleotide sequences. “BLAST 2 Sequences” can be accessed and used interactively at the NCBI website. The “BLAST 2 Sequences” tool can be used for both blastn and blastp (discussed above).

Percent identity may be measured over the length of an entire defined polynucleotide sequence or may be measured over a shorter length, for example, over the length of a

fragment taken from a larger, defined sequence, for instance, a fragment of at least 20, at least 30, at least 40, at least 50, at least 70, at least 100, or at least 200 contiguous nucleotides. Such lengths are exemplary only, and it is understood that any fragment length may be used to describe a length over which percentage identity may be measured.

A “full length” polynucleotide sequence is one containing at least a translation initiation codon (e.g., methionine) followed by an open reading frame and a translation termination codon. A “full length” polynucleotide sequence encodes a “full length” polypeptide sequence.

A “variant,” “mutant,” or “derivative” of a particular nucleic acid sequence may be defined as a nucleic acid sequence having at least 50% sequence identity to the particular nucleic acid sequence over a certain length of one of the nucleic acid sequences using blastn with the “BLAST 2 Sequences” tool available at the National Center for Biotechnology Information’s website. (See Tatiana A. Tatusova, Thomas L. Madden (1999), “Blast 2 sequences—a new tool for comparing protein and nucleotide sequences”, *FEMS Microbiol. Lett.* 174:247-250). In some embodiments a variant polynucleotide may show, for example, at least 60%, at least 70%, at least 80%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99% or greater sequence identity over a certain defined length relative to a reference polynucleotide.

Nucleic acid sequences that do not show a high degree of identity may nevertheless encode similar amino acid sequences due to the degeneracy of the genetic code. It is understood that changes in a nucleic acid sequence can be made using this degeneracy to produce multiple nucleic acid sequences that all encode substantially the same protein.

“Operably linked” refers to the situation in which a first nucleic acid sequence is placed in a functional relationship with a second nucleic acid sequence. For instance, a promoter is operably linked to a coding sequence if the promoter affects the transcription or expression of the coding sequence. Operably linked DNA sequences may be in close proximity or contiguous and, where necessary to join two protein coding regions, in the same reading frame.

A “recombinant nucleic acid” is a sequence that is not naturally occurring or has a sequence that is made by an artificial combination of two or more otherwise separated segments of sequence. This artificial combination is often accomplished by chemical synthesis or, more commonly, by the artificial manipulation of isolated segments of nucleic acids, e.g., by genetic engineering techniques such as those described in Sambrook, J. et al. (1989) *Molecular Cloning: A Laboratory Manual*, 2nd ed., vol. 1-3, Cold Spring Harbor Press, Plainview N.Y. The term recombinant includes nucleic acids that have been altered solely by addition, substitution, or deletion of a portion of the nucleic acid. Frequently, a recombinant nucleic acid may include a nucleic acid sequence operably linked to a promoter sequence. Such a recombinant nucleic acid may be part of a vector that is used, for example, to transform a cell.

“Transformation” describes a process by which exogenous DNA is introduced into a recipient cell. Transformation may occur under natural or artificial conditions according to various methods well known in the art, and may rely on any known method for the insertion of foreign nucleic acid sequences into a prokaryotic or eukaryotic host cell. The method for transformation is selected based on the type of host cell being transformed and may include, but is not limited to, bacteriophage or viral infection, electroporation, heat shock, lipofection, and particle bombardment. The term

“transformed cells” includes stably transformed cells in which the inserted DNA is capable of replication either as an autonomously replicating plasmid or as part of the host chromosome, as well as transiently transformed cells which express the inserted DNA or RNA for limited periods of time.

A “composition comprising a given polypeptide” and a “composition comprising a given polynucleotide” refer broadly to any composition containing the given polynucleotide or amino acid sequence. The composition may comprise a dry formulation or an aqueous solution. The compositions may be stored in any suitable form including, but not limited to, freeze-dried form and may be associated with a stabilizing agent such as a carbohydrate. The compositions may be aqueous solution containing salts (e.g., NaCl), detergents (e.g., sodium dodecyl sulfate; SDS), and other components.

“Substantially isolated or purified” nucleic acid or amino acid sequences are contemplated herein. The term “substantially isolated or purified” refers to nucleic acid or amino acid sequences that are removed from their natural environment, and are at least 60% free, preferably at least 75% free, and more preferably at least 90% free, even more preferably at least 95% free from other components with which they are naturally associated.

Antibodies, antigen-binding fragments of antibodies, and fusion proteins comprising fragments of antibodies are contemplated herein. For example, the ligand binding domain of the disclosed exogenous extracellular sensors may comprise an antibody, an antigen-binding fragment of an antibody, or a fusion protein comprising fragments of an antibody, where the antibody, the antigen-binding fragment of the antibody, or the fusion protein comprising fragments of an antibody binds to a ligand. Suitable antibodies, antigen-binding fragments thereof, and fusion proteins comprising fragments of an antibody may include, but are not limited to Fab fragments, single-chain variable fragments (scFvs), and single-domain antibodies (sdAb or nanobodies).

The disclosed sensors include a protease cleavage sequence. The sensors are not limited to any particular protease or corresponding protease cleavage site. In some embodiments, the protease and cleavage site are from a virus. For example, in certain embodiments, the protease and protease cleavage site are from a virus selected from: tobacco etch virus (TEV), a chymotrypsin-like serine protease and corresponding cleavage sites, alphavirus proteases and cleavage sites, Hepatitis C virus proteases (e.g., NS3 proteases) and corresponding cleavage sites, chymotrypsin-like cysteine proteases and corresponding cleavage sites, papain-like cysteine proteases and cleavage sites, picornavirus leader proteases and cleavage sites, HIV proteases and cleavage sites, Herpesvirus proteases and cleavage sites, and adenovirus proteases and cleavage sites (see, Tong, Chem. Rev. 2002, 102, 4609-4626, herein incorporated by reference in its entirety). In particular embodiments, the proteases and cleavage sites are bacterial in origin, such as, for example, from *Streptomyces griseus* protease A (SGPA), SGPB, and alpha-lytic protease and corresponding cleavage sites. In some embodiments, the proteases and cleavage sites are mammalian. For example, the proteases could be one of the five major classes of proteases known in mammals which include serine proteases, cysteine proteases, metalloproteases, aspartic proteases, and threonine proteases (see, e.g., Turk et al., The EMBO Journal, 2012, 31, 1630-1643; Lopez-Otin and Overall, 2002, Nat. Rev. Mol. Cell Biol., 2:509-519; Overall and Blobel, 2007, Nat. Rev. Mol. Cell Biol., 8: 245-257; and Lopez-Otin and Bond, 2008, J. Biol.

Chem., 283:30422-30437, all of which are herein incorporated in their entireties by references.

The disclosed exogenous extracellular sensors may be utilized to modulate expression of endogenous or exogenous genes in the presence of an extracellular ligand for the sensors. For example, the functional domain of the exogenous extracellular sensors may be targeted to a promoter region of an endogenous or exogenous gene to modulate expression of the gene. In some embodiments, the functional domain contains a transcription modulator that induces expression from a targeted promoter. In other embodiments, the functional domain contains a transcription modulator that inhibits expression from a targeted promoter. In further embodiments, the functional domain contains a subdomain that modulates the chromatin state of a targeted promoter.

The functional domain of the exogenous extracellular sensors may be targeted to an endogenous or exogenous gene via an RNA-binding subdomain and a targeting RNA. Suitable RNA-binding subdomains may include, but are not limited to, RNA-binding subdomains of bacterial proteins, such as Cas9 proteins, and suitable RNAs may include, but are not limited to, sgRNAs. Suitable RNA-binding subdomains of Cas9 proteins may include RNA-binding subdomains of Cas9 protein of *Streptococcus pyogenes* (SEQ ID NO:1) *Neisseria meningitidis* (SEQ ID NO:2), *Streptococcus thermophilus* (SEQ ID NO:3), *Treponema denticola* (SEQ ID NO:4), *Staphylococcus aureus* (SEQ ID NO: 5), *Brevibacillus laterosporus* (SEQ ID NO:6), *Campylobacter jejuni* (SEQ ID NO:7), *Listeria monocytogenes* (SEQ ID NO:8), *Lactobacillus salivarius* (SEQ ID NO:9), *Legionella pneumophila* (SEQ ID NO:10), *Francisella novicida* (SEQ ID NO:11) and the like. A suitable sgRNA may be selected and/or designed based on the RNA-binding subdomain of the functional domain and a selected target gene. For a sgRNA may be designed to bind to the RNA-binding subdomain of the functional domain and target the functional domain to a target gene, which may be an endogenous or exogenous target gene.

ILLUSTRATIVE EMBODIMENTS

The following embodiments are illustrative and do not limit the scope of the claimed subject matter.

Embodiment 1. An exogenous extracellular sensor and/or a nucleic acid sequence encoding the exogenous extracellular sensor, wherein the exogenous extracellular sensor comprises: a) a ligand binding domain; b) a transmembrane domain; c) a protease cleavage site; and d) a functional domain comprising an RNA-binding subdomain fused to a transcription regulatory subdomain (e.g., a regulatory subdomain that activates transcription from a promoter, a regulatory subdomain that repressed transcription from a promoter, and/or a regulatory subdomain that modulates chromatin state).

Embodiment 2. The exogenous extracellular sensor of embodiment 1, wherein the RNA-binding subdomain comprises a RNA-binding domain of a Cas9 protein.

Embodiment 3. The exogenous extracellular sensor of embodiment 1 or 2, wherein the exogenous extracellular sensor further comprises an extracellular spacer.

Embodiment 4. The exogenous extracellular sensor of any of the foregoing embodiments, wherein the exogenous extracellular sensor further comprises an intracellular spacer that is one, two, three, four, five, or six amino acids in length.

Embodiment 5. The exogenous extracellular sensor of any of the foregoing embodiments, wherein the transmembrane

domain of is immediately adjacent to the protease cleavage site such that there is no intracellular spacer there between.

Embodiment 6. A system comprising the exogenous extracellular sensor of any of embodiments 1-5 as a first exogenous extracellular sensor and further comprising a second exogenous extracellular sensor, wherein the second exogenous extracellular sensor comprises: e) a ligand binding domain, f) a transmembrane domain, and g) a protease domain that cleaves the protease cleavage site of the first exogenous extracellular receptor.

Embodiment 7. The system of embodiment 6 further comprising a cell wherein the first and second exogenous extracellular sensors are present in the cell as cell membrane proteins.

Embodiment 8. The system of embodiment 7, wherein the first and second exogenous sensors are located in the cell membrane such that the first and second ligand binding domains are located outside the cell and the protease cleavage site and the functional domain are located inside the cell.

Embodiment 9. The system of embodiment 8, wherein the first and second ligand binding domains bind the same ligand, and wherein the first and second exogenous sensors are configured such that the protease domain will cleave the protease cleavage site when the first and second ligand binding domains bind the same ligand.

Embodiment 10. The system of any of embodiments 7-9 further comprising a RNA or a vector that expresses the RNA wherein the RNA-binding subdomain of the functional domain binds the RNA.

Embodiment 11. The system of embodiment 10, wherein the RNA is a sgRNA or a vector that expresses the sgRNA wherein the RNA-binding subdomain of the functional domain binds the sgRNA and the functional domain is targeted to an endogenous gene of the cell or an exogenous gene introduced to the cell.

Embodiment 12. The system of embodiment 10 or 11, wherein the RNA-binding domain of the functional domain comprises a Cas9 protein domain.

Embodiment 13. The system of any of embodiment 6-12 further comprising a genetic construct, wherein the genetic construct is configured to express a gene in response to the transcription regulatory domain.

Embodiment 14. The system of any of embodiments 6-13, wherein the protease is at least partially auto-inhibited.

Embodiment 15. A method for modulating expression of an endogenous gene of a cell, the method comprising introducing the system of embodiment 11 to the cell and contacting the first and second ligand binding domains with the ligand for the first and second ligand binding domains wherein the functional domain is targeted to the endogenous gene of the cell and modulates expression of the endogenous gene of the cell in the presence of the ligand for the first and second ligand binding domains.

EXAMPLES

The following Examples are illustrative and are not intended to limit the scope of the claimed subject matter.

Example 1

Reference is made to the invention disclosure entitled "Modular Extracellular Sensor Architecture for Regulating Endogenous Genes," Josh N. Leonard, and Kelly A. Schwarz, submitted Jul. 22, 2015.

Title: Modular Extracellular Sensor Architecture for Regulating Endogenous Genes

TECHNICAL FIELD

The technology of this example comprises a general platform for building mammalian cell-based therapies and devices that interface robustly with host physiology. Specifically, this technology leverages our Modular Extracellular Sensor Architecture (MESA), a system for engineering novel receptor proteins that confer the detection of extracellular cues. (See Example 5). When MESA receptors bind an extracellular analyte, this "sensing" event is translated into the intracellular release of a previously sequestered protein, such as a transcription factor. In this example, we disclose a novel class of MESA receptor in which the sensing event releases a readily programmable protein based upon the CRISPR Cas9 protein. Cas9 is a microbial nuclease that can be programmed to bind to a DNA locus of interest via co-expression with a small guide RNA (sgRNA). Variants of Cas9 have previously been developed such that programmed DNA binding confers gene editing, transcriptional silencing, or transcriptional activation, and other modalities may be possible. This invention thus enables one to engineer a cell such that the detection of essentially any extracellular species is translated into the regulation (in various ways) of any one or more endogenous genes. This novel capability vastly expands our ability to engineer cells as diagnostics, therapeutics, and tools for fundamental biology research.

The applications of the technology of this example may include but are not limited to: engineered cell-based therapeutics: cells that sense molecules in their environment and respond via release of a therapeutic payload (e.g., custom-engineering immune cells that release immune potentiating factors only in the vicinity of tumors, serving as targeted in situ factories); cell-based diagnostics tools and assays enabling the detection and monitoring of specific molecules or analytes; and customized research reagents that investigators may use to functionally "rewire" their system of interest, comprising a type of experimental perturbation not possible via existing drugs and gene knockout approaches.

The advantages of the technology of this example may include but are not limited to: expansion of the capabilities of MESA by enabling novel capabilities including: (1) readily programmable choice of MESA "output" genes via simple DNA sequence analysis (rather than protein engineering), (2) regulation of endogenous genes, and (3) combinatorial regulation of endogenous genes; and Cas9-based gene regulation conditional upon the detection of a specific, extracellular cue.

BRIEF SUMMARY

The technology of this example comprises a general platform for building mammalian cell-based therapies and devices that interface robustly with host physiology. Specifically, this technology leverages our Modular Extracellular Sensor Architecture (MESA), a system for engineering novel receptor proteins that confer the detection of extracellular cues. When MESA receptors bind an extracellular analyte, this "sensing" event is translated into the intracellular release of a previously sequestered protein, such as a transcription factor. MESA receptors comprise of a pair of synthetic protein receptor chains that dimerize in the presence of a ligand, enabling the protease on the protease chain (PC) to cut a cognate cleavage sequence (CS) on the target

chain (TC), thus releasing a tethered transcription factor. Previously described MESA receptors release engineered transcription factors such as the tet transactivator (tTA), which binds one specific DNA sequence. Thus, MESA receptors have been used thus far only to regulate engineered promoters, and engineering novel transcriptional “outputs” required engineering novel transcription factor proteins with DNA binding specificity customized to the novel target sequence. This invention comprises engineering the TC of a MESA receptor to include a tethered version of a CRISPR Cas9-based protein. Natural Cas9 is a nuclease that hones to specific DNA sequences via interaction with a small guide RNA (sgRNA) that is complementary to the target DNA sequence of interest. Variants of Cas9 have been developed such that programmed DNA binding confers gene editing, transcriptional silencing, or transcriptional activation, and other modalities may be possible. Because sgRNAs are easy to design using extant bioinformatic tools, this invention enables one to engineer novel MESA receptors (those that regulate one or more endogenous genes) in an expedient fashion that circumvents the need for slow, costly, difficult, protein engineering.

Technical Description

We have developed and validated a MESA receptor that releases a Cas9-based transcription activator from the target chain upon ligand binding. (See FIG. 1). This catalytically inactive Cas9 (dCas9) was fused to VP64 (dCas9-VP64) to create a transcriptional activator, although in principle any such Cas9-based protein could be similarly incorporated into MESA. For example, fusion of dCas9 to a transcriptional repressive protein, such as KRAB, has been reported to generate transcriptional repressors; a MESA receptor including such a protein would be expected to confer gene repression in a manner that is conditional upon sensing of extracellular cues. Our initial demonstrations have utilized the dCas9-VP64 approach to confer conditional transcriptional activation of target genes.

In a proof-of-concept experiment, we generated sgRNAs targeting the promoter of a reporter construct (a DNA sequence in which transcriptional activation leads to expression of a fluorescent protein), and we showed that a MESA-dCas9-VP64 receptor conferred ligand-inducible expression of the reporter gene. (See FIG. 2.) We then demonstrated that it was also possible to activate endogenous genes via MESA. By expressing MESA-dCas9-VP64 along with sgRNAs targeting the endogenous human IL-2 promoter, we observed that upon addition of ligand, significant increases in IL-2 gene expression were induced. (See FIG. 3). Furthermore, we demonstrated that novel Cas9 technologies can be readily incorporated into Cas9-based MESA; it was recently reported that dCas9-based transcriptional activators may be potentiated by co-expression of an engineered protein that recruits multiple transcriptional activation domains (P65 and HSF1) to the dCas9-VP64-sgRNA complex. We observed that this strategy also enhanced MESA-dCas9-VP64-mediated transcriptional activation, emphasizing the power of this invention’s modular.

The technology of this example enables one to rapidly engineer novel receptor proteins that regulate one or more endogenous genes in a readily defined fashion. Such a technology meets an unmet need in the general field of cell-based therapies. For example, engineering local, sustained production of immune modulating factors is a widely-acknowledged need in the field of cancer immunotherapy; such a capability could complement and vastly expand the

range of cancers amenable to treatment using the rapidly-expanding commercial field of engineered cell-based cancer therapies. Ultimately this invention could improve both the safety and efficacy of such immunotherapy approaches. An additional commercial opportunity comprises the provision of customized reagents for fundamental biological research. Much like genetic knockout mice have proven to be an invaluable tool for studying organismal biology and using such model animals for translational research, this invention would enable researchers to “rewire” organismal biology in a manner not currently possible. Such experimental perturbations are therefore uniquely valuable, and much like experimental animal lines, could be commercialized as customized products for basic research (in various potential commercial models, such as design and provision of customized reagents and services).

The technology of this example could enable a range of new therapeutic capabilities including therapeutic applications such as cancer immunotherapy. The technology of this example also relates to a novel class of experimental reagent with wide-ranging potential utility.

Example 2

Reference is made to the Abstract entitled “Rewiring Cellular Input-Output to Engineer Cell-Based Therapies That Interface Robustly with Human Physiology,” Rachel M. Dudek, Nichole M. Daringer, and Joshua N. Leonard,” presented at the 2015 AIChE Annual Meeting, Session: Emerging Frontiers in Systems and Synthetic Biology, Nov. 8-13, 2015, which is incorporated by reference herein in its entirety.

Title: Rewiring Cellular Input-Output to Engineer Cell-Based Therapies that Interface Robustly with Human Physiology

Cell-based therapies have proven to be useful for treating a wide variety of diseases including autoimmune disease, infectious diseases, and cancer. A particularly promising frontier is the use of engineered cell therapies, wherein cells are programmed to carry out custom functions such as harnessing the immune system to find and destroy cancerous cells. Such therapies have now achieved robust clinical successes for some applications. However, controlling or modulating the activity of these therapies post-implantation remains both attractive, to enhance both safety and therapeutic efficacy, and challenging, since the tools necessary to engineer custom functional programs are limited. In particular, engineering cells to sense and recognize specific combinations of environmental cues would be desirable, but no such technology has yet been reported. Toward this goal, our lab has developed a platform for engineering novel protein biosensors, termed modular extracellular sensor architecture (MESA), for detecting exclusively extracellular cues. Upon binding extracellular ligand, MESA receptors release sequestered transcription factors from the cytoplasmic face of the plasma membrane, freeing the factor to regulate expression of an “output” gene or genes within the cell. Here we present three areas in which we have recently expanded upon the capabilities of the core MESA technology: (1) achieving novel ligand recognition by integrating modular ligand-binding domains, (2) integrating MESA with intracellular gene circuits that enable the cell to “process” sensory information by logical evaluation, and (3) rewiring cellular input-output by coupling extracellular sensing to intracellular modulation of endogenous gene expression.

First, as modular ligand binding domains, we used camelid single domain antibody fragments termed “nanobod-

ies". Each nanobody's small size enables multiple nanobodies to recognize distinct, non-overlapping epitopes on a ligand of interest. Thus nano body MESA are able to recognize monomeric or asymmetric ligand inputs. After demonstrating the feasibility of nano body MESA, we integrated multiple such receptors into a logical processing circuit that enables the multiplexed logical evaluation of distinct protein ligand "inputs". Next, we demonstrated that human antibody-derived scFv could also be used as ligand-binding domains in MESA, and VEGF-MESA were thus designed to sense the tumor-associated cytokine, VEGF. In order to regulate endogenous genes, the MESA transcription factor domain was replaced with the dCas9 protein, which is able to act as a 'pseudo-transcription factor' by tethering transactivation domains to the dCas9 protein and providing small guide RNAs in trans to direct the Cas9 protein to a specific genomic locus. As proof of principle, we developed VEGF-MESA that induce expression of IL-2, a cytokine important for T-cell growth and proliferation. In principle, the dCas9 protein can be guided to any genomic locus of interest by simply delivering a different small guide RNA. Cells expressing VEGF-MESA exhibited minimal background signaling in the absence of ligand, and such cells showed significant increases in IL-2 gene expression upon exposure to ligand. Thus, this initial study demonstrates that MESA may be harnessed to effectively rewire cellular input-output behavior. This work lays the foundation for leveraging MESA receptors to boost the safety and efficacy of cancer immunotherapies as well as cell-based therapies for a range of clinical applications.

Example 3

Reference is made to the Abstract entitled "Modular Receptor Engineering for Programming Cell-Based Therapies to Interface with Host Physiology," Nichole M. Daringer, Rachel M. Dudek, and Joshua N. Leonard," presented at the 5th ICBE—International Conference on Biomolecular Engineering, Jan. 11-14, 2015, which is incorporated by reference herein in its entirety.

Title: Modular Receptor Engineering for Programming Cell-Based Therapies to Interface with Host Physiology

Engineered cell-based therapies represent an emerging frontier in medicine, and the promise of this approach has recently been demonstrated through curative eradication of B cell malignancy from a growing list of patients. However, extending this success to other types of cancer and to other applications in health and medicine will require new tools enabling bioengineers to custom program the manner in which engineered cells interface with host physiology. Until recently, we lacked the ability to construct customizable cell-based "devices" that detect and respond to exclusively extracellular cues, which include many species of biological relevance including cytokines, chemokines, cell-surface antigens, and pathogens. To meet this need, we developed a Modular Extracellular Sensing Architecture. This protein engineering platform comprises a self-contained receptor and signal transduction system, wherein ligand binding at the cell surface is transduced into a change in intracellular state (i.e., induction of gene expression) without requiring involvement of native intracellular signaling mediators, and therefore without being subject to native mechanisms regulating signal transduction. Here, we report our expansion upon this platform to enable the engineering of receptors that recognize essentially any external ligand and regulate essentially any native gene. We have developed general and rapidly employable engineering approaches for incorporat-

ing modular ligand binding domains (including both single-chain variable fragments, scFv, and single domain Nanobodies) to create new functional receptors.

We have also engineered MESA receptors to generate "output" via Cas9-based gene regulation systems. Cas9-based transcriptional activators and repressors are easily targeted to regulate either native genes or engineered transgenes by co-expression with single-guide RNA (sgRNA) tailored to target a specific gene or set of genes. Finally, we have demonstrated that these approaches may be combined to functionally rewire mammalian cell input-output behavior. Ultimately, this capability provides a powerful tool for experimental systems biology, and in the context of medicine, enables the rapid evaluation of multiple potential strategies for functionally rewiring multicellular networks to achieve a therapeutic objective. We will present our application of this approach to modulating immune function, toward the goal of therapeutically modulating local immune states to overcome persistent barriers to immunotherapy of cancer.

Example 4

Reference is made to the Abstract and Poster entitled "Rewiring cellular input-output using modular extracellular sensors," Kelly A. Schwarz, Nichole M. Daringer, Taylor B. Dolberg, and Joshua N. Leonard, presented at the 2016 Synthetic Biology: Engineering, Evolution & Design (SEED) Meeting, Jul. 18-21, 2016, Chicago, IL, which is incorporated by reference herein in its entirety.

Title: Rewiring Cellular Input-Output to Engineer Cell-Based Therapies that Interface Robustly with Human Physiology

Cell-based therapies have proven to be useful diagnostic and therapeutic tools for treating a wide variety of diseases such as arthritis, infectious diseases, and cancer. Treating cancer with genetically engineered T cells is particularly promising, and for patients with certain types of "liquid" cancers, this approach has already delivered astounding clinical benefits. However, current clinical experience indicates that the extending these benefits to other types of cancer, as well as to other diseases, will require new therapeutic capabilities. In particular, there yet exists a need for technologies enabling one to program cells to sense and respond to their environment in a defined fashion. Toward this goal, our lab has developed a platform for engineered novel protein biosensors and extracellular cues, which we have termed modular extracellular sensor architecture (MESA). MESA comprises a self-contained receptor-signal transduction system in which ligand binding induces receptor dimerization, which then releases a sequestered transcription factor to regulate expression of an "output" gene or genes.

Here, we presented two expansions of the basic MESA capabilities, which make key strides towards meeting the needs identified above: 1) integrating ligand-specific MESA receptors with intracellular gene circuits to enable the cell to "process" multiparametric environmental cues, and 2) rewiring cellular input-output by coupling the sensing of biologically relevant extracellular signals to modulation of endogenous gene expression. To engineer biosensing of novel input ligands, we demonstrated that both camelid antibody analogs termed "nanobodies" and scFv antibody fragments could be incorporated as novel MESA ligand-binding domains. To achieve MESA-regulated induction of endogenous target gene expression, we integrated dCas9-based transcription factors as receptor "outputs". Altogether, these

key proof-of-principle studies identified readily generalizable strategies for rewiring cellular input-output behavior in a use-defined fashion. This work lays the foundation for leveraging MESA receptors for a range of applications in both fundamental research as well as advanced cell-based therapies.

Example 5

Reference is made to Schwarz et al., “Rewiring Human Cellular Input-Output Using Modular Extracellular Sensors,” *Nat. Chem. Biol.* 2017 February; 13(2):202-209, electronically published on Dec. 12, 2016, the content of which is incorporated by reference herein in its entirety. Title: Rewiring Human Cellular Input-Output Using Modular Extracellular Sensors

Abstract

Engineered cell-based therapies comprise a promising emerging strategy for treating diverse diseases. Realizing this promise requires new tools for engineering cells to sense and respond to soluble extracellular factors, which provide information about both physiological state and the local environment. Here, we report such a biosensor engineering strategy, leveraging a self-contained receptor-signal transduction system termed Modular Extracellular Sensor Architecture (MESA). We developed MESA receptors enabling cells to sense vascular endothelial growth factor (VEGF), and in response, secrete interleukin 2 (IL-2). By implementing these receptors in human T cells, we created a customized function not observed in nature—an immune cell that responds to a normally immunosuppressive cue (VEGF) by producing an immunostimulatory factor (IL-2). Because this platform utilizes modular, engineerable domains for ligand binding (antibodies) and output (programmable transcription factors based upon Cas9), this approach may be readily extended to novel inputs and outputs. This generalizable approach for rewiring cellular functions could enable both translational applications and fundamental biological research.

Introduction

Mammalian cell-based therapies comprise a promising and rapidly expanding technology, with over 1300 clinical trials currently underway worldwide^{1,2}. Most notably, engineered chimeric antigen receptor (CAR)-expressing T cells harness the capabilities of these immune cells to detect and destroy cancer cells, which has proven transformative for the treatment of B-cell malignancies³⁻⁵. While extending these successes to other types of cancers poses a range of challenges, many refinements to the basic CAR approach have now been achieved⁶. However, there exist many applications for which an engineered T cell receptor cannot be utilized to achieve the desired functionality, and thus additional sensing modalities are required⁷.

Technologies for engineering customized cellular functions that respond to extracellular ligands—those which are not transported into the cytoplasm—enable cells to respond in defined ways to both the local environment and overall physiological state. Such approaches generally comprise engineering native receptors to confer responsiveness to new ligands⁸⁻¹⁰ or coopting native receptor/signal transduction systems to drive induction of engineered signaling cascades¹¹ or gene circuits¹². While the use of native pathways is facile and leverages the performance of components tuned

through evolution, these approaches are also subject to regulation and cross-talk with other native cellular functions.

By contrast, orthogonal receptor systems may couple detection of extracellular ligands to engineered functions in a more programmable fashion. For example, synNotch receptors enable orthogonal detection of surface-bound extracellular ligands^{13,14}. Each synNotch receptor comprises a core Notch domain flanked by modular extracellular and intracellular domains. Binding of a surface-bound ligand triggers protease-mediated cleavage and release of an exogenous transcription factor via a mechanism that requires ligand binding-mediated mechanical forces, and thus synNotch receptors do not respond to soluble ligands. We recently developed a distinct self-contained receptor-signal transduction system, the Modular Extracellular Sensor Architecture (MESA)¹⁵ (FIG. 1a), which is also orthogonal to native cellular pathways but is designed to enable the detection of soluble ligands. Each MESA receptor comprises two transmembrane chains—the target chain (TC) and the protease chain (PC). In this system, ligand binding-induced dimerization of receptor extracellular domains promotes intracellular trans-cleavage of the target chain by the protease chain, which releases a sequestered engineered transcription factor into the cytoplasm. While this previous work demonstrated the feasibility of the MESA mechanism, ligand recognition was limited to detection of a model small molecule analyte.

In this study, we investigated whether we could leverage the MESA design to generate a general platform for rewiring cellular functions in response to physiologically relevant cues. We evaluated strategies for incorporating both highly modular ligand-binding domains as MESA inputs and readily programmable transcription regulators as MESA outputs. By exploiting the modularity of MESA design, this systematic investigation both elucidated the key factors impacting the performance of these novel receptors and ultimately validated a powerful, generalizable new technology for rewiring cellular input-output functions for therapeutic applications and as a tool for fundamental research.

Results

VEGF-MESA Receptor Development. In this study, we systematically developed a generalizable strategy for engineering MESA receptors that sense physiologically relevant cues. To this end, we first investigated whether we could generate ligand recognition domains by coopting the small chain variable fragment (scFv) derived from monoclonal antibodies, which confer high affinity ligand binding and are very stable¹⁶. scFvs have been developed against a wide range of antigens, providing a large pool of potential “parts” for receptor engineering, and such domains have been utilized successfully in other engineered receptors such as CAR T cells^{3,5} and synNotch receptors^{13,14}. As a target analyte, we selected vascular endothelial growth factor (VEGF). VEGF promotes angiogenesis and is present at high levels in the tumor microenvironment in many cancers¹⁷. Monoclonal antibodies targeting VEGF have been developed as therapeutics^{18,19}, and several are well-characterized structurally and biophysically^{20,21}. Finally, VEGF is an attractive initial target for developing MESA receptors because this ligand is a homodimer (composed of two identical protein domains), such that any scFv that binds to VEGF could conceivably be engineered into MESA chains to confer ligand binding-induced receptor dimerization.

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To initially develop MESA receptors responsive to VEGF (VEGF-MESA), we utilized a general strategy to assess receptor design and functionality. We initially evaluated two different anti-VEGF scFv clones, which we termed G6 and B20, after the names of the monoclonal antibodies from which we derived these scFvs²² (see Methods). These antibodies bind VEGF with different affinities and orientations, although each clone binds to the same face of VEGF^{20,21} (FIG. 1*a*). We fused each scFv clone to the extracellular portions of a MESA receptor scaffold (these constructs are hereafter termed V1-MESA and V2-MESA, respectively), including a range of flexible linkers (Gly2Ser)_x in between the scFv and transmembrane domain.

The VEGF scFV amino acid sequence of V1-MESA is as follows:

```
Signal Sequence
(MAWTSLILSLALCSGASS (SEQ ID NO: 12));

VL
(DIQMTQSPSSLSASVGRVTTITCRASQDVSTAVAWYQQKPGKA
PKLLIYSASFLYSGVPSRFSGSGSGTDFLTITSSLPEDFATYYCQQGYG
NPFTFGQGTKVEIKR (SEQ ID NO: 13));

GS Linker
(GGGGSGGGSGGGGS (SEQ ID NO:14));
and
VH
(EVQLVESGGGLVQPGGSLRLSCAASGFTISDYWIHWVRQAPGK
GLEWVAGITPAGGYTYADSVKGRFTISADTSKNTAYLQMNSLRADTAV
YYCARFVFFLPYAMD YWGQGLTV (SEQ ID NO:15)).
```

The VEGF scFV amino acid sequence of V2-MESA is as follows:

```
Signal Sequence
(MAWTSLILSLALCSGASS (SEQ ID NO: 1);

VL
(MAWTSLILSLALCSGASSDIQMTQSPSSLSASVGRVTTITCRAS
QVIRRLAWYQQKPGKAPKLLIYAASNLASGVPSRFSGSGSGTDFLTIS
SLQPEDFATYYCQQSNTSPLTFTGQGTKVEIKR (SEQ ID NO: 16));

GS Linker
(GGGGSGGGSGGGGS (SEQ ID NO: 14));
and
VH
(EVQLVESGGGLVQPGGSLRLSCAASGFSINGSWIFWVRQAPGK
GLEWVAGIWPFGGYTHYADSVKGRFTISADTSKNTAYLQMNSLRADTAV
YYCARWGHSTSPWAMDYWGQGLTV (SEQ ID NO: 17)).
```

The intracellular domains comprised a Tobacco Etch Virus (TEV) protease on the PC, and the cognate cleavage sequence followed by an engineered transcription factor (tTA) on the TC, as in the initial MESA design¹⁵. We carried this receptor library forward for evaluation.

We first investigated whether VEGF-MESA receptors bound VEGF at the cell surface. When expressed in HEK 293FT cells, both V1-MESA and V2-MESA were expressed on the cell surface so long as the extracellular linker exceeded a minimal length of approximately 26 amino acids (FIG. 1*b*), FIG. 4*a*). In our initial constructs, the TCs were expressed at the cell surface to a substantially greater extent

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than were the PCs. Given that these two chains are identical on the extracellular face and transmembrane regions, we hypothesized that the inner membrane-proximal sequence on the PC may destabilize these proteins. To test this theory, we inserted the two membrane-proximal amino acids derived from the TC (which exhibited robust surface expression) into the same location on the PC, and as predicted, this modification conferred robust surface expression of the PC (FIG. 5). Therefore, we used such a linker for all PC constructs. All V1-MESA and V2-MESA expressed at the cell surface were capable of binding recombinant VEGF, and binding did not vary substantially with extracellular linker length beyond the minimum required to achieve cell surface expression of the MESA chains (FIG. 1*c*, FIG. 4). Altogether, we discovered a subset of VEGF-MESA that was expressed on the cell surface and was capable of binding extracellular VEGF, and we carried this library forward for functional evaluation.

We next tested whether exposure to VEGF induced VEGF-MESA signaling. For V1-MESA, signaling in the absence of ligand increased as extracellular linker length increased from 16 to 30 amino acids, and lengthening linkers beyond this point led to no further increases in background signaling (FIG. 1*d*, left). Upon addition of VEGF, V1-MESA did not exhibit any ligand-inducible signaling at any linker length. The background signaling of V2-MESA was generally constant across all linker lengths evaluated and was consistently lower than the background signaling level conferred by V1-MESA (FIG. 1*d*, right). Most notably, for several V2-MESA configurations, we observed significant ligand-inducible signaling, with up to a 2-fold increase in reporter output upon exposure to VEGF. This observation established the fundamental feasibility of engineering MESA receptors that sense exclusively extracellular cues.

We next evaluated how modifying the inter-chain cleavage kinetics might modulate VEGF-MESA signaling. Because we developed the initial intracellular VEGF-MESA architecture in order to limit background signaling in the absence of ligand¹⁵, we hypothesized that altering the protease cleavage kinetics, by changing the final residue of the TEV protease cleavage sequence from methionine (M) to glycine (G) (increases k_{cat} by ~50%²³), could enhance signaling. VEGF-MESA with the G cleavage sequence were indeed expressed on the surface (FIG. 6*a*) and bound recombinant VEGF (FIG. 6*b*), but neither V1 or V2-MESA bearing this cleavage sequence modification exhibited ligand-inducible signaling (FIG. 6*c*). We inferred that for these G cleavage sequence MESA, transient diffusive encounters between the TC and PC resulted in cleavage and background signaling, thereby removing the dependence on the VEGF ligand to induce signaling. These results suggest a model wherein once one introduces a new ligand-binding modality, tuning MESA cleavage kinetics is neither necessary nor especially fruitful.

VEGF-MESA Receptor Implementation. Having developed a functional VEGF-MESA architecture, we next turned from protein engineering to investigate how to best implement such a receptor in order to achieve desirable performance characteristics. These could include low background signaling in the absence of ligand and large fold-induction upon ligand addition, and we hypothesized that there may exist tradeoffs between these characteristics which depend on the manner in which the receptor is implemented. To frame this investigation, we posited that during the course of receptor expression, trafficking, and ligand-binding, MESA receptors may enter a number of distinct complexes (FIG. 2*a*). Induction of signaling requires the formation of pro-

ductive dimers, where one TC and one PC dimerize at the cell surface to mediate trans-cleavage. However, since the ectodomains of both chains are identical, VEGF-binding may also induce the formation of homotypic dimers, in which two TCs or two PCs dimerize, which would not comprise functional signaling complexes. Moreover, it is also possible for TCs and PCs to encounter one another in the ER, secretory pathway, or cell surface, all of which would contribute to background signaling and generate “dead” TCs that can subsequently act as competitive inhibitors of VEGF-induced MESA signaling. We hypothesized that each of these processes would render MESA performance dependent upon the rate(s) and ratio with which each receptor chain is expressed.

To investigate these questions, we systematically varied the expression of each MESA chain in a combinatorial fashion (FIG. 2*b*, FIG. 7(*a-d*)). Signaling in the absence of ligand was quite low for all VEGF-MESA receptors, although background signaling diminished somewhat for low amounts of MESA plasmids. Most notably, for both V1-MESA and V2-MESA, we observed regimes that resulted in ligand-inducible signaling. Generally, we observed the greatest ligand-inducible signaling when TC:PC expression ratios were high, and within the signaling-competent regime, increasing TC expression levels enhanced both background and ligand-inducible signaling. Using these high performing receptor expression conditions, we next investigated how VEGF-MESA signaling varied with VEGF dose (FIG. 2*c*). Interestingly, the dose response curves for V1-MESA and V2-MESA were somewhat distinct. V1-MESA exhibited a substantial response to VEGF at doses as low as 25 ng/mL (FIG. 2*c*, left), whereas V2-MESA required a higher VEGF dose (150 ng/mL) to exhibit a substantial response (FIG. 2*c*, right). Moreover, V2-MESA exhibited a large dynamic range of differential responses to VEGF dose, whereas V1-MESA signaling reached a maximum at a relatively low concentration of VEGF. Possible contributors to such differences include ligand-binding affinity (V1-MESA is predicted to bind VEGF tighter than does V2-MESA, FIG. 1*a*) and differences between the geometric orientations in which ligand binding occurs (FIG. 1*a*). For the sake of comparability, we opted to use a VEGF dose of 250 ng/mL for future analyses, the lowest concentration that resulted in maximal reporter induction for both V1- and V2-MESA. Altogether, V1-MESA and V2-MESA responded to VEGF with different sensitivities and dynamic ranges, suggesting that these properties may be tuned based upon the desired application.

Since tuning MESA expression levels by varying plasmid dose may not be feasible for some applications, we also investigated a method for tuning expression by varying translation rates. To this end, we introduced synthetic elements into the 5' UTR of MESA expression constructs, termed synthetic upstream open reading frames (uORF), to vary both the rate and frequency of translation initiation²⁴. We screened a small library of these elements and observed that different uORF sequences did confer differential levels of MESA surface expression (FIG. 8(*a,b*)). We used a uORF leading to high gene expression for the TC, and we co-transfected this plasmid at a 1:1 ratio with PC plasmids encoding a variety of uORFs. Interestingly, background signaling in the absence of ligand varied with uORF choice following the same trend with which the level of MESA surface expression varied with uORF choice (FIG. 8(*c*)). Moreover, multiple combinations of MESA receptors containing a variety of uORFs exhibited significant VEGF-mediated reporter induction. This result confirms that tuning

the relative surface expression levels of the TC and PC can confer VEGF-inducible MESA signaling and that this tuning may be implemented via multiple strategies to achieve similar MESA performance.

Cellular Rewiring via VEGF-MESAs. We next investigated whether we could modify the MESA output to regulate endogenous gene expression. To this end, we replaced tTA with a readily reprogrammable transcription factor based upon CRISPR/Cas9²⁵⁻²⁷. In this system, catalytically inactive Cas9 (dCas9) is genetically fused to the VP64²⁸ transcriptional activation domain (dCas9-TF), and expression of dCas9-TF with a small guide RNA (sgRNA) complementary to a genomic target leads to transcriptional activation of the target gene (FIG. 3*a*). As a proof-of-principle, we chose to target dCas9-TF to the human interleukin 2 (IL-2) gene, for two main reasons. First, IL-2 performs diverse functions *in vivo* including preserving peripheral immune tolerance by stimulating regulatory T cells, although high doses of IL-2 promote immune activation via stimulation of cytotoxic T cell proliferation²⁹. Thus, achieving VEGF-induced expression of IL-2 is a non-natural cellular function that could be of use for immunotherapy. Second, transcription factors based upon TALE domains can reactivate the normally silenced IL-2 locus in HEK 293FT cells³⁰, such that this output comprises a suitable test of the proposed functional rewiring.

Since no previous studies had systematically evaluated sgRNAs for achieving dCas9-TF-mediated transcriptional activation of IL-2 in human cells, we first developed a library of sgRNAs targeted to a region spanning upstream and downstream of the IL-2 transcription start site (positions -800 to +250, see FIG. 9) using three different *in silico* sgRNA design tools³¹⁻³³. We co-expressed dCas9-TF with 15 different sgRNAs, either individually or in combination, in HEK 293FT cells, and we evaluated the induction of IL-2 mRNA expression by qPCR (FIG. 3*a*). Although many sgRNAs conferred minimal impacts on transcription, sgRNA 7 and 17 each individually resulted in significant induction of IL-2 transcription, as did several combinations of sgRNAs (5-10 fold). We carried the most promising sgRNAs (7, 17, 4+9+10+11) forward for further investigation.

We next investigated whether recently reported methods for enhancing dCas9-TF-mediated transcription could be harnessed to enhance transcription from the endogenous IL-2 gene. To do so, we used the SAM system³⁴, in which the sgRNAs were modified to contain three binding loops derived from the bacteriophage MS2 coat-protein, and two transcriptional activators (p65 and HSF1) were fused to the MS2 protein (FIG. 3*b*). Targeting IL-2 via SAM+dCas9-TF substantially enhanced transcription (10-15 fold) versus dCas9-TF alone (FIG. 3*b*), and thus we next evaluated SAM in the context of MESA.

Combining the functional components identified above, we next evaluated whether VEGF-MESA output could be redirected to induce transcription of endogenous IL-2 (FIG. 3*c*). First, we replaced the tTA domain of the TC with dCas9-TF (V1- or V2-MESA-dCas9-TF), and we confirmed that these constructs were expressed at the cell surface (FIG. 10). To evaluate the functional performance of these receptors, we transfected V1-MESA-dCas9-TF or V2-MESA-dCas9-TF along with sgRNA 7 and SAM components into HEK 293FT cells, exposed cells to VEGF, and quantified IL-2 secretion (FIG. 3(*e*), FIG. 11). Monitoring IL-2 protein expression provides a robust, time-integrated metric of MESA output by circumventing potential artifacts associated with mRNA expression dynamics. We observed sig-

nificant VEGF-induced IL-2 secretion for cells expressing either V1- or V2-MESA-dCas9-TF, with the V1 receptor conferring a somewhat higher overall response (~20 pg/mL IL-2 protein) and a better fold induction (5.2 fold) compared to the V2 receptor. In the presence of VEGF, both V1- and V2-MESA drove secretion of IL-2 at levels similar to that achieved by transient transfection of soluble dCas9-TF along with sgRNA 7 (FIG. 3d). Although sgRNA 17 drove high levels of IL-2 mRNA expression when co-expressed with the SAM components, transfecting sgRNA 17 in combination with V1- or V2-MESA-dCas9-TF did not result in any VEGF-induced IL-2 secretion (FIG. 12(a)). We do not yet have an explanation for this result, which was repeatable across multiple experiments. Further increasing the TC:PC ratio did not result in inducible signaling for either receptor (FIG. 12(b)). Thus, the VEGF-MESA platform successfully conferred a novel, programmed functionality in HEK 293FT cells.

Finally, we investigated whether our MESA reprogramming strategy could be extended to rewire a distinct, physiologically relevant cell type. To this end, we chose to evaluate the Jurkat human T cell line, which is capable of secreting IL-2 but does not do so in response to VEGF¹⁵. V1- and V2-MESA-dCas9-TF were expressed on the surface of Jurkat cells (FIG. 10). We then transfected Jurkats with MESA chains, at the same TC:PC ratio used in HEK 293FT cells. We observed substantial VEGF-inducible secretion of IL-2 for both V1 and V2 receptors, and we did not observe IL-2 secretion in the absence of the MESA components (FIG. 3(g), FIG. 11). Furthermore, we confirmed that in the presence of VEGF, IL-2 accumulated only in transfected cells (FIG. 13). As we observed in HEK 293FT cells, the V1 receptor again outperformed the V2 receptor in terms of both level of IL-2 secretion and fold-induction, and when induced, each receptor drove secretion of IL-2 at levels comparable to that conferred by co-transfection of soluble dCas9-TF with sgRNA 7 (FIG. 3(f)). Notably, this observation represents the first functional rewiring of human immune cells to secrete an immunopotentiating molecule in response to an immunosuppressive cue, which is a response not observed in nature. Moreover, although we observed a lower IL-2 production level for Jurkat cells than for HEK 293FT cells, transfection efficiency of Jurkat cells was also substantially lower (~5% for Jurkat cells vs. 80% for HEK 293FT cells), and only the transfected Jurkats exhibited IL-2 accumulation (FIG. 13). Taking this factor into account, we estimated the VEGF-induced production rate of IL-2 to be 5 pg IL-2/10⁵ transfected cells/day for HEK 293FT cells and 15 pg IL-2/10⁵ transfected cells/day for Jurkat cells, such that on a per cell basis, functional rewiring may perform similarly in these two cell types. Altogether, these data demonstrate that the MESA platform comprises a generalizable synthetic biology technology for engineering cells to manifest novel, programmable, input-output functions that could be of use for both translational applications and fundamental research.

Discussion

In this study, we developed a novel strategy for constructing customized cellular functions by functionally rewiring cellular input-output. Here, we consider key lessons learned in this investigation of MESA receptor design and implementation and suggest strategies for utilizing this technology in future applications.

A central conclusion is that MESA's modular design and mechanism enabled the rational development of receptors

recognizing novel inputs (soluble extracellular ligands). First, consider the changes required to convert the original model MESA receptors, which signal in response to rapamycin-induced dimerization of the ectodomains³⁶, into MESA receptors that recognize physiologically-relevant ligands. Achieving robust cell surface expression of MESA receptors utilizing scFvs as ligand-binding domains required two key modifications: (i) inserting additional amino acids into the intracellular membrane-proximal portion of the PC and (ii) extending the nonstructured extracellular linker between the transmembrane and scFv domains of each receptor chain. We hypothesize that both of these requirements may be interpreted as providing the geometric flexibility required to enable folding of all MESA domains. Notably, extending the extracellular linker domains beyond the minimum length required to achieve surface expression did not incur observable deficits in cell surface expression, ligand binding, or ligand-inducible signaling. This is consistent with prior observations made using the model MESA receptors, all of which confirm that, as intended, MESA signaling occurs via a mechanism by which ligand binding-induced dimerization of TC and PC mediates trans-cleavage by boosting the contact frequency between these chains, rather than by constraining these chains in a geometrically defined configuration that "causes" trans-cleavage³⁶. Therefore, we anticipate that MESA receptors incorporating novel scFvs need to include an extracellular linker of "reasonable" length. Notably, the scFvs on the TC and PC must bind to non-overlapping epitopes on a single ligand molecule, whether in a homotypic fashion, as was the case in this study, or in a heterotypic fashion. Moreover, the binding of the TC and PC to the analyte must bring the two chains together in a manner that enables the intracellular domains of these receptor chains to contact one another. The fact that both V1 MESA and V2 MESA exhibited ligand-inducible signaling supports the expectation that multiple geometries may meet this requirement, since the C-termini of the V1 and V2 antibodies (which abut the flexible nonstructured linkers) are separated by substantially different distances when bound to VEGF (85 and 100 angstroms separation, respectively³⁷). For other scFv/ligand complexes, longer extracellular linkers may be required, and for other scFv/linker complexes in which scFvs are sufficiently separated in space when simultaneously bound to ligand, it might not be possible to accommodate both ligand binding and efficient signaling. In any event, the rules identified in this study provide relatively straightforward guidance as to how design-driven modifications may be used to efficiently evaluate novel receptor designs.

The modular nature of MESA design also enabled the straightforward implementation of novel receptor output (release of Cas9-based transcription factors). Since replacing tTA with dCas9-TF resulted in functional receptors with minimal adjustments, this study did not identify specific challenges associated with substituting novel transcription factors (TF) into the intracellular domain of the TC. Interestingly, we observed a higher fold induction for VEGF-MESA based upon dCas9-TF compared to those based upon tTA. One likely explanation is that tTA/TetO-based reporters are sensitive to low amounts of tTA (i.e. small amounts of transcription factor lead to moderate levels of reporter induction), which may be especially true in the context of transfection, in which many copies of the reporter exist in each cell. Conversely, dCas9-TFs are less potent and require high levels of dCas9-TF to drive robust transcription from endogenous genes²⁵. Therefore, we hypothesize that the background level of TC cleavage by the PC may be com-

parable in each receptor system and that differences in the sensitivities of the reporter genes to released transcription factors results in higher fold-induction for the Cas9-based system when compared to the tTA-based system. We cannot exclude the possibility that some TFs may sterically occlude TEV-mediated cleavage of the target chain (suppressing ligand-inducible signaling) or that other TFs may confer less steric hindrance than does either tTA or dCas9-TF (leading to higher ligand-independent background signaling). Overall, our observations suggest that MESA output may be diversified to accommodate at least a reasonably broad range of TF structures.

The manner in which a MESA receptor is implemented (i.e., expressed) also substantially impacts functional performance, in part by impacting the balance between desirable and undesirable receptor complexes (FIG. 2a). Indeed, this statement is also true of natural receptor signaling—overexpression of natural receptors via transient or stable expression of transgenes can also lead to high background (ligand-independent) signaling. Although VEGF-MESA surface expression varied relatively little when changing either plasmid dose or the uORF used, we nonetheless observed substantial changes in MESA signaling both with and without ligand.

Chromosomal integration of expression vectors may better enable tuning levels and distributions of MESA expression. Since lentiviral transgene expression varies greatly with integration site, stable expression of MESA chains from a single genomic landing pad may be a preferable strategy³⁸. Notably, using the homotypic binding mechanism evaluated here, ligand binding may form PC-PC and TC-TC complexes, such that a maximum of 50% of the ligand-induced dimers are signaling competent. Utilizing a heterotypic binding mechanism may circumvent this challenge, although maximizing ligand-inducible signaling would likely still require balancing TC and PC expression. Overall, implementing MESA receptors via the approaches reported here already achieved functional performance (e.g., fold-induction) that meets or exceeds that which has previously been observed with other engineered receptor systems³⁹. Further exploring strategies for tuning expression within specific cellular contexts may improve functional performance without necessitating additional protein engineering.

Ultimately, rewiring cellular input-output may drive both translational applications and fundamental systems biology research. By enabling one to engineer cells that respond to a target extracellular cue via the expression of either transgenes (e.g., tTA-driven) or endogenous genes (e.g., dCas9-TF-driven), MESA could facilitate the rapid implementation and evaluation of diverse therapeutic strategies. Additionally, the MESA platform could provide unique capabilities for fundamental research, particularly in the context of multicellular networks and whole organisms, including the ability to monitor, in a spatially and temporally-resolved fashion, the presence (and perhaps concentration) of VEGF in a living animal. This information would complement that observable using existing reporter strategies, which could identify where the VEGF gene is being expressed but not where the VEGF protein accumulates in the extracellular environment. Furthermore, the MESA platform could complement genetic tools such as knockouts/ins to enable the testing of novel hypotheses pertaining to multicellular network function. Thus, MESA comprises a powerful and generalizable technology in the mammalian synthetic biology toolbox, which expands our ability to build programmable cellular functions for myriad applications.

Methods

DNA construct sources and engineering. Anti-VEGF scFvs and corresponding signal sequences were designed based upon the sequences of monoclonal antibodies raised against VEGF⁴⁰, the sequences of which were generously shared by Germaine Fuh. Inter-chain linkers of 15 non-structured amino acids were utilized to generate scFvs with strong preference for forming monovalent binding domains rather than higher order complexes⁴¹. These genes were synthesized (Invitrogen) and cloned into an expression vector using previously characterized MESA parts¹⁵. The extracellular linker (ECL) library for V1 and V2-MESA was created using PCR, digesting the product with NheI and AgeI, and inserting it back into the original backbone. The development of the tTA-responsive YFP reporter plasmid was described previously¹⁵. Source plasmids for Cas9 components included lenti dCas9-VP64_Blast (Addgene #61425), lenti MS2-P65-HSF1_Hygro (Addgene #61426), lenti sgRNA(MS2)_zeo backbone (Addgene #61427)³⁴, MLM3636 (Addgene #43860), pMLM3705 (Addgene #47754)²⁵. dCas9-VP64 (referred to here as dCas9-TF) was cloned into the MESA TC backbone in place of tTA using PCR, and digesting the product with NotI. sgRNAs were developed using overlapped primers (FIG. 7), which were inserted into either the MLM3636 or lenti sgRNA(MS2) backbone digested with BsmBI.

Cell culture and transfection. HEK 293FT cells (Life Technologies/Thermo) and Jurkat cells (ATTC® TIB-152™) were maintained at 37° C. and 5% CO₂. HEK 293FT cells were cultured in Dulbecco's modified growth medium and Jurkat cells were cultured in RPMI-1640 (Life Technologies), each supplemented with 10% FBS, 1% penicillin-streptomycin, and 2 mM L-glutamine (Life Technologies). For HEK 293FT cells, transfections were performed in 6-well plates seeded at 7.5×10⁵ cells in 2 mL media (immunohistochemistry and RNA experiments) or 24 well plates seeded at 1.5×10⁵ cells in 0.5 mL media. 6-8 h post-seeding, cells were transfected using the CaCl₂-HEPES-buffered saline (HEBS) method with a total DNA content of 1-2 µg DNA per mL of media. All experiments included a plasmid expressing DsRed or BFP as a control to assess transfection efficiency. For functional experiments, 12 h post-transfection, recombinant mouse VEGF-164 (Biolegend, carrier-free) was added to cells upon media change. For Jurkat cells, transfections were performed in 6-well plates seeded at 5×10⁵ cells in 2 mL media. These cells were transfected using Lipofectamine® LTX™ with Plus Reagent, as per manufacturer's instructions. For intracellular staining of IL-2, VEGF was added 24 h post-transfection in combination with 10 µg/mL Brefeldin-A (Life Technologies).

Flow cytometry, reporter assays, and immunolabeling. Adherent cells were suspended in PBS with 2 mM EDTA (PBS-EDTA) and 5% bovine serum albumin (flow buffer). Suspension cells were washed into the same flow buffer. Approximately 1×10⁴ single, live cells from each sample were analyzed using a LSRII flow cytometer (BD Biosciences) running FACSDiva™ software. Data were compensated and analyzed using FlowJo Software (Tree Star). Live, single cells were gated based on scatter and transfected cells were gated based on the presence of a DsRed or BFP transfection control for all experiments. "Reporter activity" was calculated as described in the caption of FIG. 1. For immunolabeling on the cell surface, 1×10⁶ cells were harvested and subsequently labeled with 0.5 µg allophycocyanin(APC)-conjugated anti-Flag (anti-DDDDK) tag antibody (Abcam #ab72569) for 30 minutes at 4° C., and washed 3 times with flow buffer prior to analysis by flow cytometry.

For intracellular staining of IL-2, cells were first fixed and permeabilized with 4% paraformaldehyde (Sigma) for 20 minutes at 4° C., and then cells were resuspended in PBS with 0.2% BSA and 0.5% saponin (Sigma). Immunolabeling with 5 µL of an anti-IL-2 antibody (Thermo RHCIL205) was performed as described for surface staining.

Statistical analysis. For most statistical analyses, two-tailed Student's t-tests were used. This test was chosen to evaluate whether there exists a significant difference between two groups of samples, and the reported comparisons meet the two requirements of this test: (1) the values compared are expected to be derived from a normal distribution and (2) the variance of each group is expected to be comparable to that of the comparison group, since the same transfection methodologies and data collection methods were used for all samples that were compared. A p value less than or equal to 0.05 was considered statistically significant. To specifically evaluate whether two flow cytometry distributions were significantly different (i.e., FIG. 1c), a chi-squared test was applied, with a significance threshold of $p < 0.05$.

VEGF binding assays. VEGF binding assays used the VEGF biotinylated fluorokine kit (R&D Systems) as per the manufacturer's instructions. Briefly, 4×10^6 cells were harvested, incubated with biotinylated recombinant human VEGF and avidin-FITC reagents, washed to remove excess reagents, and finally re-suspended in flow buffer prior to flow cytometry analysis.

Endogenous gene expression assays. RNA was isolated using Trizol® (Life Technologies), and total RNA content was determined using a Nanodrop 2000 Spectrophotometer (Thermo Scientific). 500 ng of RNA was used for cDNA synthesis using the qScript™ cDNA synthesis kit (Quanta Biosciences). 2 µL of cDNA was mixed with 12.5 µL iQ™ SYBR® Green qPCR supermix (BioRad Laboratories), forward and reverse primers at a final concentration of 0.05 µM, and water to a final volume of 25 µL. qPCR reactions were run on a CFX Real-Time PCR Detection System (BioRad Laboratories). All samples were quantified in technical triplicate using IL-2 and GAPDH primers, the C_q values were averaged for each primer set, and the "no reverse transcriptase" control was subtracted from the mean. IL-2 and GAPDH standard curves were used to convert from C_q values to mRNA copies, and IL-2 values were subsequently normalized by the corresponding GAPDH values.

Protein secretion assays. Secreted IL-2 was quantified in supernatant using the Ready-Set-Go! IL-2 ELISA kit (eBiosciences) by following the manufacturer's instructions. Undiluted supernatant was assayed in technical triplicate and a BioTek® Synergy H1 plate reader was utilized to analyze the resulting signal. Each sample was zeroed by subtracting the absorbance at 570 nm from absorbance at 450 nm, and a standard curve was used to convert between zeroed absorbance and protein concentration.

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- In the foregoing description, it will be readily apparent to one skilled in the art that varying substitutions and modifications may be made to the invention disclosed herein without departing from the scope and spirit of the invention. The invention illustratively described herein suitably may be practiced in the absence of any element or elements, limitation or limitations which is not specifically disclosed herein. The terms and expressions which have been employed are used as terms of description and not of limitation, and there is no intention that in the use of such terms and expressions of excluding any equivalents of the features shown and described or portions thereof, but it is recognized that various modifications are possible within the scope of the invention. Thus, it should be understood that although the present invention has been illustrated by specific embodiments and optional features, modification and/or variation of the concepts herein disclosed may be resorted to by those skilled in the art, and that such modifications and variations are considered to be within the scope of this invention.

Citations to a number of patent and non-patent references are made herein. The cited references are incorporated by reference herein in their entireties. In the event that there is an inconsistency between a definition of a term in the specification as compared to a definition of the term in a cited reference, the term should be interpreted based on the definition in the specification.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 16

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<212> TYPE: PRT

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Lys	Val	Leu	Gly 35	Asn	Thr	Asp	Arg 40	His	Ser	Ile	Lys	Lys 45	Asn	Leu	Ile
Gly	Ala	Leu	Leu 50	Phe	Asp	Ser 55	Gly	Glu	Ile	Ala	Glu 60	Ala	Thr	Arg	Leu
Lys 65	Arg	Thr	Ala	Arg	Arg 70	Arg	Tyr	Thr	Arg	Arg 75	Lys	Asn	Arg	Ile	Cys 80
Tyr	Leu	Gln	Glu 85	Ile	Phe	Ser	Asn	Glu	Met 90	Ala	Lys	Val	Asp	Asp 95	Ser
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His	Glu	Arg	His 115	Pro	Ile	Phe	Gly 120	Asn	Ile	Val	Asp	Glu 125	Val	Ala	Tyr
His	Glu	Lys	Tyr 130	Pro	Thr	Ile 135	Tyr	His	Leu	Arg	Lys 140	Lys	Leu	Val	Asp
Ser 145	Thr	Asp	Lys	Ala	Asp 150	Leu	Arg	Leu	Ile	Tyr	Leu	Ala	Leu	Ala	His 160
Met	Ile	Lys	Phe 165	Arg	Gly	His	Phe	Leu	Ile 170	Glu	Gly	Asp	Leu	Asn 175	Pro
Asp	Asn	Ser	Asp 180	Val	Asp	Lys	Leu	Phe	Ile 185	Gln	Leu	Val	Gln	Thr 190	Tyr
Asn	Gln	Leu	Phe 195	Glu	Glu	Asn	Pro 200	Ile	Asn	Ala	Ser	Gly 205	Val	Asp	Ala
Lys	Ala	Ile	Leu 210	Ser	Ala	Arg 215	Leu	Ser	Lys	Ser	Arg 220	Arg	Leu	Glu	Asn
Leu 225	Ile	Ala	Gln	Leu	Pro 230	Gly	Glu	Lys	Lys	Asn 235	Gly	Leu	Phe	Gly	Asn 240
Leu	Ile	Ala	Leu 245	Ser	Leu	Gly	Leu	Thr	Pro 250	Asn	Phe	Lys	Ser	Asn 255	Phe
Asp	Leu	Ala	Glu 260	Asp	Ala	Lys	Leu	Gln	Leu 265	Ser	Lys	Asp	Thr	Tyr 270	Asp
Asp	Asp	Leu	Asp 275	Asn	Leu	Leu	Ala 280	Gln	Ile	Gly	Asp	Gln 285	Tyr	Ala	Asp
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Val	Val	Asp	Lys	Gly	Ala	Ser	Ala	Gln	Ser	Phe	Ile	Glu	Arg	Met	Thr	
				485					490					495		
Asn	Phe	Asp	Lys	Asn	Leu	Pro	Asn	Glu	Lys	Val	Leu	Pro	Lys	His	Ser	
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Val	Lys	Gln	Leu	Lys	Glu	Asp	Tyr	Phe	Lys	Lys	Ile	Glu	Cys	Phe	Asp	
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Thr	Tyr	His	Asp	Leu	Leu	Lys	Ile	Ile	Lys	Asp	Lys	Asp	Phe	Leu	Asp	
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Lys	Glu	Asp	Ile	Gln	Lys	Ala	Gln	Val	Ser	Gly	Gln	Gly	Asp	Ser	Leu	
					710					715					720	
His	Glu	His	Ile	Ala	Asn	Leu	Ala	Gly	Ser	Pro	Ala	Ile	Lys	Lys	Gly	
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Ile	Leu	Gln	Thr	Val	Lys	Val	Val	Asp	Glu	Leu	Val	Lys	Val	Met	Gly	
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Arg	His	Lys	Pro	Glu	Asn	Ile	Val	Ile	Glu							

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Lys Ala Gly Phe Ile Lys Arg Gln Leu Val Glu Thr Arg Gln Ile Thr 915 920 925		
Lys His Val Ala Gln Ile Leu Asp Ser Arg Met Asn Thr Lys Tyr Asp 930 935 940		
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Arg Val   Ile Leu Ala Asp Ala  Asn Leu Asp Lys Val  Leu Ser Ala
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Tyr Asn   Lys His Arg Asp Lys  Pro Ile Arg Glu Gln  Ala Glu Asn
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Ile Ile   His Leu Phe Thr Leu  Thr Asn Leu Gly Ala  Pro Thr Ala
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Phe Lys   Tyr Phe Asp Thr Thr  Ile Asp Arg Lys Arg  Tyr Thr Ser
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Thr Lys   Glu Val Leu Asp Ala  Thr Phe Ile His Gln  Ser Ile Thr
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<210> SEQ ID NO 2

<211> LENGTH: 1081

<212> TYPE: PRT

<213> ORGANISM: Neisseria meningitidis

<400> SEQUENCE: 2

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Arg Ala Arg Arg Leu Leu Lys Arg Glu Gly Val Leu Gln Ala Ala Asp
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Gln Ala Glu Leu Asp Leu Leu Phe Glu Lys Gln Lys Glu Phe Gly Asn
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Ser	Glu 610	Asn	Gln	Asn	Lys	Gly 615	Asn	Gln	Thr	Pro	Tyr 620	Glu	Tyr	Phe	Asn
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Lys	Ser	Lys	Gly	Lys	Asp	Gly	Met	His	Arg	Ile	Gly	Val	Lys	Thr	
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Ala	Leu	Ser	Phe	Gln	Lys	Tyr	Gln	Ile	Asp	Glu	Met	Gly	Lys	Glu	
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<212> TYPE: PRT
<213> ORGANISM: Streptococcus thermophilus

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Lys Val Leu Gly Asn Thr Ser Lys Lys Tyr Ile Lys Lys Asn Leu Leu
          35          40          45

Gly Val Leu Leu Phe Asp Ser Gly Ile Thr Ala Glu Gly Arg Arg Leu
          50          55          60

Lys Arg Thr Ala Arg Arg Tyr Thr Arg Arg Arg Asn Arg Ile Leu
          65          70          75          80

Tyr Leu Gln Glu Ile Phe Ser Thr Glu Met Ala Thr Leu Asp Asp Ala
          85          90          95

Phe Phe Gln Arg Leu Asp Asp Ser Phe Leu Val Pro Asp Asp Lys Arg
          100         105         110

Asp Ser Lys Tyr Pro Ile Phe Gly Asn Leu Val Glu Glu Lys Val Tyr
          115         120         125

His Asp Glu Phe Pro Thr Ile Tyr His Leu Arg Lys Tyr Leu Ala Asp
          130         135         140

Ser Thr Lys Lys Ala Asp Leu Arg Leu Val Tyr Leu Ala Leu Ala His
          145         150         155         160

Met Ile Lys Tyr Arg Gly His Phe Leu Ile Glu Gly Glu Phe Asn Ser
          165         170         175

Lys Asn Asn Asp Ile Gln Lys Asn Phe Gln Asp Phe Leu Asp Thr Tyr
          180         185         190

Asn Ala Ile Phe Glu Ser Asp Leu Ser Leu Glu Asn Ser Lys Gln Leu
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Glu Glu Ile Val Lys Asp Lys Ile Ser Lys Leu Glu Lys Lys Asp Arg
          210         215         220

Ile Leu Lys Leu Phe Pro Gly Glu Lys Asn Ser Gly Ile Phe Ser Glu
          225         230         235         240

Phe Leu Lys Leu Ile Val Gly Asn Gln Ala Asp Phe Arg Lys Cys Phe
          245         250         255

Asn Leu Asp Glu Lys Ala Ser Leu His Phe Ser Lys Glu Ser Tyr Asp
          260         265         270

Glu Asp Leu Glu Thr Leu Leu Gly Tyr Ile Gly Asp Asp Tyr Ser Asp
          275         280         285

Val Phe Leu Lys Ala Lys Lys Leu Tyr Asp Ala Ile Leu Leu Ser Gly
          290         295         300

Phe Leu Thr Val Thr Asp Asn Glu Thr Glu Ala Pro Leu Ser Ser Ala
          305         310         315         320

Met Ile Lys Arg Tyr Asn Glu His Lys Glu Asp Leu Ala Leu Leu Lys
          325         330         335

Glu Tyr Ile Arg Asn Ile Ser Leu Lys Thr Tyr Asn Glu Val Phe Lys
          340         345         350

Asp Asp Thr Lys Asn Gly Tyr Ala Gly Tyr Ile Asp Gly Lys Thr Asn
          355         360         365

Gln Glu Asp Phe Tyr Val Tyr Leu Lys Asn Leu Leu Ala Glu Phe Glu
          370         375         380

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Gly	Ala	Asp	Tyr	Phe	Leu	Glu	Lys	Ile	Asp	Arg	Glu	Asp	Phe	Leu	Arg	385	390	395	400
Lys	Gln	Arg	Thr	Phe	Asp	Asn	Gly	Ser	Ile	Pro	Tyr	Gln	Ile	His	Leu	405	410	415	
Gln	Glu	Met	Arg	Ala	Ile	Leu	Asp	Lys	Gln	Ala	Lys	Phe	Tyr	Pro	Phe	420	425	430	
Leu	Ala	Lys	Asn	Lys	Glu	Arg	Ile	Glu	Lys	Ile	Leu	Thr	Phe	Arg	Ile	435	440	445	
Pro	Tyr	Tyr	Val	Gly	Pro	Leu	Ala	Arg	Gly	Asn	Ser	Asp	Phe	Ala	Trp	450	455	460	
Ser	Ile	Arg	Lys	Arg	Asn	Glu	Lys	Ile	Thr	Pro	Trp	Asn	Phe	Glu	Asp	465	470	475	480
Val	Ile	Asp	Lys	Glu	Ser	Ser	Ala	Glu	Ala	Phe	Ile	Asn	Arg	Met	Thr	485	490	495	
Ser	Phe	Asp	Leu	Tyr	Leu	Pro	Glu	Glu	Lys	Val	Leu	Pro	Lys	His	Ser	500	505	510	
Leu	Leu	Tyr	Glu	Thr	Phe	Asn	Val	Tyr	Asn	Glu	Leu	Thr	Lys	Val	Arg	515	520	525	
Phe	Ile	Ala	Glu	Ser	Met	Arg	Asp	Tyr	Gln	Phe	Leu	Asp	Ser	Lys	Gln	530	535	540	
Lys	Lys	Asp	Ile	Val	Arg	Leu	Tyr	Phe	Lys	Asp	Lys	Arg	Lys	Val	Thr	545	550	555	560
Asp	Lys	Asp	Ile	Ile	Glu	Tyr	Leu	His	Ala	Ile	Tyr	Gly	Tyr	Asp	Gly	565	570	575	
Ile	Glu	Leu	Lys	Gly	Ile	Glu	Lys	Gln	Phe	Asn	Ser	Ser	Leu	Ser	Thr	580	585	590	
Tyr	His	Asp	Leu	Leu	Asn	Ile	Ile	Asn	Asp	Lys	Glu	Phe	Leu	Asp	Asp	595	600	605	
Ser	Ser	Asn	Glu	Ala	Ile	Ile	Glu	Glu	Ile	Ile	His	Thr	Leu	Thr	Ile	610	615	620	
Phe	Glu	Asp	Arg	Glu	Met	Ile	Lys	Gln	Arg	Leu	Ser	Lys	Phe	Glu	Asn	625	630	635	640
Ile	Phe	Asp	Lys	Ser	Val	Leu	Lys	Lys	Leu	Ser	Arg	Arg	His	Tyr	Thr	645	650	655	
Gly	Trp	Gly	Lys	Leu	Ser	Ala	Lys	Leu	Ile	Asn	Gly	Ile	Arg	Asp	Glu	660	665	670	
Lys	Ser	Gly	Asn	Thr	Ile	Leu	Asp	Tyr	Leu	Ile	Asp	Asp	Gly	Ile	Ser	675	680	685	
Asn	Arg	Asn	Phe	Met	Gln	Leu	Ile	His	Asp	Asp	Ala	Leu	Ser	Phe	Lys	690	695	700	
Lys	Lys	Ile	Gln	Lys	Ala	Gln	Ile	Ile	Gly	Asp	Glu	Asp	Lys	Gly	Asn	705	710	715	720
Ile	Lys	Glu	Val	Val	Lys	Ser	Leu	Pro	Gly	Ser	Pro	Ala	Ile	Lys	Lys	725	730	735	
Gly	Ile	Leu	Gln	Ser	Ile	Lys	Ile	Val	Asp	Glu	Leu	Val	Lys	Val	Met	740	745	750	
Gly	Gly	Arg	Lys	Pro	Glu	Ser	Ile	Val	Val	Glu	Met	Ala	Arg	Glu	Asn	755	760	765	
Gln	Tyr	Thr	Asn	Gln	Gly	Lys	Ser	Asn	Ser	Gln	Gln	Arg	Leu	Lys	Arg	770	775	780	
Leu	Glu	Lys	Ser	Leu	Lys	Glu	Leu	Gly	Ser	Lys	Ile	Leu	Lys	Glu	Asn	785	790	795	800

Ile	Pro	Ala	Lys	Leu	Ser	Lys	Ile	Asp	Asn	Asn	Ala	Leu	Gln	Asn	Asp	
				805						810		815				
Arg	Leu	Tyr	Leu	Tyr	Tyr	Leu	Gln	Asn	Gly	Lys	Asp	Met	Tyr	Thr	Gly	
				820						825		830				
Asp	Asp	Leu	Asp	Ile	Asp	Arg	Leu	Ser	Asn	Tyr	Asp	Ile	Asp	His	Ile	
				835						840		845				
Ile	Pro	Gln	Ala	Phe	Leu	Lys	Asp	Asn	Ser	Ile	Asp	Asn	Lys	Val	Leu	
				850						855		860				
Val	Ser	Ser	Ala	Ser	Asn	Arg	Gly	Lys	Ser	Asp	Asp	Phe	Pro	Ser	Leu	
				865						870		875				
Glu	Val	Val	Lys	Lys	Arg	Lys	Thr	Phe	Trp	Tyr	Gln	Leu	Leu	Lys	Ser	
				885						890		895				
Lys	Leu	Ile	Ser	Gln	Arg	Lys	Phe	Asp	Asn	Leu	Thr	Lys	Ala	Glu	Arg	
				900						905		910				
Gly	Gly	Leu	Leu	Pro	Glu	Asp	Lys	Ala	Gly	Phe	Ile	Gln	Arg	Gln	Leu	
				915						920		925				
Val	Glu	Thr	Arg	Gln	Ile	Thr	Lys	His	Val	Ala	Arg	Leu	Leu	Asp	Glu	
				930						935		940				
Lys	Phe	Asn	Asn	Lys	Lys	Asp	Glu	Asn	Asn	Arg	Ala	Val	Arg	Thr	Val	
				945						950		955				
Lys	Ile	Ile	Thr	Leu	Lys	Ser	Thr	Leu	Val	Ser	Gln	Phe	Arg	Lys	Asp	
				965						970		975				
Phe	Glu	Leu	Tyr	Lys	Val	Arg	Glu	Ile	Asn	Asp	Phe	His	His	Ala	His	
				980						985		990				
Asp	Ala	Tyr	Leu	Asn	Ala	Val	Ile	Ala	Ser	Ala	Leu	Leu	Lys	Lys	Tyr	
				995						1000		1005				
Pro	Lys	Leu	Glu	Pro	Glu	Phe	Val	Tyr	Gly	Asp	Tyr	Pro	Lys	Tyr		
				1010		1015						1020				
Asn	Ser	Phe	Arg	Glu	Arg	Lys	Ser	Ala	Thr	Glu	Lys	Val	Tyr	Phe		
				1025		1030						1035				
Tyr	Ser	Asn	Ile	Met	Asn	Ile	Phe	Lys	Lys	Ser	Ile	Ser	Leu	Ala		
				1040		1045						1050				
Asp	Gly	Arg	Val	Ile	Glu	Arg	Pro	Leu	Ile	Glu	Val	Asn	Glu	Glu		
				1055		1060						1065				
Thr	Gly	Glu	Ser	Val	Trp	Asn	Lys	Glu	Ser	Asp	Leu	Ala	Thr	Val		
				1070		1075						1080				
Arg	Arg	Val	Leu	Ser	Tyr	Pro	Gln	Val	Asn	Val	Val	Lys	Lys	Val		
				1085		1090						1095				
Glu	Glu	Gln	Asn	His	Gly	Leu	Asp	Arg	Gly	Lys	Pro	Lys	Gly	Leu		
				1100		1105						1110				
Phe	Asn	Ala	Asn	Leu	Ser	Ser	Lys	Pro	Lys	Pro	Asn	Ser	Asn	Glu		
				1115		1120						1125				
Asn	Leu	Val	Gly	Ala	Lys	Glu	Tyr	Leu	Asp	Pro	Lys	Lys	Tyr	Gly		
				1130		1135						1140				
Gly	Tyr	Ala	Gly	Ile	Ser	Asn	Ser	Phe	Ala	Val	Leu	Val	Lys	Gly		
				1145		1150						1155				
Thr	Ile	Glu	Lys	Gly	Ala	Lys	Lys	Lys	Ile	Thr	Asn	Val	Leu	Glu		
				1160		1165						1170				
Phe	Gln	Gly	Ile	Ser	Ile	Leu	Asp	Arg	Ile	Asn	Tyr	Arg	Lys	Asp		
				1175		1180						1185				
Lys	Leu	Asn	Phe	Leu	Leu	Glu	Lys	Gly	Tyr	Lys	Asp	Ile	Glu	Leu		

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1205	1210	1215
Ser Arg Arg Met Leu Ala Ser Ile Leu Ser Thr Asn Asn Lys Arg		
1220	1225	1230
Gly Glu Ile His Lys Gly Asn Gln Ile Phe Leu Ser Gln Lys Phe		
1235	1240	1245
Val Lys Leu Leu Tyr His Ala Lys Arg Ile Ser Asn Thr Ile Asn		
1250	1255	1260
Glu Asn His Arg Lys Tyr Val Glu Asn His Lys Lys Glu Phe Glu		
1265	1270	1275
Glu Leu Phe Tyr Tyr Ile Leu Glu Phe Asn Glu Asn Tyr Val Gly		
1280	1285	1290
Ala Lys Lys Asn Gly Lys Leu Leu Asn Ser Ala Phe Gln Ser Trp		
1295	1300	1305
Gln Asn His Ser Ile Asp Glu Leu Cys Ser Ser Phe Ile Gly Pro		
1310	1315	1320
Thr Gly Ser Glu Arg Lys Gly Leu Phe Glu Leu Thr Ser Arg Gly		
1325	1330	1335
Ser Ala Ala Asp Phe Glu Phe Leu Gly Val Lys Ile Pro Arg Tyr		
1340	1345	1350
Arg Asp Tyr Thr Pro Ser Ser Leu Leu Lys Asp Ala Thr Leu Ile		
1355	1360	1365
His Gln Ser Val Thr Gly Leu Tyr Glu Thr Arg Ile Asp Leu Ala		
1370	1375	1380
Lys Leu Gly Glu Gly		
1385		

<210> SEQ ID NO 4

<211> LENGTH: 1395

<212> TYPE: PRT

<213> ORGANISM: Treponema denticola

<400> SEQUENCE: 4

Met Lys Lys Glu Ile Lys Asp Tyr Phe Leu Gly Leu Asp Val Gly Thr		
1	5	10
Gly Ser Val Gly Trp Ala Val Thr Asp Thr Asp Tyr Lys Leu Leu Lys		
20	25	30
Ala Asn Arg Lys Asp Leu Trp Gly Met Arg Cys Phe Glu Thr Ala Glu		
35	40	45
Thr Ala Glu Val Arg Arg Leu His Arg Gly Ala Arg Arg Arg Ile Glu		
50	55	60
Arg Arg Lys Lys Arg Ile Lys Leu Leu Gln Glu Leu Phe Ser Gln Glu		
65	70	75
Ile Ala Lys Ile Asp Glu Gly Phe Phe Gln Arg Met Lys Glu Ser Pro		
85	90	95
Phe Tyr Ala Glu Asp Lys Thr Ile Leu Gln Glu Asn Thr Leu Phe Asn		
100	105	110
Asp Lys Asp Phe Thr Asp Lys Thr Tyr His Lys Ala Tyr Pro Thr Ile		
115	120	125
Asn His Leu Ile Lys Ala Trp Ile Glu Asn Lys Val Lys Pro Asp Pro		
130	135	140
Arg Leu Leu Tyr Leu Ala Cys His Asn Ile Ile Lys Lys Arg Gly His		
145	150	155
Phe Leu Phe Glu Gly Asp Phe Asp Ser Glu Asn Gln Phe Asp Thr Ser		
165	170	175

Ile	Gln	Ala	Leu	Phe	Glu	Tyr	Leu	Arg	Glu	Asp	Met	Glu	Val	Asp	Ile	
			180				185						190			
Asp	Ala	Asp	Ser	Gln	Lys	Val	Lys	Glu	Ile	Leu	Lys	Asp	Ser	Ser	Leu	
			195				200						205			
Lys	Asn	Ser	Glu	Lys	Gln	Ser	Arg	Leu	Asn	Lys	Ile	Leu	Gly	Leu	Lys	
			210				215						220			
Pro	Ser	Asp	Lys	Gln	Lys	Lys	Ala	Ile	Thr	Asn	Leu	Ile	Ser	Gly	Asn	
			225				230						235			
Lys	Ile	Asn	Phe	Ala	Asp	Leu	Tyr	Asp	Asn	Pro	Asp	Leu	Lys	Asp	Ala	
			245				250						255			
Glu	Lys	Asn	Ser	Ile	Ser	Phe	Ser	Lys	Asp	Asp	Phe	Asp	Ala	Leu	Ser	
			260				265						270			
Asp	Asp	Leu	Ala	Ser	Ile	Leu	Gly	Asp	Ser	Phe	Glu	Leu	Leu	Leu	Lys	
			275				280						285			
Ala	Lys	Ala	Val	Tyr	Asn	Cys	Ser	Val	Leu	Ser	Lys	Val	Ile	Gly	Asp	
			290				295						300			
Glu	Gln	Tyr	Leu	Ser	Phe	Ala	Lys	Val	Lys	Ile	Tyr	Glu	Lys	His	Lys	
			305				310						315			
Thr	Asp	Leu	Thr	Lys	Leu	Lys	Asn	Val	Ile	Lys	Lys	His	Phe	Pro	Lys	
			325				330						335			
Asp	Tyr	Lys	Lys	Val	Phe	Gly	Tyr	Asn	Lys	Asn	Glu	Lys	Asn	Asn	Asn	
			340				345						350			
Asn	Tyr	Ser	Gly	Tyr	Val	Gly	Val	Cys	Lys	Thr	Lys	Ser	Lys	Lys	Leu	
			355				360						365			
Ile	Ile	Asn	Asn	Ser	Val	Asn	Gln	Glu	Asp	Phe	Tyr	Lys	Phe	Leu	Lys	
			370				375						380			
Thr	Ile	Leu	Ser	Ala	Lys	Ser	Glu	Ile	Lys	Glu	Val	Asn	Asp	Ile	Leu	
			385				390						395			
Thr	Glu	Ile	Glu	Thr	Gly	Thr	Phe	Leu	Pro	Lys	Gln	Ile	Ser	Lys	Ser	
			405				410						415			
Asn	Ala	Glu	Ile	Pro	Tyr	Gln	Leu	Arg	Lys	Met	Glu	Leu	Glu	Lys	Ile	
			420				425						430			
Leu	Ser	Asn	Ala	Glu	Lys	His	Phe	Ser	Phe	Leu	Lys	Gln	Lys	Asp	Glu	
			435				440						445			
Lys	Gly	Leu	Ser	His	Ser	Glu	Lys	Ile	Ile	Met	Leu	Leu	Thr	Phe	Lys	
			450				455						460			
Ile	Pro	Tyr	Tyr	Ile	Gly	Pro	Ile	Asn	Asp	Asn	His	Lys	Lys	Ser	Phe	
			465				470						475			
Pro	Asp	Arg	Cys	Trp	Val	Ile	Lys	Lys	Glu	Lys	Ser	Pro	Ser	Gly	Lys	
			485				490						495			
Thr	Thr	Pro	Trp	Asn	Phe	Phe	Asp	His	Ile	Asp	Lys	Asp	Lys	Thr	Ala	
			500				505						510			
Glu	Ala	Phe	Ile	Thr	Ser	Arg	Thr	Asn	Phe	Cys	Thr	Tyr	Leu	Val	Gly	
			515				520						525			
Glu	Ser	Val	Leu	Pro	Lys	Ser	Ser	Leu	Leu	Tyr	Ser	Glu	Tyr	Thr	Val	
			530				535						540			
Leu	Asn	Glu	Ile	Asn	Asn	Leu	Gln	Ile	Ile	Ile	Asp	Gly	Lys	Asn	Ile	
			545				550						555			
Cys	Asp	Ile	Lys	Leu	Lys	Gln	Lys	Ile	Tyr	Glu	Glu	Leu	Phe	Lys	Lys	
			565				570						575			
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595	600	605
Glu Cys Thr Ser Ser Leu Lys Ser Tyr Ile Glu Leu Lys Asn Ile Phe 610 615 620		
Gly Lys Gln Val Asp Glu Ile Ser Thr Lys Asn Met Leu Glu Glu Ile 625 630 635 640		
Ile Arg Trp Ala Thr Ile Tyr Asp Glu Gly Glu Gly Lys Thr Ile Leu 645 650 655		
Lys Thr Lys Ile Lys Ala Glu Tyr Gly Lys Tyr Cys Ser Asp Glu Gln 660 665 670		
Ile Lys Lys Ile Leu Asn Leu Lys Phe Ser Gly Trp Gly Arg Leu Ser 675 680 685		
Arg Lys Phe Leu Glu Thr Val Thr Ser Glu Met Pro Gly Phe Ser Glu 690 695 700		
Pro Val Asn Ile Ile Thr Ala Met Arg Glu Thr Gln Asn Asn Leu Met 705 710 715 720		
Glu Leu Leu Ser Ser Glu Phe Lys Phe Thr Glu Asn Ile Lys Lys Ile 725 730 735		
Asn Ser Gly Phe Glu Asp Ala Glu Lys Gln Phe Ser Tyr Asn Gly Leu 740 745 750		
Val Lys Pro Leu Phe Leu Ser Pro Ser Val Lys Lys Met Leu Trp Gln 755 760 765		
Thr Leu Lys Leu Val Lys Glu Ile Ser His Ile Thr Gln Ala Pro Pro 770 775 780		
Lys Lys Ile Phe Ile Glu Met Ala Lys Gly Ala Glu Leu Glu Pro Ala 785 790 795 800		
Arg Thr Lys Thr Arg Leu Lys Ile Leu Gln Asp Leu Tyr Asn Asn Cys 805 810 815		
Lys Asn Asp Ala Asp Ala Phe Ser Ser Glu Ile Lys Asp Leu Ser Gly 820 825 830		
Lys Ile Glu Asn Glu Asp Asn Leu Arg Leu Arg Ser Asp Lys Leu Tyr 835 840 845		
Leu Tyr Tyr Thr Gln Leu Gly Lys Cys Met Tyr Cys Gly Lys Pro Ile 850 855 860		
Glu Ile Gly His Val Phe Asp Thr Ser Asn Tyr Asp Ile Asp His Ile 865 870 875 880		
Tyr Pro Gln Ser Lys Ile Lys Asp Asp Ser Ile Ser Asn Arg Val Leu 885 890 895		
Val Cys Ser Ser Cys Asn Lys Asn Lys Glu Asp Lys Tyr Pro Leu Lys 900 905 910		
Ser Glu Ile Gln Ser Lys Gln Arg Gly Phe Trp Asn Phe Leu Gln Arg 915 920 925		
Asn Asn Phe Ile Ser Leu Glu Lys Leu Asn Arg Leu Thr Arg Thr Thr 930 935 940		
Pro Ile Ser Asp Asp Glu Thr Ala Lys Phe Ile Ala Arg Gln Leu Val 945 950 955 960		
Glu Thr Arg Gln Ala Thr Lys Val Ala Ala Lys Val Leu Glu Lys Met 965 970 975		
Phe Pro Glu Thr Lys Ile Val Tyr Ser Lys Ala Glu Thr Val Ser Met 980 985 990		
Phe Arg Asn Lys Phe Asp Ile Ile Lys Cys Arg Glu Ile Asn Asp Phe 995 1000 1005		
His His Ala His Asp Ala Tyr Leu Asn Ile Val Val Gly Asn Val 1010 1015 1020		

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Tyr Asn Thr Lys Phe Thr Asn Asn Pro Trp Asn Phe Ile Lys Glu	
1025 1030 1035	
Lys Arg Asp Asn Pro Lys Ile Ala Asp Thr Tyr Asn Tyr Tyr Lys	
1040 1045 1050	
Val Phe Asp Tyr Asp Val Lys Arg Asn Asn Ile Thr Ala Trp Glu	
1055 1060 1065	
Lys Gly Lys Thr Ile Ile Thr Val Lys Asp Met Leu Lys Arg Asn	
1070 1075 1080	
Thr Pro Ile Tyr Thr Arg Gln Ala Ala Cys Lys Lys Gly Gly Leu	
1085 1090 1095	
Phe Asp Gln Thr Ile Met Lys Lys Gly Leu Gly Gln His Pro Leu	
1100 1105 1110	
Lys Lys Glu Gly Ala Phe Ser Asn Ile Ser Lys Tyr Gly Gly Tyr	
1115 1120 1125	
Asn Lys Val Ser Ala Ala Tyr Tyr Thr Leu Ile Glu Tyr Glu Glu	
1130 1135 1140	
Lys Gly Asn Lys Ile Arg Ser Leu Glu Thr Ile Pro Leu Tyr Leu	
1145 1150 1155	
Val Lys Asp Ile Gln Lys Asp Gln Asp Val Leu Lys Ser Tyr Leu	
1160 1165 1170	
Thr Asp Leu Leu Gly Lys Lys Glu Ile Lys Ile Leu Val Pro Lys	
1175 1180 1185	
Ile Lys Ile Asn Ser Leu Leu Lys Ile Asn Gly Phe Pro Cys His	
1190 1195 1200	
Ile Thr Gly Lys Thr Gly Asp Ser Phe Val Leu Arg Pro Ala Val	
1205 1210 1215	
Gln Phe Cys Cys Ser Asn Asp Glu Val Leu Tyr Phe Lys Lys Ile	
1220 1225 1230	
Ile Arg Phe Asn Glu Ile Arg Ser Gln Arg Glu Lys Met Gly Lys	
1235 1240 1245	
Thr Ile Ser Pro Tyr Glu Asp Leu Ser Phe Arg Ser Tyr Ile Lys	
1250 1255 1260	
Glu Asn Leu Cys Lys Lys Thr Lys Ala Asp Glu Ile Gly Glu Lys	
1265 1270 1275	
Glu Phe Tyr Asp Leu Leu Gln Lys Lys Asn Leu Glu Ile Tyr Asp	
1280 1285 1290	
Met Leu Leu Thr Lys His Lys Asp Thr Ile Tyr Asn Lys Arg Pro	
1295 1300 1305	
Asn Ser Thr Thr Leu Gly Met Leu Thr Lys Gly Arg Tyr Glu Phe	
1310 1315 1320	
Ile Asn Leu Lys Pro Lys Asp Gln Ile Ile Val Met Leu Glu Ile	
1325 1330 1335	
Leu Lys Leu Phe Cys Thr Thr Arg Glu Ala Ile Asp Leu Gly Leu	
1340 1345 1350	
Ile Lys Gly Lys Pro Ala Ala Gly Val Ala Thr Leu Gly Lys Lys	
1355 1360 1365	
Ile Ser Asn Pro Asp Asn Cys Ile Leu Ile Tyr Gln Ser Ile Thr	
1370 1375 1380	
Gly Ile Phe Glu Lys Arg Ile Asp Leu Leu Lys Val	
1385 1390 1395	

<210> SEQ ID NO 5

<211> LENGTH: 749

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<212> TYPE: PRT

<213> ORGANISM: Staphylococcus aureus

<400> SEQUENCE: 5

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Met Asn Glu Glu Asp Ile Lys Gly Tyr Arg Val Thr Ser Thr Gly Lys
 1          5          10          15

Pro Glu Phe Thr Asn Leu Lys Val Tyr His Asp Ile Lys Asp Ile Thr
 20          25          30

Ala Arg Lys Glu Ile Ile Glu Asn Ala Glu Leu Leu Asp Gln Ile Ala
 35          40          45

Lys Ile Leu Thr Ile Tyr Gln Ser Ser Glu Asp Ile Gln Glu Glu Leu
 50          55          60

Thr Asn Leu Asn Ser Glu Leu Thr Gln Glu Glu Ile Lys Gln Ile Ser
 65          70          75          80

Lys Leu Lys Gly Tyr Thr Gly Thr His Asn Leu Ser Leu Lys Ala Ile
 85          90          95

Asn Leu Ile Leu Asp Glu Leu Trp His Thr Asn Asp Asn Gln Met Ala
100          105          110

Ile Phe Asn Arg Leu Lys Leu Val Pro Lys Lys Val Asp Leu Ser Gln
115          120          125

Gln Lys Glu Ile Pro Thr Thr Leu Val Asp Asp Phe Ile Leu Ser Pro
130          135          140

Val Val Lys Arg Ser Phe Ile Gln Ser Ile Lys Val Ile Asn Ala Ile
145          150          155          160

Ile Lys Lys Tyr Gly Leu Pro Asn Asp Ile Ile Ile Glu Leu Ala Arg
165          170          175

Glu Lys Asn Ser Lys Asp Ala Gln Lys Met Ile Asn Glu Met Gln Lys
180          185          190

Arg Asn Arg Gln Thr Asn Glu Arg Ile Glu Glu Ile Ile Arg Thr Thr
195          200          205

Gly Lys Glu Asn Ala Lys Tyr Leu Ile Glu Lys Ile Lys Leu His Asp
210          215          220

Met Gln Glu Gly Lys Cys Leu Tyr Ser Leu Glu Ala Ile Pro Leu Glu
225          230          235          240

Asp Leu Leu Asn Asn Pro Phe Asn Tyr Glu Val Asp His Ile Ile Pro
245          250          255

Arg Ser Val Ser Phe Asp Asn Ser Phe Asn Asn Lys Val Leu Val Lys
260          265          270

Gln Glu Glu Asn Ser Lys Lys Gly Asn Arg Thr Pro Phe Gln Tyr Leu
275          280          285

Ser Ser Ser Asp Ser Lys Ile Ser Tyr Glu Ile Phe Lys Lys His Ile
290          295          300

Leu Asn Leu Ala Lys Asp Lys Gly Arg Ile Ser Lys Thr Lys Lys Glu
305          310          315          320

Tyr Leu Leu Glu Glu Arg Asp Ile Asn Arg Phe Ser Val Gln Lys Asp
325          330          335

Phe Ile Asn Arg Asn Leu Val Asp Thr Arg Tyr Ala Thr Arg Glu Leu
340          345          350

Met Asn Leu Leu Arg Ser Tyr Phe Arg Val Asn Asn Leu Asp Val Lys
355          360          365

Val Lys Ser Ile Asn Gly Gly Phe Thr Ser Phe Leu Arg Arg Lys Trp
370          375          380

Lys Phe Lys Lys Glu Arg Asn Lys Gly Tyr Lys His His Ala Glu Asp
385          390          395          400

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Ala	Leu	Ile	Ile	Ala	Asn	Ala	Asp	Phe	Ile	Phe	Lys	Glu	Trp	Lys	Lys	
				405					410					415		
Leu	Asp	Lys	Ala	Lys	Lys	Val	Met	Glu	Asn	Gln	Thr	Phe	Glu	Glu	Lys	
				420					425					430		
Gln	Ala	Glu	Ser	Met	Pro	Glu	Ile	Glu	Thr	Glu	Gln	Glu	Tyr	Lys	Glu	
				435					440					445		
Ile	Phe	Ile	Thr	Ser	His	Gln	Ile	Lys	His	Ile	Lys	Asp	Phe	Lys	Asp	
				450					455					460		
Tyr	Lys	Tyr	Ser	His	Arg	Val	Asp	Lys	Lys	Pro	Asn	Arg	Glu	Leu	Ile	
				465					470					475		
Asn	Asp	Thr	Leu	Tyr	Ser	Thr	Arg	Asn	Asp	Asp	Lys	Gly	Asn	Thr	Leu	
				485					490					495		
Ile	Val	Asn	Asn	Leu	Asn	Gly	Leu	Tyr	Asp	Lys	Asp	Asn	Asp	Lys	Leu	
				500					505					510		
Lys	Lys	Leu	Ile	Asn	Lys	Ser	Pro	Glu	Lys	Leu	Leu	Met	Tyr	His	His	
				515					520					525		
Asp	Pro	Gln	Thr	Tyr	Gln	Lys	Leu	Lys	Leu	Ile	Met	Glu	Gln	Tyr	Gly	
				530					535					540		
Asp	Glu	Lys	Asn	Pro	Leu	Tyr	Lys	Tyr	Tyr	Glu	Glu	Thr	Gly	Asn	Tyr	
				545					550					555		
Leu	Thr	Lys	Tyr	Ser	Lys	Lys	Asp	Asn	Arg	Pro	Val	Ile	Lys	Lys	Ile	
				565					570					575		
Lys	Tyr	Tyr	Gly	Asn	Lys	Leu	Asn	Ala	His	Leu	Asp	Ile	Thr	Asp	Asp	
				580					585					590		
Tyr	Pro	Asn	Ser	Arg	Asn	Lys	Val	Val	Lys	Leu	Ser	Leu	Lys	Pro	Tyr	
				595					600					605		
Arg	Phe	Asp	Val	Tyr	Leu	Asp	Asn	Gly	Val	Tyr	Lys	Phe	Val	Thr	Val	
				610					615					620		
Lys	Asn	Leu	Asp	Val	Ile	Lys	Lys	Glu	Asn	Tyr	Tyr	Glu	Val	Asn	Ser	
				625					630					635		
Lys	Cys	Tyr	Glu	Glu	Ala	Lys	Lys	Leu	Lys	Lys	Ile	Ser	Asn	Gln	Ala	
				645					650					655		
Glu	Phe	Ile	Ala	Ser	Phe	Tyr	Asn	Asn	Asp	Leu	Ile	Lys	Ile	Asn	Gly	
				660					665					670		
Glu	Leu	Tyr	Arg	Val	Ile	Gly	Val	Asn	Ser	Asp	Leu	Leu	Asn	Arg	Ile	
				675					680					685		
Glu	Val	Asn	Met	Ile	Asp	Ile	Thr	Tyr	Arg	Glu	Tyr	Leu	Glu	Asn	Met	
				690					695					700		
Asn	Asp	Lys	Arg	Thr	Pro	Arg	Ile	Ile	Lys	Thr	Ile	Ala	Ser	Lys	Thr	
				705					710					715		
Gln	Ser	Ile	Lys	Lys	Tyr	Ser	Thr	Asp	Ile	Leu	Gly	Asn	Leu	Tyr	Glu	
				725					730					735		
Val	Lys	Ser	Lys	Lys	Tyr	Pro	Gln	Ile	Lys	Lys	Gly					
				740					745							

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<210> SEQ ID NO 6
<211> LENGTH: 984
<212> TYPE: PRT
<213> ORGANISM: Campylobacter jejuni
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<400> SEQUENCE: 6

Met Ala Arg Ile Leu Ala Phe Asp Ile Gly Ile Ser Ser Ile Gly Trp
1 5 10 15

Ala Phe Ser Glu Asn Asp Glu Leu Lys Asp Cys Gly Val Arg Ile Phe

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20						25						30					
Thr	Lys	Ala	Glu	Asn	Pro	Lys	Thr	Gly	Glu	Ser	Leu	Ala	Leu	Pro	Arg		
		35				40					45						
Arg	Leu	Ala	Arg	Ser	Ala	Arg	Lys	Arg	Leu	Ala	Arg	Arg	Lys	Ala	Arg		
		50				55					60						
Leu	Asn	His	Leu	Lys	His	Leu	Ile	Ala	Asn	Glu	Phe	Lys	Leu	Asn	Tyr		
		65			70					75					80		
Glu	Asp	Tyr	Gln	Ser	Phe	Asp	Glu	Ser	Leu	Ala	Lys	Ala	Tyr	Lys	Gly		
			85						90					95			
Ser	Leu	Ile	Ser	Pro	Tyr	Glu	Leu	Arg	Phe	Arg	Ala	Leu	Asn	Glu	Leu		
			100				105						110				
Leu	Ser	Lys	Gln	Asp	Phe	Ala	Arg	Val	Ile	Leu	His	Ile	Ala	Lys	Arg		
		115				120						125					
Arg	Gly	Tyr	Asp	Asp	Ile	Lys	Asn	Ser	Asp	Asp	Lys	Glu	Lys	Gly	Ala		
		130			135						140						
Ile	Leu	Lys	Ala	Ile	Lys	Gln	Asn	Glu	Glu	Lys	Leu	Ala	Asn	Tyr	Gln		
		145			150				155						160		
Ser	Val	Gly	Glu	Tyr	Leu	Tyr	Lys	Glu	Tyr	Phe	Gln	Lys	Phe	Lys	Glu		
			165						170				175				
Asn	Ser	Lys	Glu	Phe	Thr	Asn	Val	Arg	Asn	Lys	Lys	Glu	Ser	Tyr	Glu		
			180				185						190				
Arg	Cys	Ile	Ala	Gln	Ser	Phe	Leu	Lys	Asp	Glu	Leu	Lys	Leu	Ile	Phe		
		195				200						205					
Lys	Lys	Gln	Arg	Glu	Phe	Gly	Phe	Ser	Phe	Ser	Lys	Lys	Phe	Glu	Glu		
		210				215						220					
Glu	Val	Leu	Ser	Val	Ala	Phe	Tyr	Lys	Arg	Ala	Leu	Lys	Asp	Phe	Ser		
		225			230				235						240		
His	Leu	Val	Gly	Asn	Cys	Ser	Phe	Phe	Thr	Asp	Glu	Lys	Arg	Ala	Pro		
			245						250				255				
Lys	Asn	Ser	Pro	Leu	Ala	Phe	Met	Phe	Val	Ala	Leu	Thr	Arg	Ile	Ile		
			260				265						270				
Asn	Leu	Leu	Asn	Asn	Leu	Lys	Asn	Thr	Glu	Gly	Ile	Leu	Tyr	Thr	Lys		
		275				280						285					
Asp	Asp	Leu	Asn	Ala	Leu	Leu	Asn	Glu	Val	Leu	Lys	Asn	Gly	Thr	Leu		
		290				295				300							
Thr	Tyr	Lys	Gln	Thr	Lys	Lys	Leu	Leu	Gly	Leu	Ser	Asp	Asp	Tyr	Glu		
		305			310				315						320		
Phe	Lys	Gly	Glu	Lys	Gly	Thr	Tyr	Phe	Ile	Glu	Phe	Lys	Lys	Tyr	Lys		
			325						330				335				
Glu	Phe	Ile	Lys	Ala	Leu	Gly	Asp	His	Ser	Leu	Ser	Gln	Asp	Asp	Leu		
		340						345				350					
Asn	Glu	Ile	Ala	Lys	Asp	Ile	Thr	Leu	Ile	Lys	Asp	Glu	Ile	Lys	Leu		
		355				360						365					
Lys	Lys	Ala	Leu	Ala	Lys	Tyr	Asp	Leu	Asn	Gln	Asn	Gln	Ile	Asp	Ser		
		370				375				380							
Leu	Ser	Lys	Leu	Glu	Phe	Lys	Asp	His	Leu	Asn	Ile	Ser	Phe	Lys	Ala		
		385			390				395						400		
Leu	Lys	Leu	Ile	Thr	Pro	Leu	Met	Leu	Glu	Gly	Lys	Lys	Tyr	Asp	Glu		
			405						410				415				
Ala	Cys	Asn	Glu	Leu	Asn	Leu	Lys	Val	Ala	Ile	Asn	Glu	Asp	Lys	Lys		
		420						425				430					
Asp	Phe	Leu	Pro	Ala	Phe	Asn	Glu	Thr	Tyr	Tyr	Lys	Asp	Glu	Val	Thr		
		435				440						445					

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Asn Pro Val Val Leu Arg	Ala Ile Lys Glu Tyr Arg Lys Ile Leu Asn
450	455 460
Ala Leu Leu Lys Lys Tyr Gly Lys Val His Lys Ile Asn Ile Glu Leu	
465	470 475 480
Ala Arg Glu Val Gly Lys Asn His Ser Gln Arg Ala Lys Ile Glu Lys	
	485 490 495
Glu Gln Asn Glu Asn Tyr Lys Ala Lys Lys Asp Ala Glu Ile Glu Cys	
	500 505 510
Glu Lys Leu Gly Leu Lys Ile Asn Ser Lys Asn Ile Leu Lys Leu Arg	
	515 520 525
Leu Phe Lys Glu Gln Lys Glu Phe Cys Ala Tyr Ser Gly Glu Lys Ile	
	530 535 540
Lys Leu Ser Asp Leu Gln Asp Glu Lys Met Leu Glu Ile Asp His Ile	
545	550 555 560
Tyr Pro Tyr Ser Arg Ser Phe Asp Asp Ser Tyr Met Asn Lys Val Leu	
	565 570 575
Val Phe Thr Lys Gln Asn Gln Glu Lys Leu Asn Lys Thr Pro Phe Glu	
	580 585 590
Ala Phe Gly Asn Asp Ser Thr Lys Trp Gln Lys Ile Glu Val Leu Ala	
	595 600 605
Lys Asn Leu Pro Glu Lys Lys Gln Lys Arg Ile Leu Asp Lys Asn Tyr	
	610 615 620
Lys Asp Lys Glu Gln Lys Asp Phe Lys Asp Arg Asn Leu Asn Asp Thr	
625	630 635 640
Arg Tyr Ile Ala Arg Leu Val Leu Asn Tyr Thr Lys Asp Tyr Leu Asp	
	645 650 655
Phe Leu Pro Leu Ser Asp Asp Glu Asn Thr Lys Leu Asn Asp Thr Gln	
	660 665 670
Lys Gly Ser Lys Val His Val Glu Ala Lys Ser Gly Met Leu Thr Ser	
	675 680 685
Ala Leu Arg His Thr Trp Gly Phe Ser Ala Lys Asp Arg Asn Asn His	
	690 695 700
Leu His His Ala Ile Asp Ala Ala Ile Ile Ala Tyr Ala Asn Asn Ser	
705	710 715 720
Ile Val Lys Ala Phe Ser Asp Phe Lys Lys Glu Gln Glu Ser Asn Ser	
	725 730 735
Val Glu Leu Tyr Ala Lys Lys Ile Ser Glu Leu Asp Tyr Lys Asn Lys	
	740 745 750
Arg Lys Phe Phe Glu Pro Phe Ser Gly Phe Arg Gln Lys Val Leu Asp	
	755 760 765
Lys Ile Asp Glu Ile Phe Val Ser Lys Pro Glu Arg Lys Lys Pro Ser	
	770 775 780
Gly Ala Leu His Glu Glu Thr Phe Arg Lys Glu Glu Glu Phe Tyr Gln	
785	790 795 800
Ser Tyr Gly Gly Lys Glu Gly Val Leu Lys Ala Leu Glu Leu Gly Lys	
	805 810 815
Ile Arg Lys Val Asn Gly Lys Ile Val Lys Asn Gly Asp Met Phe Arg	
	820 825 830
Val Asp Ile Phe Lys His Lys Lys Thr Asn Lys Phe Tyr Ala Val Pro	
	835 840 845
Ile Tyr Thr Met Asp Phe Ala Leu Lys Val Leu Pro Asn Lys Ala Val	
850	855 860

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Ala Arg Ser Lys Lys Gly Glu Ile Lys Asp Trp Ile Leu Met Asp Glu
865 870 875 880

Asn Tyr Glu Phe Cys Phe Ser Leu Tyr Lys Asp Ser Leu Ile Leu Ile
885 890 895

Gln Thr Lys Asp Met Gln Glu Pro Glu Leu Val Tyr Phe Asn Ala Phe
900 905 910

Thr Ser Ser Thr Val Ser Leu Ile Val Ser Lys His Asp Asn Lys Phe
915 920 925

Glu Thr Leu Ser Lys Asn Gln Lys Ile Leu Phe Lys Asn Ala Asn Glu
930 935 940

Lys Glu Val Ile Ala Lys Ser Ile Gly Ile Gln Asn Leu Lys Val Phe
945 950 955 960

Glu Lys Tyr Ile Val Ser Ala Leu Gly Glu Val Thr Lys Ala Glu Phe
965 970 975

Arg Gln Arg Glu Asp Phe Lys Lys
980

<210> SEQ ID NO 7
 <211> LENGTH: 1334
 <212> TYPE: PRT
 <213> ORGANISM: Listeria monocytogenes

<400> SEQUENCE: 7

Met Lys Asn Pro Tyr Thr Ile Gly Leu Asp Ile Gly Thr Asn Ser Val
1 5 10 15

Gly Trp Ala Val Leu Thr Asn Gln Tyr Asp Leu Val Lys Arg Lys Met
20 25 30

Lys Val Ala Gly Asn Ser Asp Lys Lys Gln Ile Lys Lys Asn Phe Trp
35 40 45

Gly Val Arg Leu Phe Asp Asp Gly Gln Thr Ala Val Asp Arg Arg Met
50 55 60

Asn Arg Thr Ala Arg Arg Arg Ile Glu Arg Arg Arg Asn Arg Ile Ser
65 70 75 80

Tyr Leu Gln Glu Ile Phe Ala Val Glu Met Ala Asn Ile Asp Ala Asn
85 90 95

Phe Phe Cys Arg Leu Asn Asp Ser Phe Tyr Val Asp Ser Glu Lys Arg
100 105 110

Asn Ser Arg His Pro Phe Phe Ala Thr Ile Glu Glu Glu Val Ala Tyr
115 120 125

His Lys Asn Tyr Arg Thr Ile Tyr His Leu Arg Glu Glu Leu Val Asn
130 135 140

Ser Ser Glu Lys Ala Asp Leu Arg Leu Val Tyr Leu Ala Leu Ala His
145 150 155 160

Ile Ile Lys Tyr Arg Gly Asn Phe Leu Ile Glu Gly Ala Leu Asp Thr
165 170 175

Lys Asn Thr Ser Val Asp Gly Val Tyr Glu Gln Phe Ile Gln Thr Tyr
180 185 190

Asn Gln Val Phe Met Ser Asn Ile Glu Glu Gly Thr Leu Ala Lys Val
195 200 205

Glu Glu Asn Ile Glu Val Ala Asn Ile Leu Ala Gly Lys Phe Thr Arg
210 215 220

Arg Glu Lys Phe Glu Arg Ile Leu Gln Leu Tyr Pro Gly Glu Lys Ser
225 230 235 240

Thr Gly Met Phe Ala Gln Phe Ile Ser Leu Ile Val Gly Ser Lys Gly
245 250 255

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Asn	Phe	Gln	Lys	Val	Phe	Asp	Leu	Ile	Glu	Lys	Thr	Asp	Ile	Glu	Cys
		260						265					270		
Ala	Lys	Asp	Ser	Tyr	Glu	Glu	Asp	Leu	Glu	Ala	Leu	Leu	Ala	Ile	Ile
		275						280					285		
Gly	Asp	Glu	Tyr	Ala	Glu	Leu	Phe	Val	Ala	Ala	Lys	Asn	Thr	Tyr	Asn
		290						295					300		
Ala	Val	Val	Leu	Ser	Ser	Ile	Ile	Thr	Val	Thr	Ala	Thr	Glu	Thr	Asn
		305						310					315		320
Ala	Lys	Leu	Ser	Ala	Ser	Met	Ile	Glu	Arg	Phe	Asp	Ala	His	Glu	Lys
				325						330				335	
Asp	Leu	Gly	Glu	Leu	Lys	Ala	Phe	Ile	Lys	Leu	His	Leu	Pro	Lys	Gln
			340						345					350	
Tyr	Gln	Glu	Ile	Phe	Asn	Asn	Ala	Ala	Ile	Asp	Gly	Tyr	Ala	Gly	Tyr
			355						360				365		
Ile	Asp	Gly	Lys	Thr	Lys	Gln	Val	Asp	Phe	Tyr	Lys	Tyr	Leu	Lys	Thr
		370						375					380		
Ile	Leu	Glu	Asn	Ile	Glu	Gly	Ala	Asp	Tyr	Phe	Ile	Ala	Lys	Ile	Glu
		385						390					395		400
Glu	Glu	Asn	Phe	Leu	Arg	Lys	Gln	Arg	Thr	Phe	Asp	Asn	Gly	Ala	Ile
				405						410					415
Pro	His	Gln	Leu	His	Leu	Glu	Glu	Leu	Glu	Ala	Ile	Ile	His	Gln	Gln
				420						425				430	
Ala	Lys	Tyr	Tyr	Pro	Phe	Leu	Arg	Glu	Asp	Tyr	Glu	Lys	Ile	Lys	Ser
				435						440			445		
Leu	Val	Thr	Phe	Arg	Ile	Pro	Tyr	Phe	Val	Gly	Pro	Leu	Ala	Lys	Gly
		450						455				460			
Gln	Ser	Glu	Phe	Ala	Trp	Leu	Thr	Arg	Lys	Ala	Asp	Gly	Glu	Ile	Arg
		465						470				475			480
Pro	Trp	Asn	Ile	Glu	Glu	Lys	Val	Asp	Phe	Gly	Lys	Ser	Ala	Val	Asp
				485						490				495	
Phe	Ile	Glu	Lys	Met	Thr	Asn	Lys	Asp	Thr	Tyr	Leu	Pro	Lys	Glu	Asn
				500					505					510	
Val	Leu	Pro	Lys	His	Ser	Leu	Cys	Tyr	Gln	Lys	Tyr	Met	Val	Tyr	Asn
				515					520				525		
Glu	Leu	Thr	Lys	Val	Arg	Tyr	Ile	Asp	Asp	Gln	Gly	Lys	Thr	Asn	Tyr
		530						535				540			
Phe	Ser	Gly	Gln	Glu	Lys	Gln	Gln	Ile	Phe	Asn	Asp	Leu	Phe	Lys	Gln
		545						550				555			560
Lys	Arg	Lys	Val	Lys	Lys	Lys	Asp	Leu	Glu	Leu	Phe	Leu	Arg	Asn	Ile
				565						570				575	
Asn	His	Ile	Glu	Ser	Pro	Thr	Ile	Glu	Gly	Leu	Glu	Asp	Ser	Phe	Asn
				580					585					590	
Ala	Ser	Tyr	Ala	Thr	Tyr	His	Asp	Leu	Leu	Lys	Val	Gly	Met	Lys	Gln
				595					600				605		
Glu	Ile	Leu	Asp	Asn	Pro	Leu	Asn	Thr	Glu	Met	Leu	Glu	Asp	Ile	Val
		610						615				620			
Lys	Ile	Leu	Thr	Val	Phe	Glu	Asp	Lys	Pro	Met	Ile	Lys	Glu	Gln	Leu
		625						630				635			640
Gln	Gln	Phe	Ser	Asp	Val	Leu	Asp	Gly	Gly	Val	Leu	Lys	Lys	Leu	Glu
				645						650				655	
Arg	Arg	His	Tyr	Thr	Gly	Trp	Gly	Arg	Leu	Ser	Ala	Lys	Leu	Leu	Val
				660					665					670	

Gly	Ile	Arg	Glu	Lys	Gln	Ser	His	Leu	Thr	Ile	Leu	Asp	Tyr	Leu	Met	
675																
Asn	Asp	Asp	Gly	Leu	Asn	Arg	Asn	Leu	Met	Gln	Leu	Ile	Asn	Asp	Ser	
690																
Asn	Leu	Ser	Phe	Lys	Ser	Ile	Ile	Glu	Lys	Glu	Gln	Val	Ser	Thr	Thr	
705																
					710							715				
Asp	Lys	Asp	Leu	Gln	Ser	Ile	Val	Ala	Asp	Leu	Ala	Gly	Ser	Pro	Ala	
725																
					725							730				
Ile	Lys	Lys	Gly	Ile	Leu	Gln	Ser	Leu	Lys	Ile	Val	Asp	Glu	Leu	Val	
740																
					740							745				
Ser	Ile	Met	Gly	Tyr	Pro	Pro	Gln	Thr	Ile	Val	Val	Glu	Met	Ala	Arg	
755																
					755							760				
Glu	Asn	Gln	Thr	Thr	Gly	Lys	Gly	Lys	Asn	Asn	Ser	Lys	Pro	Arg	Tyr	
770																
					770							775				
Lys	Ser	Leu	Glu	Lys	Ala	Ile	Lys	Glu	Phe	Gly	Ser	Lys	Ile	Leu	Lys	
785																
					785							790				
Glu	His	Pro	Thr	Asp	Asn	Gln	Glu	Leu	Lys	Asn	Asn	Arg	Leu	Tyr	Leu	
805																
					805							810				
Tyr	Tyr	Leu	Gln	Asn	Gly	Lys	Asp	Met	Tyr	Thr	Gly	Gln	Glu	Leu	Asp	
820																
					820							825				
Ile	His	Asn	Leu	Ser	Asn	Tyr	Asp	Ile	Asp	His	Ile	Val	Pro	Gln	Ser	
835																
					835							840				
Phe	Ile	Thr	Asp	Asn	Ser	Ile	Asp	Asn	Leu	Val	Leu	Thr	Ser	Ser	Ala	
850																
					850							855				
Gly	Asn	Arg	Glu	Lys	Gly	Gly	Asp	Val	Pro	Pro	Leu	Glu	Ile	Val	Arg	
865																
					865							870				
Lys	Arg	Lys	Val	Phe	Trp	Glu	Lys	Leu	Tyr	Gln	Gly	Asn	Leu	Met	Ser	
885																
					885							890				
Lys	Arg	Lys	Phe	Asp	Tyr	Leu	Thr	Lys	Ala	Glu	Arg	Gly	Gly	Leu	Thr	
900																
					900							905				
Asp	Ala	Asp	Lys	Ala	Arg	Phe	Ile	His	Arg	Gln	Leu	Val	Glu	Thr	Arg	
915																
					915							920				
Gln	Ile	Thr	Lys	Asn	Val	Ala	Asn	Ile	Leu	His	Gln	Arg	Phe	Asn	Asn	
930																
					930							935				
Glu	Thr	Asp	Asn	His	Gly	Asn	Thr	Met	Glu	Gln	Val	Arg	Ile	Val	Thr	
945																
					945							950				
Leu	Lys	Ser	Ala	Leu	Val	Ser	Gln	Phe	Arg	Lys	Gln	Phe	Gln	Leu	Tyr	
965																
					965							970				
Lys	Val	Arg	Glu	Val	Asn	Asp	Tyr	His	His	Ala	His	Asp	Ala	Tyr	Leu	
980																
					980							985				
Asn	Gly	Val	Val	Ala	Asn	Thr	Leu	Leu	Lys	Val	Tyr	Pro	Gln	Leu	Glu	
995																
					995							1000				
Pro	Glu	Phe	Val	Tyr	Gly	Glu	Tyr	His	Gln	Phe	Asp	Trp	Phe	Lys		
1010																
					1010							1015				
Ala	Asn	Lys	Ala	Thr	Ala	Lys	Lys	Gln	Phe	Tyr	Thr	Asn	Ile	Met		
1025																
					1025							1030				
Leu	Phe	Phe	Gly	Gln	Lys	Glu	Arg	Ile	Ile	Asp	Glu	Asn	Gly	Glu		
1040																
					1040							1045				
Ile	Leu	Trp	Asp	Lys	Lys	Tyr	Leu	Glu	Thr	Ile	Lys	Lys	Val	Leu		
1055																
					1055							1060				

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1085	1090	1095
Lys Leu Ile Pro Arg Lys Glu Asn Trp Asp Pro Met Lys Tyr Gly		
1100	1105	1110
Gly Leu Asp Ser Pro Asn Met Ala Tyr Ala Val Ile Ile Glu His		
1115	1120	1125
Ala Lys Gly Lys Lys Lys Leu Ile Phe Glu Lys Lys Ile Ile Arg		
1130	1135	1140
Ile Thr Ile Met Glu Arg Lys Met Phe Glu Lys Asp Glu Glu Ala		
1145	1150	1155
Phe Leu Glu Glu Lys Gly Tyr Arg His Pro Lys Val Leu Thr Lys		
1160	1165	1170
Leu Pro Lys Tyr Thr Leu Tyr Glu Cys Glu Lys Gly Arg Arg Arg		
1175	1180	1185
Met Leu Ala Ser Ala Asn Glu Ala Gln Lys Gly Asn Gln Leu Val		
1190	1195	1200
Leu Ser Asn His Leu Val Ser Leu Leu Tyr His Ala Lys Asn Cys		
1205	1210	1215
Glu Ala Ser Asp Gly Lys Ser Leu Lys Tyr Ile Glu Ala His Arg		
1220	1225	1230
Glu Thr Phe Ser Glu Leu Leu Ala Gln Val Ser Glu Phe Ala Thr		
1235	1240	1245
Arg Tyr Thr Leu Ala Asp Ala Asn Leu Ser Lys Ile Asn Asn Leu		
1250	1255	1260
Phe Glu Gln Asn Lys Glu Gly Asp Ile Gln Ala Ile Ala Gln Ser		
1265	1270	1275
Phe Val Asp Leu Met Ala Phe Asn Ala Met Gly Ala Pro Ala Ser		
1280	1285	1290
Phe Lys Phe Phe Glu Ala Thr Ile Asp Arg Lys Arg Tyr Thr Asn		
1295	1300	1305
Leu Lys Glu Leu Leu Ser Ser Thr Ile Ile Tyr Gln Ser Ile Thr		
1310	1315	1320
Gly Leu Tyr Glu Ser Arg Lys Arg Leu Asp Asp		
1325	1330	

<210> SEQ ID NO 8

<211> LENGTH: 1346

<212> TYPE: PRT

<213> ORGANISM: Lactobacillus salivarius

<400> SEQUENCE: 8

Met	Glu	Arg	Tyr	His	Ile	Gly	Leu	Asp	Ile	Gly	Thr	Ser	Ser	Val	Gly
1				5					10					15	
Trp	Ala	Val	Ile	Gly	Asp	Asp	Phe	Lys	Ile	Lys	Arg	Lys	Lys	Gly	Lys
		20					25						30		
Asn	Leu	Ile	Gly	Val	Arg	Leu	Phe	Lys	Glu	Gly	Asp	Thr	Ala	Ala	Glu
		35				40						45			
Arg	Arg	Gly	Phe	Arg	Thr	Gln	Arg	Arg	Arg	Leu	Asn	Arg	Arg	Lys	Trp
		50			55					60					
Arg	Leu	Lys	Leu	Leu	Glu	Glu	Ile	Phe	Asp	Pro	Tyr	Met	Ala	Glu	Val
	65				70				75					80	
Asp	Glu	Tyr	Phe	Phe	Ala	Arg	Leu	Lys	Glu	Ser	Asn	Leu	Ser	Pro	Lys
			85					90					95		
Asp	Ser	Asn	Lys	Lys	Tyr	Leu	Gly	Ser	Leu	Leu	Phe	Pro	Asp	Val	Ser
			100					105					110		

Asp	Ser	Asn	Phe	Tyr	Asp	Lys	Tyr	Pro	Thr	Ile	Tyr	His	Leu	Arg	Arg
115						120				125					
Asp	Leu	Met	Glu	Lys	Asp	Lys	Lys	Phe	Asp	Leu	Arg	Glu	Ile	Tyr	Leu
130						135				140					
Ala	Ile	His	His	Ile	Val	Lys	Tyr	Arg	Gly	Asn	Phe	Leu	Glu	Lys	Val
145				150						155		160			
Pro	Ala	Lys	Asn	Tyr	Lys	Asn	Ser	Gly	Ala	Ser	Ile	Gly	Phe	Leu	Leu
				165				170						175	
Glu	Glu	Val	Asn	Ser	Leu	Tyr	Lys	Asp	Ile	Ile	Gly	Asp	Glu	Ser	Val
		180						185				190			
Ala	Ile	Leu	Asn	Ser	Gly	Lys	Phe	Glu	Asp	Val	Glu	Lys	Ile	Ile	Leu
195						200				205					
Asp	Glu	Glu	Thr	Arg	Asn	Leu	Asp	Lys	Gln	Lys	Ser	Val	Gly	Lys	Leu
210						215				220					
Leu	Val	Glu	Asp	Lys	Lys	Lys	Lys	Asn	Ile	Val	Thr	Ala	Phe	Ser	Lys
225				230						235		240			
Ala	Ile	Leu	Gly	Tyr	Lys	Phe	Asn	Ile	Glu	Asp	Leu	Leu	Leu	Ile	Lys
				245				250						255	
Asp	Ser	Thr	Leu	Ile	Glu	Ser	Asp	Lys	Asn	Lys	Lys	Ile	Thr	Phe	Asn
		260						265				270			
Asp	Glu	Asn	Ile	Asp	Asp	Ile	Phe	Asn	Glu	Leu	Ser	His	Ser	Leu	Asp
275						280				285					
Asp	Asn	Gln	Met	Asp	Leu	Leu	Thr	Lys	Thr	Arg	Glu	Ile	Tyr	Phe	Lys
290						295				300					
Phe	Lys	Leu	Asn	Met	Ile	Val	Pro	Thr	Gly	Tyr	Thr	Leu	Ser	Glu	Ser
305				310						315		320			
Met	Ile	Glu	Lys	Tyr	Glu	Met	His	Lys	Ala	His	Leu	Lys	Met	Tyr	Lys
				325				330						335	
Glu	Phe	Ile	Asn	Thr	Leu	Asn	Ala	Lys	Asp	Arg	Lys	Ile	Leu	Lys	Asn
		340				345				350					
Ala	Tyr	Ser	Asp	Tyr	Ile	Asn	Asn	Glu	Lys	Ala	Lys	Ala	Ala	Asn	Ala
355						360				365					
Gln	Glu	Asn	Phe	Tyr	Lys	Thr	Val	Lys	Lys	Thr	Ile	Lys	Asp	Asn	Asn
370						375				380					
Ser	Asp	Thr	Ala	Lys	Lys	Ile	Ile	Gly	Leu	Ile	Asp	Glu	Gly	Asn	Phe
385				390						395		400			
Met	Pro	Lys	Gln	Arg	Thr	Gly	Glu	Asn	Ser	Val	Ile	Pro	His	Gln	Ile
				405				410						415	
His	Gln	Ile	Glu	Leu	Asp	Arg	Ile	Ile	Glu	Asn	Gln	Ala	Lys	Tyr	Tyr
		420						425				430			
Pro	Trp	Leu	Ala	Glu	Glu	Asn	Pro	Val	Glu	Lys	Asn	Arg	Lys	Phe	Ala
435						440				445					
Lys	Tyr	Lys	Leu	Asp	Glu	Leu	Val	Thr	Phe	Arg	Val	Pro	Tyr	Tyr	Val
450						455				460					
Gly	Pro	Leu	Val	Asp	Lys	Thr	Glu	Ser	Asn	Lys	Asn	Glu	Lys	Glu	Thr
465				470						475		480			
Lys	Phe	Ala	Trp	Met	Val	Arg	Lys	Ala	Lys	Gly	Thr	Ile	Thr	Pro	Trp

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530					535					540					
Asn 545	Asn	Ile	Lys	Val	Asn 550	Glu	Lys	Lys	Leu	Asp 555	Ile	Glu	Gln	Lys	Gln 560
Asp	Ile	Tyr	Leu	Asn 565	Leu	Phe	Thr	Thr	Ala 570	Lys	Asn	Val	Thr	Lys 575	Lys
Ser	Leu	Ala	Thr	Tyr 580	Leu	Asn	Cys	Ser	Thr 585	Asp	Ser	Ile	Ser	Gly	Leu
Ser	Asp	Gly	Glu	Lys 595	Phe	Asn	Ser	Ser	Leu	Ser	Ser	Tyr 605	Ile	Asp	Leu
Lys	Ala 610	Ile	Leu	Gly	Asn 615	Ile	Val	Asp	Asp	Cys	Asn 620	Lys	Asn	Glu	Asn
Leu 625	Glu	Lys	Ile	Ile	Glu 630	Tyr	Ser	Thr	Val	Phe 635	Glu	Asp	Gly	Asn	Ile 640
Tyr	Lys	Glu	Lys	Leu 645	Ser	Glu	Ile	Ser	Trp 650	Leu	Thr	Asp	Glu	Gln	Ile
Glu	Lys	Leu	Ser	Asn 660	Ile	His	Phe	Lys 665	Gly	Trp	Gly	Arg	Leu	Ser	Lys
Lys	Leu	Leu	Thr	Gln 675	Ile	Thr	Asn 680	Glu	Asn	Gly	Glu	Arg 685	Ile	Ile	Asp
Ala	Leu 690	Trp	Asn	Thr	Ser	Asn 695	Asn	Phe	Met	Gln	Ile 700	Ile	Asn	Asp	Glu
Ser 705	Ile	Gln	Ala	Lys	Leu 710	Ala	Glu	Ile	Asn 715	Ser	Glu	Tyr	Thr	Asp	Thr
Asp	Lys	Tyr	Asp	Leu 725	Glu	Asp	Ile	Leu	Asp 730	Lys	Val	Tyr	Thr	Ser	Pro
Gln	Asn	Lys	Lys	Ala 740	Ile	Arg	Gln	Val 745	Met	Lys	Val	Val	Glu	Asp	Ile
Glu	Lys	Ala	Met	Lys 755	Cys	Glu	Pro 760	Thr	Ser	Ile	Ala	Ile 765	Glu	Phe	Thr
Arg 770	Glu	Lys	Arg	Lys	Ser	Lys 775	Leu	Thr	Asn	Thr	Arg 780	Tyr	Lys	Lys	Ile
Ser 785	Glu	Thr	Tyr	Glu	Glu 790	Ile	Thr	Asp	Glu	Leu 795	Ile	Ser	Glu	Tyr	Glu 800
Leu	Gly	Lys	Leu	Gln 805	Ser	Glu	Leu	Asp	Ser 810	Lys	Ala	Asn	Asn	Met	Arg
Asp	Arg	Tyr	Tyr	Leu 820	Tyr	Phe	Met	Gln 825	Leu	Gly	Arg	Asp 830	Met	Tyr	Thr
Gly	Glu	Lys	Ile	Asn 835	Ile	Asp	Glu 840	Leu	His	Gln	Lys	Tyr 845	Asp	Ile	Asp
His 850	Ile	Leu	Pro	Gln	Ser	Phe 855	Ile	Lys	Asp	Asp	Ser 860	Leu	Asn	Asn	Arg
Val 865	Leu	Thr	Ser	Lys	Gly 870	Val	Asn	Ile	Lys	Glu 875	Lys	Ser	Asp	Lys	Thr 880
Ala	Ala	Asp	Leu	Tyr 885	Ala	Ala	Lys	Met	Gly 890	Asp	Phe	Trp	Arg	Lys	Leu
Arg	Lys	Gln	Gly	Leu 900	Met	Thr	Glu	Gln 905	Lys	Tyr	Lys	Asn 910	Leu	Leu	Thr
Arg	Thr	Asp	Ser	Ile 915	Asn	Lys	Tyr 920	Thr	Lys	Gln	Ser	Phe 925	Ile	Lys	Arg
Gln 930	Leu	Val	Glu	Thr	Ser	Gln 935	Val	Val	Lys	Leu	Ala 940	Ala	Asn	Ile	Leu
Gln 945	Asp	Lys	Tyr	Ser	Asn 950	Thr	Lys	Ile	Ile	Glu 955	Val	Arg	Ala	Arg	Leu 960

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Asn Ser Asp Leu Arg Lys Glu Tyr Glu Leu Ile Lys Asn Arg Glu Val	965	970	975
Asn Asp Tyr His His Ala Ile Asp Gly Tyr Leu Thr Thr Phe Val Gly	980	985	990
Gln Tyr Leu Tyr Lys Val Tyr Pro Lys Leu Arg Ser Tyr Phe Val Tyr	995	1000	1005
Asp Asp Phe Lys Lys Leu Asp Ser Asn Tyr Leu Lys His Met Asp	1010	1015	1020
Lys Phe Asn Phe Ile Trp Lys Leu Glu Asp Lys Lys Ala Glu Asp	1025	1030	1035
Val Tyr Asp Asn Val Asn Asn Glu Phe Ile Leu Asn Val Pro Lys	1040	1045	1050
Met Lys Asp Tyr Ile Gln Lys Ile Tyr Asn Tyr Lys Tyr Met Leu	1055	1060	1065
Val Ser Lys Glu Val Thr Thr Glu Ser Ser Ala Phe Tyr Lys Glu	1070	1075	1080
Thr Lys Tyr Lys Ala Gly Gly Met Asn Leu Ile Pro Ile Lys Gln	1085	1090	1095
Asn Lys Pro Ile Asn Ile Tyr Gly Gly Tyr Lys Glu Lys Arg Asn	1100	1105	1110
Ser Tyr Met Met Leu Val Lys Ile Lys Lys Lys Lys Glu Thr Ile	1115	1120	1125
Tyr Lys Ile Val Gly Ile Pro Arg Leu Leu Ser Asp Glu Leu Asp	1130	1135	1140
Arg Leu Asn Thr Met Ser Glu Lys Gln Leu Leu Leu Glu Thr Ile	1145	1150	1155
Ala Arg Thr Ser Leu Ser Lys Thr Glu Gln Asn Phe Lys Ile Ile	1160	1165	1170
Leu Asp Lys Val Tyr Tyr Gly Gln Leu Val Ile Asp Gly Asn Gln	1175	1180	1185
Lys Tyr Thr Leu Gly Ser Val Ile Tyr Lys His Asn Ala Met Gln	1190	1195	1200
Leu His Leu Ser Lys Gln Ser Leu Asp Thr Leu Ala Val Gly Lys	1205	1210	1215
Asn Lys Ser Arg Glu Val Ser Asp Glu Glu Leu Met Ala Val Tyr	1220	1225	1230
Lys Glu Ile Leu Leu Val Val Asn Lys Tyr Phe Glu Leu Tyr Asp	1235	1240	1245
Ile Ser Lys Phe Arg Gln Lys Leu Asn Asn Gly Leu Glu Ile Phe	1250	1255	1260
Lys Glu Leu Pro Ile Tyr Asn Ile Tyr Glu Asn Asn Lys Ile Lys	1265	1270	1275
Lys Ile Gly Lys Phe Glu Val Leu Asn Arg Ile Leu Ile Gly Leu	1280	1285	1290
His Ala Asn Ala Ala Arg Ala Asp Leu Ser Val Leu Gly Phe Lys	1295	1300	1305
Asp Leu Gly Lys Leu Gln Val Asn Gly Gly Ile Lys Leu Ser Pro	1310	1315	1320
Asp Ala Lys Leu Ile Tyr Gln Ser Pro Thr Gly Ile Phe Ser Arg	1325	1330	1335
Ala Val Arg Val Lys Asp Leu Gly	1340	1345	

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<210> SEQ ID NO 9
<211> LENGTH: 1372
<212> TYPE: PRT
<213> ORGANISM: Legionella pneumophila

<400> SEQUENCE: 9

Met Glu Ser Ser Gln Ile Leu Ser Pro Ile Gly Ile Asp Leu Gly Gly
1      5      10      15
Lys Phe Thr Gly Val Cys Leu Ser His Leu Glu Ala Phe Ala Glu Leu
20     25     30
Pro Asn His Ala Asn Thr Lys Tyr Ser Val Ile Leu Ile Asp His Asn
35     40     45
Asn Phe Gln Leu Ser Gln Ala Gln Arg Arg Ala Thr Arg His Arg Val
50     55     60
Arg Asn Lys Lys Arg Asn Gln Phe Val Lys Arg Val Ala Leu Gln Leu
65     70     75     80
Phe Gln His Ile Leu Ser Arg Asp Leu Asn Ala Lys Glu Glu Thr Ala
85     90     95
Leu Cys His Tyr Leu Asn Asn Arg Gly Tyr Thr Tyr Val Asp Thr Asp
100    105    110
Leu Asp Glu Tyr Ile Lys Asp Glu Thr Thr Ile Asn Leu Leu Lys Glu
115    120    125
Leu Leu Pro Ser Glu Ser Glu His Asn Phe Ile Asp Trp Phe Leu Gln
130    135    140
Lys Met Gln Ser Ser Glu Phe Arg Lys Ile Leu Val Ser Lys Val Glu
145    150    155    160
Glu Lys Lys Asp Asp Lys Glu Leu Lys Asn Ala Val Lys Asn Ile Lys
165    170    175
Asn Phe Ile Thr Gly Phe Glu Lys Asn Ser Val Glu Gly His Arg His
180    185    190
Arg Lys Val Tyr Phe Glu Asn Ile Lys Ser Asp Ile Thr Lys Asp Asn
195    200    205
Gln Leu Asp Ser Ile Lys Lys Lys Ile Pro Ser Val Cys Leu Ser Asn
210    215    220
Leu Leu Gly His Leu Ser Asn Leu Gln Trp Lys Asn Leu His Arg Tyr
225    230    235    240
Leu Ala Lys Asn Pro Lys Gln Phe Asp Glu Gln Thr Phe Gly Asn Glu
245    250    255
Phe Leu Arg Met Leu Lys Asn Phe Arg His Leu Lys Gly Ser Gln Glu
260    265    270
Ser Leu Ala Val Arg Asn Leu Ile Gln Gln Leu Glu Gln Ser Gln Asp
275    280    285
Tyr Ile Ser Ile Leu Glu Lys Thr Pro Pro Glu Ile Thr Ile Pro Pro
290    295    300
Tyr Glu Ala Arg Thr Asn Thr Gly Met Glu Lys Asp Gln Ser Leu Leu
305    310    315    320
Leu Asn Pro Glu Lys Leu Asn Asn Leu Tyr Pro Asn Trp Arg Asn Leu
325    330    335
Ile Pro Gly Ile Ile Asp Ala His Pro Phe Leu Glu Lys Asp Leu Glu
340    345    350
His Thr Lys Leu Arg Asp Arg Lys Arg Ile Ile Ser Pro Ser Lys Gln
355    360    365
Asp Glu Lys Arg Asp Ser Tyr Ile Leu Gln Arg Tyr Leu Asp Leu Asn
370    375    380

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Lys Lys Ile Asp Lys Phe Lys Ile Lys Lys Gln Leu Ser Phe Leu Gly
 385 390 395 400
 Gln Gly Lys Gln Leu Pro Ala Asn Leu Ile Glu Thr Gln Lys Glu Met
 405 410 415
 Glu Thr His Phe Asn Ser Ser Leu Val Ser Val Leu Ile Gln Ile Ala
 420 425 430
 Ser Ala Tyr Asn Lys Glu Arg Glu Asp Ala Ala Gln Gly Ile Trp Phe
 435 440 445
 Asp Asn Ala Phe Ser Leu Cys Glu Leu Ser Asn Ile Asn Pro Pro Arg
 450 455 460
 Lys Gln Lys Ile Leu Pro Leu Leu Val Gly Ala Ile Leu Ser Glu Asp
 465 470 475 480
 Phe Ile Asn Asn Lys Asp Lys Trp Ala Lys Phe Lys Ile Phe Trp Asn
 485 490 495
 Thr His Lys Ile Gly Arg Thr Ser Leu Lys Ser Lys Cys Lys Glu Ile
 500 505 510
 Glu Glu Ala Arg Lys Asn Ser Gly Asn Ala Phe Lys Ile Asp Tyr Glu
 515 520 525
 Glu Ala Leu Asn His Pro Glu His Ser Asn Asn Lys Ala Leu Ile Lys
 530 535 540
 Ile Ile Gln Thr Ile Pro Asp Ile Ile Gln Ala Ile Gln Ser His Leu
 545 550 555 560
 Gly His Asn Asp Ser Gln Ala Leu Ile Tyr His Asn Pro Phe Ser Leu
 565 570 575
 Ser Gln Leu Tyr Thr Ile Leu Glu Thr Lys Arg Asp Gly Phe His Lys
 580 585 590
 Asn Cys Val Ala Val Thr Cys Glu Asn Tyr Trp Arg Ser Gln Lys Thr
 595 600 605
 Glu Ile Asp Pro Glu Ile Ser Tyr Ala Ser Arg Leu Pro Ala Asp Ser
 610 615 620
 Val Arg Pro Phe Asp Gly Val Leu Ala Arg Met Met Gln Arg Leu Ala
 625 630 635 640
 Tyr Glu Ile Ala Met Ala Lys Trp Glu Gln Ile Lys His Ile Pro Asp
 645 650 655
 Asn Ser Ser Leu Leu Ile Pro Ile Tyr Leu Glu Gln Asn Arg Phe Glu
 660 665 670
 Phe Glu Glu Ser Phe Lys Lys Ile Lys Gly Ser Ser Ser Asp Lys Thr
 675 680 685
 Leu Glu Gln Ala Ile Glu Lys Gln Asn Ile Gln Trp Glu Glu Lys Phe
 690 695 700
 Gln Arg Ile Ile Asn Ala Ser Met Asn Ile Cys Pro Tyr Lys Gly Ala
 705 710 715 720
 Ser Ile Gly Gly Gln Gly Glu Ile Asp His Ile Tyr Pro Arg Ser Leu
 725 730 735
 Ser Lys Lys His Phe Gly Val Ile Phe Asn Ser Glu Val Asn Leu Ile
 740 745 750
 Tyr Cys Ser Ser Gln Gly Asn Arg Glu Lys Lys Glu Glu His Tyr Leu
 755 760 765
 Leu Glu His Leu Ser Pro Leu Tyr Leu Lys His Gln Phe Gly Thr Asp
 770 775 780
 Asn Val Ser Asp Ile Lys Asn Phe Ile Ser Gln Asn Val Ala Asn Ile
 785 790 795 800

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Lys	Lys	Tyr	Ile	Ser	Phe	His	Leu	Leu	Thr	Pro	Glu	Gln	Gln	Lys	Ala	805	810	815
Ala	Arg	His	Ala	Leu	Phe	Leu	Asp	Tyr	Asp	Asp	Glu	Ala	Phe	Lys	Thr	820	825	830
Ile	Thr	Lys	Phe	Leu	Met	Ser	Gln	Gln	Lys	Ala	Arg	Val	Asn	Gly	Thr	835	840	845
Gln	Lys	Phe	Leu	Gly	Lys	Gln	Ile	Met	Glu	Phe	Leu	Ser	Thr	Leu	Ala	850	855	860
Asp	Ser	Lys	Gln	Leu	Gln	Leu	Glu	Phe	Ser	Ile	Lys	Gln	Ile	Thr	Ala	865	870	875
Glu	Glu	Val	His	Asp	His	Arg	Glu	Leu	Leu	Ser	Lys	Gln	Glu	Pro	Lys	885	890	895
Leu	Val	Lys	Ser	Arg	Gln	Gln	Ser	Phe	Pro	Ser	His	Ala	Ile	Asp	Ala	900	905	910
Thr	Leu	Thr	Met	Ser	Ile	Gly	Leu	Lys	Glu	Phe	Pro	Gln	Phe	Ser	Gln	915	920	925
Glu	Leu	Asp	Asn	Ser	Trp	Phe	Ile	Asn	His	Leu	Met	Pro	Asp	Glu	Val	930	935	940
His	Leu	Asn	Pro	Val	Arg	Ser	Lys	Glu	Lys	Tyr	Asn	Lys	Pro	Asn	Ile	945	950	955
Ser	Ser	Thr	Pro	Leu	Phe	Lys	Asp	Ser	Leu	Tyr	Ala	Glu	Arg	Phe	Ile	965	970	975
Pro	Val	Trp	Val	Lys	Gly	Glu	Thr	Phe	Ala	Ile	Gly	Phe	Ser	Glu	Lys	980	985	990
Asp	Leu	Phe	Glu	Ile	Lys	Pro	Ser	Asn	Lys	Glu	Lys	Leu	Phe	Thr	Leu	995	1000	1005
Leu	Lys	Thr	Tyr	Ser	Thr	Lys	Asn	Pro	Gly	Glu	Ser	Leu	Gln	Glu		1010	1015	1020
Leu	Gln	Ala	Lys	Ser	Lys	Ala	Lys	Trp	Leu	Tyr	Phe	Pro	Ile	Asn		1025	1030	1035
Lys	Thr	Leu	Ala	Leu	Glu	Phe	Leu	His	His	Tyr	Phe	His	Lys	Glu		1040	1045	1050
Ile	Val	Thr	Pro	Asp	Asp	Thr	Thr	Val	Cys	His	Phe	Ile	Asn	Ser		1055	1060	1065
Leu	Arg	Tyr	Tyr	Thr	Lys	Lys	Glu	Ser	Ile	Thr	Val	Lys	Ile	Leu		1070	1075	1080
Lys	Glu	Pro	Met	Pro	Val	Leu	Ser	Val	Lys	Phe	Glu	Ser	Ser	Lys		1085	1090	1095
Lys	Asn	Val	Leu	Gly	Ser	Phe	Lys	His	Thr	Ile	Ala	Leu	Pro	Ala		1100	1105	1110
Thr	Lys	Asp	Trp	Glu	Arg	Leu	Phe	Asn	His	Pro	Asn	Phe	Leu	Ala		1115	1120	1125
Leu	Lys	Ala	Asn	Pro	Ala	Pro	Asn	Pro	Lys	Glu	Phe	Asn	Glu	Phe		1130	1135	1140
Ile	Arg	Lys	Tyr	Phe	Leu	Ser	Asp	Asn	Asn	Pro	Asn	Ser	Asp	Ile		1145	1150	1155
Pro	Asn	Asn	Gly	His	Asn	Ile	Lys	Pro	Gln	Lys	His	Lys	Ala	Val		1160	1165	1170
Arg	Lys	Val	Phe	Ser	Leu	Pro	Val	Ile	Pro	Gly	Asn	Ala	Gly	Thr		1175	1180	1185
Met	Met	Arg	Ile	Arg	Arg	Lys	Asp	Asn	Lys	Gly	Gln	Pro	Leu	Tyr		1190	1195	1200
Gln	Leu	Gln	Thr	Ile	Asp	Asp	Thr	Pro	Ser	Met	Gly	Ile	Gln	Ile				

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1205	1210	1215
Asn Glu Asp Arg Leu Val Lys Gln Glu Val Leu Met Asp Ala Tyr		
1220	1225	1230
Lys Thr Arg Asn Leu Ser Thr Ile Asp Gly Ile Asn Asn Ser Glu		
1235	1240	1245
Gly Gln Ala Tyr Ala Thr Phe Asp Asn Trp Leu Thr Leu Pro Val		
1250	1255	1260
Ser Thr Phe Lys Pro Glu Ile Ile Lys Leu Glu Met Lys Pro His		
1265	1270	1275
Ser Lys Thr Arg Arg Tyr Ile Arg Ile Thr Gln Ser Leu Ala Asp		
1280	1285	1290
Phe Ile Lys Thr Ile Asp Glu Ala Leu Met Ile Lys Pro Ser Asp		
1295	1300	1305
Ser Ile Asp Asp Pro Leu Asn Met Pro Asn Glu Ile Val Cys Lys		
1310	1315	1320
Asn Lys Leu Phe Gly Asn Glu Leu Lys Pro Arg Asp Gly Lys Met		
1325	1330	1335
Lys Ile Val Ser Thr Gly Lys Ile Val Thr Tyr Glu Phe Glu Ser		
1340	1345	1350
Asp Ser Thr Pro Gln Trp Ile Gln Thr Leu Tyr Val Thr Gln Leu		
1355	1360	1365
Lys Lys Gln Pro		
1370		

<210> SEQ ID NO 10

<211> LENGTH: 1632

<212> TYPE: PRT

<213> ORGANISM: Francisella novicida

<400> SEQUENCE: 10

Gly Ser His Met Asn Phe Lys Ile Leu Pro Ile Ala Ile Asp Leu Gly		
1	5	10 15
Val Lys Asn Thr Gly Val Phe Ser Ala Phe Tyr Gln Lys Gly Thr Ser		
20	25	30
Leu Glu Arg Leu Asp Asn Lys Asn Gly Lys Val Tyr Glu Leu Ser Lys		
35	40	45
Asp Ser Tyr Thr Leu Leu Met Asn Asn Arg Thr Ala Arg Arg His Gln		
50	55	60
Arg Arg Gly Ile Asp Arg Lys Gln Leu Val Lys Arg Leu Phe Lys Leu		
65	70	75 80
Ile Trp Thr Glu Gln Leu Asn Leu Glu Trp Asp Lys Asp Thr Gln Gln		
85	90	95
Ala Ile Ser Phe Leu Phe Asn Arg Arg Gly Phe Ser Phe Ile Thr Asp		
100	105	110
Gly Tyr Ser Pro Glu Tyr Leu Asn Ile Val Pro Glu Gln Val Lys Ala		
115	120	125
Ile Leu Met Asp Ile Phe Asp Asp Tyr Asn Gly Glu Asp Asp Leu Asp		
130	135	140
Ser Tyr Leu Lys Leu Ala Thr Glu Gln Glu Ser Lys Ile Ser Glu Ile		
145	150	155 160
Tyr Asn Lys Leu Met Gln Lys Ile Leu Glu Phe Lys Leu Met Lys Leu		
165	170	175
Cys Thr Asp Ile Lys Asp Asp Lys Val Ser Thr Lys Thr Leu Lys Glu		
180	185	190

Ile	Thr	Ser	Tyr	Glu	Phe	Glu	Leu	Leu	Ala	Asp	Tyr	Leu	Ala	Asn	Tyr
195							200					205			
Ser	Glu	Ser	Leu	Lys	Thr	Gln	Lys	Phe	Ser	Tyr	Thr	Asp	Lys	Gln	Gly
210						215					220				
Asn	Leu	Lys	Glu	Leu	Ser	Tyr	Tyr	His	His	Asp	Lys	Tyr	Asn	Ile	Gln
225						230				235					240
Glu	Phe	Leu	Lys	Arg	His	Ala	Thr	Ile	Asn	Asp	Arg	Ile	Leu	Asp	Thr
				245					250					255	
Leu	Leu	Thr	Asp	Asp	Leu	Asp	Ile	Trp	Asn	Phe	Asn	Phe	Glu	Lys	Phe
			260					265					270		
Asp	Phe	Asp	Lys	Asn	Glu	Glu	Lys	Leu	Gln	Asn	Gln	Glu	Asp	Lys	Asp
		275					280					285			
His	Ile	Gln	Ala	His	Leu	His	His	Phe	Val	Phe	Ala	Val	Asn	Lys	Ile
290						295					300				
Lys	Ser	Glu	Met	Ala	Ser	Gly	Gly	Arg	His	Arg	Ser	Gln	Tyr	Phe	Gln
305					310					315					320
Glu	Ile	Thr	Asn	Val	Leu	Asp	Glu	Asn	Asn	His	Gln	Glu	Gly	Tyr	Leu
				325					330					335	
Lys	Asn	Phe	Cys	Glu	Asn	Leu	His	Asn	Lys	Lys	Tyr	Ser	Asn	Leu	Ser
			340					345					350		
Val	Lys	Asn	Leu	Val	Asn	Leu	Ile	Gly	Asn	Leu	Ser	Asn	Leu	Glu	Leu
			355				360					365			
Lys	Pro	Leu	Arg	Lys	Tyr	Phe	Asn	Asp	Lys	Ile	His	Ala	Lys	Ala	Asp
370						375					380				
His	Trp	Asp	Glu	Gln	Lys	Phe	Thr	Glu	Thr	Tyr	Cys	His	Trp	Ile	Leu
385					390					395					400
Gly	Glu	Trp	Arg	Val	Gly	Val	Lys	Asp	Gln	Asp	Lys	Lys	Asp	Gly	Ala
				405					410					415	
Lys	Tyr	Ser	Tyr	Lys	Asp	Leu	Cys	Asn	Glu	Leu	Lys	Gln	Lys	Val	Thr
			420					425					430		
Lys	Ala	Gly	Leu	Val	Asp	Phe	Leu	Leu	Glu	Leu	Asp	Pro	Cys	Arg	Thr
		435				440						445			
Ile	Pro	Pro	Tyr	Leu	Asp	Asn	Asn	Asn	Arg	Lys	Pro	Pro	Lys	Cys	Gln
450						455					460				
Ser	Leu	Ile	Leu	Asn	Pro	Lys	Phe	Leu	Asp	Asn	Gln	Tyr	Pro	Asn	Trp
465					470				475						480
Gln	Gln	Tyr	Leu	Gln	Glu	Leu	Lys	Lys	Leu	Gln	Ser	Ile	Gln	Asn	Tyr
				485					490					495	
Leu	Asp	Ser	Phe	Glu	Thr	Asp	Leu	Lys	Val	Leu	Lys	Ser	Ser	Lys	Asp
			500					505					510		
Gln	Pro	Tyr	Phe	Val	Glu	Tyr	Lys	Ser	Ser	Asn	Gln	Gln	Ile	Ala	Ser
		515					520					525			
Gly	Gln	Arg	Asp	Tyr	Lys	Asp	Leu	Asp	Ala	Arg	Ile	Leu	Gln	Phe	Ile
530						535					540				
Phe	Asp	Arg													

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610	615	620
Arg Asp Ser Arg Leu Tyr Ile Met Pro Glu Tyr Arg Tyr Asp Lys Lys		
625	630	635 640
Leu His Lys Tyr Asn Asn Thr Gly Arg Phe Asp Asp Asp Asn Gln Leu		
645	650	655
Leu Thr Tyr Cys Asn His Lys Pro Arg Gln Lys Arg Tyr Gln Leu Leu		
660	665	670
Asn Asp Leu Ala Gly Val Leu Gln Val Ser Pro Asn Phe Leu Lys Asp		
675	680	685
Lys Ile Gly Ser Asp Asp Asp Leu Phe Ile Ser Lys Trp Leu Val Glu		
690	695	700
His Ile Arg Gly Phe Lys Lys Ala Cys Glu Asp Ser Leu Lys Ile Gln		
705	710	715 720
Lys Asp Asn Arg Gly Leu Leu Asn His Lys Ile Asn Ile Ala Arg Asn		
725	730	735
Thr Lys Gly Lys Cys Glu Lys Glu Ile Phe Asn Leu Ile Cys Lys Ile		
740	745	750
Glu Gly Ser Glu Asp Lys Lys Gly Asn Tyr Lys His Gly Leu Ala Tyr		
755	760	765
Glu Leu Gly Val Leu Leu Phe Gly Glu Pro Asn Glu Ala Ser Lys Pro		
770	775	780
Glu Phe Asp Arg Lys Ile Lys Lys Phe Asn Ser Ile Tyr Ser Phe Ala		
785	790	795 800
Gln Ile Gln Gln Ile Ala Phe Ala Glu Arg Lys Gly Asn Ala Asn Thr		
805	810	815
Cys Ala Val Cys Ser Ala Asp Asn Ala His Arg Met Gln Gln Ile Lys		
820	825	830
Ile Thr Glu Pro Val Glu Asp Asn Lys Asp Lys Ile Ile Leu Ser Ala		
835	840	845
Lys Ala Gln Arg Leu Pro Ala Ile Pro Thr Arg Ile Val Asp Gly Ala		
850	855	860
Val Lys Lys Met Ala Thr Ile Leu Ala Lys Asn Ile Val Asp Asp Asn		
865	870	875 880
Trp Gln Asn Ile Lys Gln Val Leu Ser Ala Lys His Gln Leu His Ile		
885	890	895
Pro Ile Ile Thr Glu Ser Asn Ala Phe Glu Phe Glu Pro Ala Leu Ala		
900	905	910
Asp Val Lys Gly Lys Ser Leu Lys Asp Arg Arg Lys Lys Ala Leu Glu		
915	920	925
Arg Ile Ser Pro Glu Asn Ile Phe Lys Asp Lys Asn Asn Arg Ile Lys		
930	935	940
Glu Phe Ala Lys Gly Ile Ser Ala Tyr Ser Gly Ala Asn Leu Thr Asp		
945	950	955 960
Gly Asp Phe Asp Gly Ala Lys Glu Glu Leu Asp His Ile Ile Pro Arg		
965	970	975
Ser His Lys Lys Tyr Gly Thr Leu Asn Asp Glu Ala Asn Leu Ile Cys		
980	985	990
Val Thr Arg Gly Asp Ala Lys Asn Lys Gly Asn Arg Ile Phe Cys Leu		
995	1000	1005
Arg Asp Leu Ala Asp Asn Tyr Lys Leu Lys Gln Phe Glu Thr Thr		
1010	1015	1020
Asp Asp Leu Glu Ile Glu Lys Lys Ile Ala Asp Thr Ile Trp Asp		
1025	1030	1035

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Ala Asn	Lys Lys Asp Phe	Lys	Phe Gly Asn Tyr	Arg	Ser Phe Ile
1040		1045		1050	
Asn Leu	Thr Pro Gln Glu	Gln	Lys Ala Phe Arg	His	Ala Leu Phe
1055		1060		1065	
Leu Ala	Asp Glu Asn Pro	Ile	Lys Gln Ala Val	Ile	Arg Ala Ile
1070		1075		1080	
Asn Asn	Arg Asn Arg Thr	Phe	Val Asn Gly Thr	Gln	Arg Tyr Phe
1085		1090		1095	
Ala Glu	Val Leu Ala Asn	Asn	Ile Tyr Leu Arg	Ala	Lys Lys Glu
1100		1105		1110	
Asn Leu	Asn Thr Asp Lys	Ile	Ser Phe Asp Tyr	Phe	Gly Ile Pro
1115		1120		1125	
Thr Ile	Gly Asn Gly Arg	Gly	Ile Ala Glu Ile	Arg	Gln Leu Tyr
1130		1135		1140	
Glu Lys	Val Asp Ser Asp	Ile	Gln Ala Tyr Ala	Lys	Gly Asp Lys
1145		1150		1155	
Pro Gln	Ala Ser Tyr Ser	His	Leu Ile Asp Ala	Met	Leu Ala Phe
1160		1165		1170	
Cys Ile	Ala Ala Asp Glu	His	Arg Asn Asp Gly	Ser	Ile Gly Leu
1175		1180		1185	
Glu Ile	Asp Lys Asn Tyr	Ser	Leu Tyr Pro Leu	Asp	Lys Asn Thr
1190		1195		1200	
Gly Glu	Val Phe Thr Lys	Asp	Ile Phe Ser Gln	Ile	Lys Ile Thr
1205		1210		1215	
Asp Asn	Glu Phe Ser Asp	Lys	Lys Leu Val Arg	Lys	Lys Ala Ile
1220		1225		1230	
Glu Gly	Phe Asn Thr His	Arg	Gln Met Thr Arg	Asp	Gly Ile Tyr
1235		1240		1245	
Ala Glu	Asn Tyr Leu Pro	Ile	Leu Ile His Lys	Glu	Leu Asn Glu
1250		1255		1260	
Val Arg	Lys Gly Tyr Thr	Trp	Lys Asn Ser Glu	Glu	Ile Lys Ile
1265		1270		1275	
Phe Lys	Gly Lys Lys Tyr	Asp	Ile Gln Gln Leu	Asn	Asn Leu Val
1280		1285		1290	
Tyr Cys	Leu Lys Phe Val	Asp	Lys Pro Ile Ser	Ile	Asp Ile Gln
1295		1300		1305	
Ile Ser	Thr Leu Glu Glu	Leu	Arg Asn Ile Leu	Thr	Thr Asn Asn
1310		1315		1320	
Ile Ala	Ala Thr Ala Glu	Tyr	Tyr Tyr Ile Asn	Leu	Lys Thr Gln
1325		1330		1335	
Lys Leu	His Glu Tyr Tyr	Ile	Glu Asn Tyr Asn	Thr	Ala Leu Gly
1340		1345		1350	
Tyr Lys	Lys Tyr Ser Lys	Glu	Met Glu Phe Leu	Arg	Ser Leu Ala
1355		1360		1365	
Tyr Arg	Ser Glu Arg Val	Lys	Ile Lys Ser Ile	Asp	Asp Val Lys
1370		1375		1380	
Gln Val	Leu Asp Lys Asp	Ser	Asn Phe Ile Ile	Gly	Lys Ile Thr
1385		1390		1395	
Leu Pro	Phe Lys Lys Glu	Trp	Gln Arg Leu Tyr	Arg	Glu Trp Gln
1400		1405		1410	
Asn Thr	Thr Ile Lys Asp	Asp	Tyr Glu Phe Leu	Lys	Ser Phe Phe
1415		1420		1425	

Asn	Val	Lys	Ser	Ile	Thr	Lys	Leu	His	Lys	Lys	Val	Arg	Lys	Asp
1430						1435					1440			
Phe	Ser	Leu	Pro	Ile	Ser	Thr	Asn	Glu	Gly	Lys	Phe	Leu	Val	Lys
1445						1450					1455			
Arg	Lys	Thr	Trp	Asp	Asn	Asn	Phe	Ile	Tyr	Gln	Ile	Leu	Asn	Asp
1460						1465					1470			
Ser	Asp	Ser	Arg	Ala	Asp	Gly	Thr	Lys	Pro	Phe	Ile	Pro	Ala	Phe
1475						1480					1485			
Asp	Ile	Ser	Lys	Asn	Glu	Ile	Val	Glu	Ala	Ile	Ile	Asp	Ser	Phe
1490						1495					1500			
Thr	Ser	Lys	Asn	Ile	Phe	Trp	Leu	Pro	Lys	Asn	Ile	Glu	Leu	Gln
1505						1510					1515			
Lys	Val	Asp	Asn	Lys	Asn	Ile	Phe	Ala	Ile	Asp	Thr	Ser	Lys	Trp
1520						1525					1530			
Phe	Glu	Val	Glu	Thr	Pro	Ser	Asp	Leu	Arg	Asp	Ile	Gly	Ile	Ala
1535						1540					1545			
Thr	Ile	Gln	Tyr	Lys	Ile	Asp	Asn	Asn	Ser	Arg	Pro	Lys	Val	Arg
1550						1555					1560			
Val	Lys	Leu	Asp	Tyr	Val	Ile	Asp	Asp	Asp	Ser	Lys	Ile	Asn	Tyr
1565						1570					1575			
Phe	Met	Asn	His	Ser	Leu	Leu	Lys	Ser	Arg	Tyr	Pro	Asp	Lys	Val
1580						1585					1590			
Leu	Glu	Ile	Leu	Lys	Gln	Ser	Thr	Ile	Ile	Glu	Phe	Glu	Ser	Ser
1595						1600					1605			
Gly	Phe	Asn	Lys	Thr	Ile	Lys	Glu	Met	Leu	Gly	Met	Lys	Leu	Ala
1610						1615					1620			
Gly	Ile	Tyr	Asn	Glu	Thr	Ser	Asn	Asn						
1625						1630								

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<210> SEQ ID NO 11
<211> LENGTH: 19
<212> TYPE: PRT
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: Signal Sequence for VEGF scFV

<400> SEQUENCE: 11

Met Ala Trp Thr Ser Leu Ile Leu Ser Leu Leu Ala Leu Cys Ser Gly
1             5             10             15
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Ala Ser Ser

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<210> SEQ ID NO 12
<211> LENGTH: 108
<212> TYPE: PRT
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: VL for VEGF scFV
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<400> SEQUENCE: 12

Asp	Ile	Gln	Met	Thr	Gln	Ser	Pro	Ser	Ser	Leu	Ser	Ala	Ser	Val	Gly
1				5					10					15	
Asp	Arg	Val	Thr	Ile	Thr	Cys	Arg	Ala	Ser	Gln	Asp	Val	Ser	Thr	Ala
			20					25					30		
Val	Ala	Trp	Tyr	Gln	Gln	Lys	Pro	Gly	Lys	Ala	Pro	Lys	Leu	Leu	Ile
		35				40						45			
Tyr	Ser	Ala	Ser	Phe	Leu	Tyr	Ser	Gly	Val	Pro	Ser	Arg	Phe	Ser	Gly
	50					55					60				

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Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser Leu Gln Pro
65 70 75 80

Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Gly Tyr Gly Asn Pro Phe
85 90 95

Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg
100 105

<210> SEQ ID NO 13
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: Glycine-Serine Linker for VEGF scFV

<400> SEQUENCE: 13

Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Ser
1 5 10 15

<210> SEQ ID NO 14
<211> LENGTH: 116
<212> TYPE: PRT
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: VH for VEGF scFV

<400> SEQUENCE: 14

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly
1 5 10 15

Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Thr Ile Ser Asp Tyr
20 25 30

Trp Ile His Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val
35 40 45

Ala Gly Ile Thr Pro Ala Gly Gly Tyr Thr Tyr Tyr Ala Asp Ser Val
50 55 60

Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr
65 70 75 80

Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys
85 90 95

Ala Arg Phe Val Phe Phe Leu Pro Tyr Ala Met Asp Tyr Trp Gly Gln
100 105 110

Gly Thr Leu Val
115

<210> SEQ ID NO 15
<211> LENGTH: 127
<212> TYPE: PRT
<213> ORGANISM: Artificial
<220> FEATURE:
<223> OTHER INFORMATION: VL for VEGF scFV

<400> SEQUENCE: 15

Met Ala Trp Thr Ser Leu Ile Leu Ser Leu Leu Ala Leu Cys Ser Gly
1 5 10 15

Ala Ser Ser Asp Ile Gln Met Thr Gln Ser Pro Ser Ser Leu Ser Ala
20 25 30

Ser Val Gly Asp Arg Val Thr Ile Thr Cys Arg Ala Ser Gln Val Ile
35 40 45

Arg Arg Ser Leu Ala Trp Tyr Gln Gln Lys Pro Gly Lys Ala Pro Lys
50 55 60

Leu Leu Ile Tyr Ala Ala Ser Asn Leu Ala Ser Gly Val Pro Ser Arg

-continued

65	70	75	80
Phe Ser Gly Ser Gly Ser Gly Thr Asp Phe Thr Leu Thr Ile Ser Ser			
	85	90	95
Leu Gln Pro Glu Asp Phe Ala Thr Tyr Tyr Cys Gln Gln Ser Asn Thr			
	100	105	110
Ser Pro Leu Thr Phe Gly Gln Gly Thr Lys Val Glu Ile Lys Arg			
	115	120	125

<210> SEQ ID NO 16
 <211> LENGTH: 117
 <212> TYPE: PRT
 <213> ORGANISM: Artificial
 <220> FEATURE:
 <223> OTHER INFORMATION: VH for VEGF scFV

<400> SEQUENCE: 16

Glu Val Gln Leu Val Glu Ser Gly Gly Gly Leu Val Gln Pro Gly Gly			
1	5	10	15
Ser Leu Arg Leu Ser Cys Ala Ala Ser Gly Phe Ser Ile Asn Gly Ser			
	20	25	30
Trp Ile Phe Trp Val Arg Gln Ala Pro Gly Lys Gly Leu Glu Trp Val			
	35	40	45
Gly Ala Ile Trp Pro Phe Gly Gly Tyr Thr His Tyr Ala Asp Ser Val			
	50	55	60
Lys Gly Arg Phe Thr Ile Ser Ala Asp Thr Ser Lys Asn Thr Ala Tyr			
65	70	75	80
Leu Gln Met Asn Ser Leu Arg Ala Glu Asp Thr Ala Val Tyr Tyr Cys			
	85	90	95
Ala Arg Trp Gly His Ser Thr Ser Pro Trp Ala Met Asp Tyr Trp Gly			
	100	105	110
Gln Gly Thr Leu Val			
	115		

We claim:

1. An exogenous extracellular sensor, comprising a target chain (TC) and a protease chain (PC),

wherein the TC comprises: a) a first scFv ligand binding domain, b) a first extracellular linker comprising 26-30 amino acids, c) a first transmembrane domain, d) a first intracellular linker comprising 2-6 amino acids, wherein the first two amino acids of the first intracellular linker are the first two amino acids of a tobacco etch virus (TEV) protease cleavage sequence, e) a protease cleavage site, and f) a functional domain comprising an RNA-binding subdomain fused to a transcription regulatory subdomain; and

wherein the PC comprises: a) a second scFv ligand binding domain, b) a second extracellular linker comprising 26-30 amino acids, c) a second transmembrane domain, d) a second intracellular linker comprising 2-6 amino acids, wherein the first two amino acids of the second intracellular linker are the first two amino acids of the TEV protease cleavage sequence, and e) a protease capable of cleaving the protease cleavage site of the TC.

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2. The exogenous extracellular sensor of claim 1, wherein the ligand binding domains of the TC and PC bind to a species of biological relevance.

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3. The exogenous extracellular sensor of claim 2, wherein the species of biological relevance comprises: a growth factor, a cytokine, a chemokine, a cell-surface antigen, or a pathogen.

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4. The exogenous extracellular sensor of claim 1, wherein the RNA-binding subdomain comprises a RNA-binding domain of a Cas9 protein.

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5. The exogenous extracellular sensor of claim 1, wherein the transcription regulatory subdomain comprises a transcription activator for a cognate targeted promoter.

6. The exogenous extracellular sensor of claim 1, wherein the transcription regulatory subdomain comprises a transcription repressor for a cognate targeted promoter.

60

7. The exogenous extracellular sensor of claim 1, wherein the transcription regulatory subdomain comprises a subdomain that modulates the chromatin state for a cognate targeted promoter.

* * * * *